Introduction to Econometrics

Course notes | MGT581 | EPFL

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Preface

You are looking at the course notes for *Introduction to Econometrics* (MGT581) at EPFL.

Time stamp

This version: December 16, 2025. (Compare this date with the latest version available on moodle.)

Additional readings

Here's a short list of references if you want to deepen you knowledge in topics covered in this course.

Probability theory. A good overview is Casella and Berger (2021). Another one is DeGroot and Schervish (2002) but is less advanced. There exist many shorter introductions that cover the background needed for this course (e.g., Rozanov 1969). Stachurski (2016) has good chapters on the topic. For measure theory, see Billingsley (1995) or Tao (2011) (this will be overkill for the content of this course, except if you plan to go further into research). On asymptotic theory, see Van der Vaart (2000) or DasGupta (2008).

Econometrics. Recent textbooks include Verbeek (2018), Stock and Watson (2011). Hayashi (2000) and Davidson and MacKinnon (2004) are a bit older but good. Wooldridge (2002) covers panel data. Peter M. Aronow and Miller (2019) is a good modern treatment. B. Hansen (2022a) covers a lot of ground. Greene (2008) is a good reference on the mechanics of econometrics. Goldberger (1991) is now quite dated, but

it does a good job building up intuition, as does Kennedy (2003).

Causal inference. Morgan and Winship (2014) covers causal inference and quasi-experimental methods. A more technical approach to causal inference is Imbens and Rubin (2015). Gailmard (2014) contains both probability theory material as well as inference. Pearl (2009) introduces (directed acyclic) graphs, going in quite some details. J. Angrist and Pischke (2008) builds bridges between econometric theory and causal inference. M. Huber (2023) connects causal inference to machine learning.

Next steps. For other topics in econometrics and applied statistics, see Theodore W. Anderson (2003) (multivariate analysis), Hamilton (2020) (time series), Gelman and Hill (2007), Jackman (2009), and McElreath (2018) (Bayesian modeling), Hastie et al. (2009), Murphy (2022), Chernozhukov et al. (2024) (machine learning), Grimmer, Roberts, and Stewart (2022) (text-as-data).

Thanks

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1 Foundations

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> ⚠ **Caveat**
>
> These notes are offered in support of *Introduction to Econometrics* (MGT 581). They were written to help students understand the logic behind important topics in econometrics. But they don't (always) replace the insights and depth that textbooks can provide. This is why we connect each session to chapters of our core textbooks. See the syllabus for more details.

1.1 Readings

This chapter provides a refresher on a few foundational concepts of this course. Some of them will be used repeatedly. Others belong to the "good to know" category.

Textbooks that cover foundational material in mathematical statistics include Casella and Berger (2021), DeGroot and Schervish (2002), or Durrett (2019). B. Hansen (2022b) covers probability in a way that's tailored for econometrics. For measure theory, an older reference is Billingsley (1995). For asymptotic theory, see Van der Vaart (2000). In French, LeGall's lecture notes *Intégration, Probabilités et Processus Aléatoire* (available online) are good, but generally beyond what we will need.

1.2 Probability

1.2.1 The building blocks of probability

We will be working a lot with probabilities. They will allow us to compute all kinds of quantities we are interested in, including:

- 'Raw' probabilities. Eg, the probability of obtaining a contract for your business.
- Sampling. Eg, the probability of obtaining a representative set of respondents to your consumer survey.
- Expected values. Eg, the expected returns from a new product.
- Inference. Eg, a range of plausible values for the results of an election.

To get started, we need to define a few key concepts that will serve as out building blocks.

Define Ω (upper-case omega) to be the **sample space**, that is, the *set* of potential outcomes. Define $\omega \in \Omega$ (lower-case omega) as an (arbitrary) **outcome** from the sample space. One can also say that $\Omega := \{\omega\}$ (i.e., "Ω is defined as the set of outcomes ω").

What we are interested in are **events**. The realization of a specific outcome ω tells us whether the event occurred or not. In that sense, an event is a true/false statement. Define A as an event. If the outcome ω is realized, and if $\omega \in A$, we say that A occurred ('true'). See box.

Examples of sample spaces, outcomes, events

Example 1.1 (Example)**.** A countable sample space from throwing a dice:

- Sample space $\Omega := \{1, 2, 3, 4, 5, 6\}$
- Arbitrary outcome $\omega_4 = 4$
- Event A: even outcomes from the dice, i.e. $\{2, 4, 6\}$. *Did the dice yield an even number? Yes/no?*

Annual sales (in USD) of Nvidia are an uncountable sample space:

- $\Omega := \mathbb{R}_+$. Or written slightly differently: $\Omega := \{\omega | \omega \in \mathbb{R}_+\}$.
- $\omega_i = 26.974$ b (USD).
- A: annual sales in billions, e.g., $[26-27)$. *Did Nvidia make sales in the* $[26b - 27b)$ *interval? Yes/no?*

Another example of an uncountable sample space:

- $\Omega :=$ surface of a dart board.
- $\omega_{i,j} =$ hitting point $\{i, j\}$.
- A: hitting a specific area (e.g., the bull's eye). *Did the dart hit the bull's eye? Yes/no?*

What we want to figure out: *what is the probability of the event of interest A?* To answer this question rigorously, we need to tool up a little bit. We will need to work with a **probability space**.

Definition 1.1 (Probability space)**.** A probability space is a triple $(\Omega, \mathcal{F}, P)$, that is, a sample space, a σ-algebra (or σ-field), and a probability measure.

- Ω: this is the **sample space**, i.e. the set of all admissible outcomes (see above).
- $\mathcal{F}$: it's a **σ-algebra** (to be defined below). Colloquially, it is the set of events A under consideration (i.e. a set of events that we want to associate to a probability). Events are defined as set of outcomes.
- P: it is a **probability measure** (to be defined below). Colloquially, it is a function that assigns a probability to each event A in $\mathcal{F}$.

At this stage, you might be puzzled by $\mathcal{F}$: why do we need this and what is it in the first place? Why not just attach probabilities to ω? There are two connected reasons for this. First, we are interested in *events* A, not *outcomes* ω. So we must build the set of events over which probabilities must be

defined. This is $\mathcal{F}$, namely a set of subsets of Ω that form the list (set) of events we are interested in.

Second, we could build up a set of events that displays some pathologies. Thus, we will need an extra step to ensure that the set of events $\mathcal{F}$ is well-behaved and that we can assign probabilities in a way that meets several important criteria. In some cases, the definition of $\mathcal{F}$ is straightforward. This will be the case when Ω is countable. When Ω is not countable, however, then things become more complicated and we need to be careful about $\mathcal{F}$. That's why we want $\mathcal{F}$ to be a σ-algebra, because that guarantees that $\mathcal{F}$ exhibits all kind of nice properties. See box for a formal definition of σ-algebra.

It's not a great analogy, but: you could think of ω as letters and $\mathcal{F}$ as a dictionary. The letters are uninformative without the latter.

See Figure 1.1 for an illustration of the relation between Ω, $\mathcal{F}$, and P.

σ-algebra

Definition 1.2 (σ-algebra)**.** We say that the set $\mathcal{F}$ of subsets of Ω is a σ-algebra if:

1. $A \in \mathcal{F} \to A^c \in \mathcal{F}$ (if you can give a probability to A, you can also give it to non-A) ("closed under the taking of complements")
2. $A_1, A_2, ... \in \mathcal{F} \to \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$ ("closed under the taking of unions")
3. $\Omega \in \mathcal{F}$

Note:

- Conditions (1) and (3) imply that $\emptyset \in \mathcal{F}$.
- Using **De Morgan's law**, we note that a σ-algebra is closed under countable *intersections* in addition to unions.
- A σ-algebra doesn't itself have to be countable.

Illustration of a σ-algebra

Example 1.2 (σ-algebra)**.** Consider a dice thrown a single time. A σ-algebra set could be as follows. Suppose the event we are interested in is whether a throw is even ($A = \{2, 4, 6\}$). Then:

$$\mathcal{F} = \{\underbrace{\{1, 2, 3, 4, 5, 6\}}_{\Omega}, \emptyset, \underbrace{\{2, 4, 6\}}_{A}, \underbrace{\{1, 3, 5\}}_{B}\}$$

This qualifies as a σ-algebra:

1. A and A^c are included in $\mathcal{F}$. A is defined as $\{2, 4, 6\}$ and A^c is $\{1, 3, 5\}$ (i.e. B).
2. The union of all sets is included. Here, the union of $\{2, 4, 6\}$ and $\{1, 3, 5\}$ is there (trivially, since it's Ω).
3. $\Omega \in \mathcal{F}$ (that's the set $\{1, 2, 3, 4, 5, 6\}$).

Obviously, a more comprehensive σ-algebra than this one would be 2^{Ω}, which contains 64 sets plus the empty set. But we might have reasons to restrict ourselves (e.g., if we don't see the dice outcome ourselves and someone only tells us if it's even or odd).
The next step is to assign probabilities to each sets.

1.2.2 Building a σ-algebra

How to build a σ-algebra depends on the nature of the outcomes Ω we are interested in. To keep things simple, we can broadly distinguish between two types of Ω. The easy case is when Ω is finite and countable. A set is countable if its elements can be placed in a one-to-one correspondence with natural numbers $\mathbb{N}$. In that case, we could build a σ-algebra from the power set of Ω (sometimes also written $\mathcal{P}(S)$).

Remember: countable is not the same as finite. A set can be countable and infinite (e.g., $\mathbb{Q}$). A set can be finite yet uncountable (e.g., $[-1, 1]$).

$$\mathcal{F} = 2^{\Omega}$$

For instance, let's suppose we are tossing two coins. For now, let's say that we don't yet know what kind of events we are

interested in. Then, the power set of Ω allows us to take into account all possible events:

$$\begin{aligned}\mathcal{F} = 2^{\Omega} =\{&\emptyset, \{(H,H)\}, \{(H,T)\}, \{(T,H)\}, \{(T,T)\},\\ &\{(H,H),(H,T)\},\\ &\{(H,H),(T,H)\},\\ &\{(H,H),(T,T)\},\\ &\{(H,T),(T,H)\},\\ &\{(H,T),(T,T)\},\\ &\{(T,H),(T,T)\},\\ &\{(H,H),(H,T),(T,H)\},\\ &\{(H,H),(H,T),(T,T)\},\\ &\{(H,H),(T,H),(T,T)\},\\ &\{(H,T),(T,H),(T,T)\},\\ &\underbrace{\{(H,H),(H,T),(T,H),(T,T)\}}_{\Omega}\}\end{aligned}$$

For instance, we could now decide that we are interested in the event of getting two identical tosses (A_1) versus not (A_2). We would define:

$$\begin{aligned}A_1 &:= \{(H,H),(T,T)\} = A_2^c\\ A_2 &:= \{(H,T),(T,H)\} = A_1^c\end{aligned}$$

As you can see, each event A_1 and A_2 and their complement is in $\mathcal{F}$ and so is their union. It is therefore a σ-algebra. More generally, here, this would be true of any event involving two coin tosses. That's why the power set is a neat way to build the σ-algebra.

Things get a bit more complicated with continuous Ω. For instance: people's height. We obviously cannot use the same strategy. Instead, we will use intervals on Ω. We restrict $\mathcal{F}$ (the set of admissible events) to only include subsets that have a determinable length. That's one of the benefits of restricting ourselves to σ-algebra! Limiting ourselves to a σ-algebra guarantees that we will be able to assign well-defined probabilities to each event in $\mathcal{F}$. The other nice thing is that the requirements for a σ-algebra are desirable:

Here it gets a bit complicated, but the short answer is that not all subsets on $\mathbb{R}$ (which is what we are using in this example) have a measurable **length** (here: the 'distance' between a and b). As a result, probabilities cannot be defined for these. Those interested in this should read more about measurement theory (Tao 2011) and problems such as the Banach-Tarski paradox.

One particularly useful class of σ-algebras in such cases, whenever we deal with $\mathbb{R}^N$ (or subsets thereof), is the **Borel σ-algebra** (or **Borel field**) of $\mathcal{F}$ (typcally written $\mathcal{B}(\Omega)$). $\mathcal{B}(\mathbb{R})$ is the σ-algebra of all open sets $(a, b) : a < b$ on $\mathbb{R}$. Since it includes all unions, intersections, and complements (per definition of a σ-algebra), it means that it includes every closed, open, and semi open sets on the real line. That is, it contains $[a, b]$, $(a, b]$, $[a, b)$, and (a, b) for all a, b. (Adapt accordingly if we go in higher dimensions $\mathbb{R}^{\mathbb{N}}$.)

The benefit of a σ-algebras in general and Borel fields in particular? We can define probabilities even for uncountable sample spaces Ω, including especially $\mathbb{R}$. Thus, probability isn't limited to coin tosses and games of cards. It can also apply to sample spaces such as (subsets of) the real line $\mathbb{R}$.[1]

To summarize: most of the time, we can typically work with one of two σ-algebras to build probability spaces:

1. $\mathcal{F} = 2^{\Omega}$ if Ω is countable.
2. $\mathcal{F} = \mathcal{B}(\Omega)$ if Ω is on the real line $\mathbb{R}$.

But sometimes that's not feasible. For instance, if we only observe the sum of two throws of a dice, we cannot always infer what each of the two throws is, and thus we are using another set.

1.2.3 Building a probability measure

We now have our set of events, $\mathcal{F}$. We can then assign probabilities to each element of $\mathcal{F}$. We define a **probability measure** as a function $\mathbb{P}$ on the (countable) sample space Ω:

$$\mathbb{P} : \mathcal{F} \to [0, 1]$$

Following the axiomatic approach to probability (Kolmogorov 1956), a probability measure must also meet the following conditions (**Axioms of Probability** or **Kolmogorov axioms**):

[1] Using Borel fields will also be useful to ensure that values of random variables can be attached to a well-defined probability (see below).

1. $\mathbb{P}(A) \geq 0$ for all $A \in \mathcal{F}$
2. $\mathbb{P}(\Omega) = 1$ for $\Omega \in \mathcal{F}$
3. If $A_1, A_2, ... \in \mathcal{F}$ are mutually disjoint, then $\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n)$.

The definition of a σ-algebra ensures that each of these three axioms can be met.

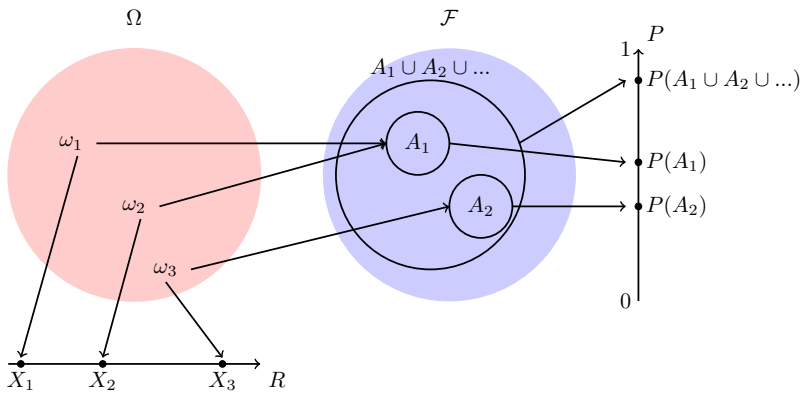

Figure 1.1: Probability space. Omega contains three outcomes that map onto two events. Each set of events maps onto a probability, i.e. a real number on the [0,1] interval.

1.2.4 What's a probability?

From Kolmogorov, we know that a probability function must satisfy a very small number of axioms ($\mathbb{P}(\Omega) = 1$, etc.). But that doesn't tell us how to conceptualize and define a probability. The most intuitive way to think about a probability is to define it as a ratio of relative frequencies (Rozanov 1969):

$$\mathbb{P}(A) = \frac{N(A)}{N},$$

where N is the number of outcomes and $N(A)$ is the number of outcomes leading to the event A. We can also think of this with an empiricist's view as a limit: the relative frequency of

A as the number of trials goes toward infinity:[2]

$$\mathbb{P}(A) = \lim_{n \to \infty} \frac{n(A)}{n},$$

where lower case n denotes the number of trials.

Once we think of probabilities as ratios, it becomes natural to think about counting the numerator and denominator. That's where textbooks introduce combinatorial analysis. See for instance Rozanov (1969, 1.2).

To connect this with what we have done so far: suppose we have a countable Ω. Say, three candidates want to be elected Mayor of Lausanne. Call them $\{A, B, C\}$. Suppose we could run 10 elections under identical and independent conditions. This would mean that we get 3^{10} combinations (59,049) of potential sequences of winners. One arbitrary combination is: $\{A, B, B, A, C, A, ...\}$. That's the denominator. Then, we would count the number of times candidate A, say, wins. Of course, to be accurate, we would want to move from 10 to an infinity number of elections.

This is the kind of stuff that made Nate Silver rich.

1.2.5 Implications and additional concepts

We can generate several implications and additional concepts from these building blocks.

Let $(\Omega, \mathcal{F}, P)$ be an arbitrary probability space. Assume $A, B \in \mathcal{F}$. If $A \subset B$, then:

1. $\mathbb{P}(B \backslash A) = \mathbb{P}(B) - \mathbb{P}(A)$
2. $\mathbb{P}(A) \leq \mathbb{P}(B)$

Note also:

3. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$

[2] There are other views about what probabilities are. Some prominent mathematicians and statisticians, such as Frank Ramsey, studied probabilities as a *subjective* belief rather than some kind of (asymptotic) property of the world. See Ramsey (1926), Savage (1951), and De Finetti (1975).

4. $\mathbb{P}(\emptyset) = 0$

Next, we can define the concept of **conditional probability**. Given two events A, B (and B with positive probability), the conditional probability of A given B is:

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

This is often a source of confusion among people. $\mathbb{P}(A|B)$ is a very different quantity compared to $\mathbb{P}(B|A)$. For instance, consider the probability of speeding, given that you are a man, versus the probability of being a man, given that you are speeding.

We can then recover **Bayes' law**:

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)}$$

One type of conditional probability that we will be interested in a lot is the following: *given* the data we obtained (our sample), what is the probability of a parameter of interest being equal to a certain value. For instance, denote $\mathbf{X}$ a set of data and τ an unknown parameter (or vector of unknown parameters) that says something about $\mathbf{X}$, we may wonder what the latter's conditional probability distribution is:

$$\mathbb{P}(\tau|\mathbf{X}).$$

And sometimes we may do the opposite: given a dataset, what is the probability of a given parameter.

$$\mathbb{P}(\mathbf{X}|\tau).$$

This logic underpins null hypothesis testing.

Next, we say that two events A, B are **independent** if and only if:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B).$$

Note that if A and B are independent, then $\mathbb{P}(A|B) = \mathbb{P}(A)$.

More generally, for independent events $A_1, A_2, ..., A_N \in \mathcal{F}$:

$$\mathbb{P}(\cap_{n=1}^{N} A_n) = \prod_{n=1}^{N} \mathbb{P}(A_n).$$

Intuitively, the notion of independence means that knowing whether B happened or not doesn't move the needle regarding our belief that A happened. For instance, knowing whether it rained in New York doesn't give me information regarding whether I decided to bike in Lausanne. However, knowing whether it rained in Geneva does change the likelihood that I decided to bike to EPFL.

1.2.6 Probability distributions

Probabilities are fairly straightforward when we have a probability space over discrete events. Things gets more complicated with continuous events. When the sample space is the real line $\mathbb{R}$, we saw that it was convenient to restrict the σ-algebra $\mathcal{F}$ to the Borel set $\mathcal{B}(\mathbb{R})$.

Still, this doesn't tell us what the probability of any subset of the real line is. Now: a probability measure defined over $\mathcal{B}(\mathbb{R})$ is called a **distribution** (or sometimes a **law**).

Some distributions are more common than others. For instance, the normal (Gaussian) probability density function (PDF) is:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

We generally write more compactly $X \sim N(\mu, \sigma^2)$ ("X is distributed normally with first moment (mean) μ and second moment (variance) σ^2"). We shall shortly talk more about this, but note that this distribution function is entirely parametrized by two variables: μ and σ^2, which correspond to the mean (or first moment) and the variance (or second moment) of the distribution.

It will sometimes useful to work with the **cumulative distribution function** (CDF).

$$\Phi(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{x} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx$$

Here, we integrate from minus infinity upward to some value $x_i \in \mathbb{R}$. This tells us how much mass under the curve is below x_i. Of course, if we integrate from minus to plus infinity, we obtain 1 (since $\mathbb{P}(\Omega) = 1$). The stuff in front of the integral is here precisely to ensure that the integral sums up to one.

For a historical account of the normal distribution, tracing back to Abraham de Moivre, see Hald (2008).

Notice that both expressions become considerably easier if we center X (so that $\mu = 0$) and if we standardize it (so that $\sigma^2 = 1$). In that case, we talk of the **standard normal distribution** $N \sim (0, 1)$.

There are many other distributions that we will run into (F distribution, χ^2 distribution, etc.). At this stage, no need to dig deeper into them, since doing so is not particularly enlighting.

1.3 Random variables

A random variable is a measurable function x that maps from the outcome space to a measurable space $\mathcal{M}$:

$$x : \Omega \to \mathcal{M}$$

Oftentimes, $\mathcal{M}$ will be defined by the real line $\mathbb{R}$. This would make it a **continuous** random variable. Sometimes, the random variable will be **discrete**, as when it maps onto $\mathbb{N}$.

We can link random variables back to σ-algebras.

Valid random variables

For a random variable X to valid, it must be **measurable** with respect to $(\Omega, \mathcal{F})$. What it means is: the set of *outcomes* (Ω) over which an event A is defined must be in $\mathcal{F}$.
For instance, suppose that $\Omega = \mathbb{R}$, and that we have just

two events: $x < 0$ (A_1) and $x \geq 0$ ($A_2 = A_1^c$). It must be that the set of outcomes (i.e., the event) that allow us to say whether $x < 0$ or $x \geq 0$ is in $\mathcal{F}$. Here, it would mean that the sets $(-\infty, 0)$ and $[0, \infty]$ must be $\mathcal{F}$ (which is indeed the case if $\mathcal{F}$ is the Borel-field).

Notice that x doesn't have to yield an actually *variable* variable. For instance, x could generate a constant c. More generally, this function x can take all kinds of shape. For instance, consider a famous case (in some circles at least), namely Bernoulli's St. Petersburg paradox. Someone tosses a coin until a head is obtained, at which stage the game ends. Suppose that we want to know the number of tails until head. Suppose that tails equal zero and heads 1. The random variable is (Stachurski 2016, 90):

$$x(\omega) = x(\text{toss } \#1, \text{ toss } \#2, ...) = \min\{n \in \mathbb{N} : \text{toss}_n = 1\}$$

1.3.1 Describing random variables

Random variables are drawn from an underlying **probability density function** (PDFs). As we saw, some are well known: the normal distribution, the Student's t distribution, etc. Other variables could follow arbitrary PDFs.

Be this as it may, we often find it useful to describe random variables based on their **moments**. These are parameters of the underlying PDF that help us describe them.

The **first moment** is **expection** of a variable (aka μ). Given a probability space $(\Omega, \mathcal{F}, P)$, the first moment is defined as:

$$\mathbb{E}[X] := \mu_x = \sum_{i=1}^{J} x_i \cdot \mathbb{P}[X = x_i] \qquad \text{discrete RV}$$

$$\mathbb{E}[X] := \mu_x = \int_{-\infty}^{\infty} x_i \cdot f_X(x_i) dx \qquad \text{continuous RV}$$

where $\mathbb{P}(x_j)$ is the probability of event x_j and $f_X(x)$ is the probability density function of X.

Note here that this is why we care a lot about well-defined probabilities (and thus why we talked about σ-algebra): without well-defined probabilities, we cannot compute any moment of X. And as we will see in a second, we care *a lot* about moments in econometrics (and any discipline that tries to make causal claims from data). See box.

We can extend this definition to expectations of functions (and similarly for discrete functions):

$$\mathbb{E}[g(X)] = \int g(x_i) \cdot f_X(x_i) dx$$

Probabilities and expectations can be conneceted in another way, namely **Markov's inequality**. For a non-negative random variable X and any $\delta > 0$:

$$\mathbb{P}(x \geq \delta) \leq \frac{\mathbb{E}[X]}{\delta}$$

Random variables and Borel subsets

The reason a probability might not be defined is a tad complicated. The short version is the following: suppose we want to know the probability $\mathbb{P}(x(\omega) \in S)$ for some subset $S \in \mathbb{R}$. For instance, we want to know the probability that $x(\omega)$ is in the $[9, 10]$ interval. The problem is that nothing guarantees that the $[9, 10]$ interval is in the σ-algebra $\mathcal{F}$. Yet probabilities are only defined for elements in $\mathcal{F}$! The solution to this is to restrict x to be defined over Borel subsets of $\mathbb{R}$ (i.e., $\mathcal{B}(\Omega)$).

Properties of expectations

Expectations have several properties that derive from their definition.

- For a constant a:

$$\mathbb{E}[a] = a$$

- For any random variables, $X, Y, ..., Z$:

$$\mathbb{E}[X + Y + ... + Z] = \mathbb{E}[X] + \mathbb{E}[Y] + ... + \mathbb{E}[Z]$$

- For a, b constant:

$$\mathbb{E}[a \cdot f(X) + b \cdot g(X)] = a\mathbb{E}[g(X)] + b\mathbb{E}[f(X)]$$

  As a result:

$$\mathbb{E}[aX + b] = \mathbb{E}[aX] + \mathbb{E}[b] = a\mathbb{E}[X] + b$$

- For two *independent* random variables X, Y:

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

- For an event A (where $A \in \{0, 1\}$), note that:

$$\begin{aligned}
\mathbb{E}[A] &= \sum_{i \in \{0,1\}} i\mathbb{P}[A = i] && \text{by def} \\
&= 0 \cdot \mathbb{P}[A = 0] + 1 \cdot \mathbb{P}[A = 1] && \text{expanding} \\
&= \mathbb{P}[A = 1] \\
&= \mathbb{P}[A]
\end{aligned}$$

  Thus, the expected value of a dichotomous event is equal to its probability.

Note. We talk of **centered moment** if we compute the moment of a centered variable, i.e. $X - \mathbb{E}[X]$. A centered first moment equals 0 by construction.

Note. Extending the definition of expected value $\mathbb{E}[A]$ to *conditional* expected value $\mathbb{E}[A|B]$ is reasonably straightforward. For instance:

$$\mathbb{E}[X|Y] = \sum_{j=1}^{J} x_j \cdot \mathbb{P}(X = x_j|Y),$$

where the last term is a conditional probability.

The first moment tells us "where" the random variable X is "anchored" on the real line. This will be a quantity that we will be extremely interested in. For instance, we could be interested in the expected salary of individuals *given* that they obtained a college degree:

$$\mathbb{E}[\text{Salary}|\text{College degree}].$$

We might then be interested in figuring out what the gains of college are:

$$\delta = \mathbb{E}[\text{Salary}|\text{College degree}] - \mathbb{E}[\text{Salary}|\text{No college degree}].$$

Or we might be interested in the expected effect of a new product on a firm's profit:

$$\delta = \mathbb{E}[\text{Profit}|\text{Innovation}] - \mathbb{E}[\text{Profit}|\text{No innovation}].$$

As we will see, we can interpret (under certain conditions) δ as an **average treatment effect** (here: of college, respectively of an innovation). We will spend a lot of the semester understanding how to **estimate** quantities like these in a credible manner. Because here's the problem: *these moments are unobserved **population** parameters.* We never observe the entire universe of observations. Thus, we will often work with **samples** from which we will estimate these parameters. Even $\mathbb{E}[X]$ must be estimated! This will create all kinds of complicated problems.

Distributions have often more than one moment. The general statement is that the kth moment of an already centered variable X is defined as:

$$\mathbb{E}[X^k],$$

otherwise the kth moment is $\mathbb{E}[(X-\mathbb{E}[X])^k]$. Thus, the **second moment** (aka the **variance**, $\mathbb{V}$, or σ^2) is:

$$\mathbb{V}[X] := \sigma_X^2 := \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2,$$

which directly gives us the standard deviation:

$$\text{SD}(X) := \sigma_X := \sqrt{\mathbb{E}[(X - \mathbb{E}[X])^2]}.$$

Both quantities are strictly positive by **Jensen's inequality** (Jensen 1906). The variance tells us how spread out the distribution of X is. This will be very helpful when we think about the uncertainty surrounding our estimates of treatment effects. Thanks to this, we will be able to engage in **hypothesis testing** and build **confidence intervals**.

Notice that if the random variable is centered (i.e., transformed such that $\mathbb{E}[X] = 0$), then this simplifies to:

$$\mathbb{V}[X] = \mathbb{E}[X^2].$$

We can continue like this. The **skewness** of a distribution (3rd moment) tells us how asymmetric the distribution of X is. If skewness is zero, then the distribution is symmetrical. If skewness is positive, then the distribution has a long right tail.

The **kurtosis** (4th moment) tells us somehting about the mass of the tails of the distribution. If it's large, then we have heavy tails ("leptokurtotic" distribution).

Skewness and kurtosis often generate less interest. But some audiences will value them. For instance, the consequences of fat tails can be dramatic in stock markets.

Note that distributions generally have a finite number of moments. The normal distribution, for instance, is entirely defined by its first and second moments. For the normal distribution higher moments are undefined (or ∞). And some variables don't have any moments (i.e. only infinite moments), like Cauchy.

1.4 Association between random variables

We are often interested in the relation between two (or more) variables. A starting point is the **covariance**, which is simply a co-first moment:

$$\mathrm{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

The problem with covariance is that its scale is not intuitive. We could end up with an estimate $\widehat{\mathrm{Cov}}(X, Y) = 8$; is this large? Small?

Thus, we often scale the covariance by the product of the standard deviations of X and Y to generate the **correlation coefficient** (aka ρ) between X and Y:

$$\mathrm{Cor}(X, Y) := \rho =: \frac{\mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]}{\sigma_x \sigma_y} \in [-1, 1].$$

Note that if X and Y are centered, this boils down to:

$$\begin{aligned}\mathrm{Cov}(X,Y) &= \mathbb{E}[XY] \\ \mathrm{Cor}(X,Y) &= \frac{\mathbb{E}[XY]}{\sqrt{\mathbb{E}[X^2]\mathbb{E}[Y^2]}}\end{aligned}$$

And note that if X and Y are independent, then the covariance and the correlation is zero, since:

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] = 0$$

Note that we don't observe the covariance or the correlation. These are population parameters (they are unobserved quantities). We will need to estimate them from a sample of observations, where population expectations are replaced with sample means.

1.5 Asymptotics

We are often interested in understanding how certain quantities of interest behave as the number of data points we have increases toward infinity. That's what we mean by **asymptotics**.

Here, a few key theorems and laws will be very helpful. But first, let's define the idea of **convergence**.

1.5.1 Convergences

The concept of **convergence** will often come up, especially when we will cover the problem of asymptotics..

There are different types of convergence.

Definition 1.3 (Convergence in probability)**.** A sequence of random variables $\{X_n\}$ converges in probability to a random variable X if:

$$\lim_{n\to\infty} \mathbb{P}(|X_n - X| \geq \varepsilon) = 0 \quad \forall \varepsilon > 0.$$

In other words, the probability of getting a value X_n further than ε from X goes toward zero as n goes to infinity.

This is also written as:

$$X_n \xrightarrow{p} X.$$

Definition 1.4 (Convergence in distribution (weak convergence))**.** A sequence of random variables $\{X_n\}$ converges in distribution to a random variable X if the CDF of X_n, denoted $F_n(x)$, converges to the CDF of X, denoted $F^*(x)$:

$$\lim_{n\to\infty} F_n(x) = F^*(x) \quad \forall x \text{ where } F^*(x) \text{ is continuous}$$

This is also reported as:

$$X_n \xrightarrow{d} X.$$

Definition 1.5 (Almost sure convergence (strong convergence))**.** A sequence of random variables $\{X_n\}$ converges almost surely to a random variable X if:

$$\mathbb{P}\left(\lim_{n\to\infty} X_n = X\right) = 1.$$

This is denoted as:

$$X_n \xrightarrow{a.s.} X.$$

Definition 1.6 (Convergence in mean square (or L^2 convergence))**.** A sequence of random variables $\{X_n\}$ converges in mean square to a random variable X if:

$$\lim_{n\to\infty} \mathbb{E}\left[(X_n - X)^2\right] = 0.$$

Convergence in mean square is written as:

$$X_n \xrightarrow{L^2} X.$$

Note:

1. **Almost sure convergence** implies **convergence in probability**.
2. **Convergence in probability** implies **convergence in distribution**.
3. **Convergence in mean square** implies **convergence in probability**

   Note: the reverse implications do not necessarily hold.

1.5.2 (Weak) Law of large numbers

The **weak law of large numbers** (WLLN) is going to be a useful result. WLLN is a result of the principle of convergence in probability.

Theorem 1.1 (Weak law of large numbers)**.** *Assume we have a sequence of random variables that are independent and identically distributed (IID)* $X_1, X_2, ...$ *with first moment* $\mathbb{E}[X] = \mu$ *and* $Var[X] = \sigma^2 < \infty$*. Define the mean of* X *up to* n *as* $\bar{X}_n = 1/n \sum_{i=1}^{n} X_i$*. Then:*

$$\lim_{n\to\infty} \mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) = 0, \qquad \forall \varepsilon > 0$$

In other words, the probability that the sample average is further from μ (true mean) than ε goes to zero. Or, put another way, the absolute value of the error (the gap between $\bar{X}_n$ and μ) is increasingly less likely to be greater than ε. Or put yet another way: the probability of getting an inaccurate value from our sample is less and less likely.

This result will be useful when we discuss the topic of **consistency**. Informally, we will be interested in figuring out whether a quantity of interest (such as an estimated treatment effect) is increasingly likely to be close to the true value of that quantity.

1.5.3 Central limit theory

The **central limit theorem** (CLT) offers a result that neighbors the WLLN. The idea here is that we could, hopefully, characterize the *distribution* of the difference between $\bar{X}_n$ and μ. That's what the CLT does.

There exists several variants of the CLT. We will primarily use the **Lindeberg–Lévy CLT**.

Theorem 1.2 (Lindeberg–Lévy CLT)**.** *Suppose we have a sequence of IID random variables $X_1, X_2, ...$ with first moment $\mathbb{E}[X] = \mu$ and second moment $Var[X] = \sigma^2 < \infty$. Then:*

$$\sqrt{N}\frac{(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0,1)$$

where $\xrightarrow{d}$ denotes convergence in distribution.

Or, equivalently:

$$\sqrt{N}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2).$$

Right now, it might not seem clear why we would care about this result (aside from its aesthetic elegance). What we will find out later is that it will help us tremendously with hypothesis testing.

Review questions

Exercise 1.1. Consider a study that wishes to establish the probability of receiving a patent for a given project, as a function of a R&D team's funding. Write down the probability space for such a hypothetical study.

Exercise 1.2. Construct a σ-algebra with respect to the set of events consisting in throwing three six-sided dices.

Exercise 1.3. Consider the following table which reports hypothetical frequencies of people using various modes of transportation (broken down by gender).

	Car	Public Transit	Bike	Total
Male	30	10	10	50
Female	20	25	5	50
Total	50	35	15	100

1. What is the probability that a randomly selected individual takes public transit?
2. What is the probability that a randomly selected female bikes to work?
3. Is biking dependent or independent from gender?

Exercise 1.4. Consider the Lindeberg-Lévy CLT (Theorem 1.2). What is the asymptotic distribution of $\bar{X}_n$?

Exercise 1.5. Consider the following sequence $\{X_n\}$ that can take one of two values:

$$X_n = \begin{cases} n & \text{with } \mathbb{P}\left(\frac{1}{n}\right), \\ 0 & \text{with } \mathbb{P}\left(1 - \frac{1}{n}\right). \end{cases}$$

Two questions:

1. Does $\{X_n\}$ converge in probability to $X = 0$?
2. Does $\{X_n\}$ converge in distribution to X?

2 Causal inference and the linear model

This version: December 16, 2025

> **Caveat**
>
> These notes are offered in support of *Introduction to Econometrics* (MGT 581). They were written to help students understand the logic behind important topics in econometrics. But they don't (always) replace the insights and depth that textbooks can provide. This is why we connect each session to chapters of our core textbooks. See the syllabus for more details.

2.1 Readings

Imbens and Rubin (2015) provides an in-depth introduction to causal effects from an econometric standpoint. Morgan and Winship (2014) does the same in a slightly less technical manner. Paul, Hall, and Collins (2004) offer philosophical insights into causality. J. Angrist and Pischke (2008) explains how identification issues arise and how they are (or were) solved in economics.

2.2 Treatment effects

A large part of econometrics consists in identifying causal effects. You can think of two ways to do so.

1. **Test hypotheses**: we use a discrete rule to assess whether some treatment has a causal effect (is there an effect: yes/no).
2. **Quantify**: we want to measure the strength of the causal effect of a cause on an effect (how strong/weak is the effect).

The notion of **cause** itself is tricky. Here, I follow Rubin (1974) and think of causal effects as **potential outcomes**. The idea is that the causal effect of a treatment is a contrast between an outcome under one state of the treatment and an outcome under a different state of the treatment.

See Lewis (1974) for a philosophical take on causation and counterfactuals. The idea is that there are "possible worlds" and causation is the 'what-if' relation from a cause to an effect that creates that counterfactual world. See also Paul, Hall, and Collins (2004).

> **Note**: people also talk about (cause): independent variable, treatement, intervention, X or D; and (effect) outcome, dependent variable, Y. Terminology will vary with time, norms, and disciplines, but they ultimately all capture the idea that there are some things that, when exogenously modified, change other things.

For an individual unit i, we will be interested in a relation between the outcome Y under treatment D versus control ($D \in \{0, 1\}$), holding fixed the distribution of treatments across all other units ($\mathbf{d}_{-i}$):

The "versus" comes from Rubin. Also, $\mathbf{d}_{-i}$ is a vector of length $N - 1$.

$$Y(D_i = 1, \mathbf{d}_{-i}) \text{ versus } Y(D_i = 0, \mathbf{d}_{-i})$$

In general, we will be interested in the difference between the two.

Definition 2.1 (Unit treatment effect)**.** We define the **unit treatment effect** (UTE) of D as:

$$\text{UTE} = Y(D_i = 1, \mathbf{d}_{-i}) - Y(D_i = 0, \mathbf{d}_{-i}) \tag{2.1}$$

Note that the treatment effect here may differ for each individual unit $i \in \{1, 2, ..., N\}$.

> **Note**: the UTE, like other quantities defined later, is an **estimand**. The rule to generate a specific estimate of the estimand from a sample is the **estimator** (Lundberg, Johnson, and Stewart 2021).

Note also that it is here a function of $\mathbf{d_{-i}}$, which is the vector of treatment for all units aside from i. Consider an experimental setting with five units. As the experimenter, you end up with 5 joint decisions to treat or not each unit. This could look like:

$$\mathbf{d} = [0, 0, 0, 1, 1]$$

In other words, the UTE for unit 1 is here a function of how all units 1 to N are assigned to treatment or control. At this stage, I cannot rule out that the my treatment effect could look different if, for instance, the researcher had randomly allocated treatments like:

$$\mathbf{d} = [0, 1, 0, 0, 1]$$

We will want to simplify this from here on. To do so, we will make our first assumption.

Assumption: SUTVA

Assumption 1 (SUTVA). Stable unit treatment treatment value assumption (SUTVA). Colloqually: the values of Y under each potential outcome are not affected by the distribution of the treatment. Y^k remain the same for all treatment values k regardless of how the treatments are distributed.

> **Note**: Consider a study on income discrimination by race, with the "treatment" being black and the control group being white. For simplicity, let's assume that there are only black and white people. Try to think about what would happen to potential outcomes if (a) everybody was, say, black vs. (b) everybody was white vs. (c) some distribution in-between. It's not clear that the potential outcomes would remain the same in all cases. It may well be

that potential outcomes depend on how many people get the 'treatment.' Thus, SUTVA might not be met here.

If we assume SUTVA, we can write:

$$\begin{aligned} Y_i^1 - Y_i^0 &= \delta_i \\ &\equiv \text{UTE} \qquad\qquad \text{with SUTVA} \end{aligned}$$

Y_i^1 could also have been written $Y(D_i = 1)$ or $Y_i(1)$. It's the vector of outcomes when all units are treated. Same for Y^0.

The next table gives us a sense of what δ_i could look like for an imaginary sample of 5 units.

i	Y^0	Y^1	δ_i	D
1	4	8	4	0
2	2	8	6	0
3	5	6	1	0
4	5	7	2	1
5	4	9	5	1

Table 2.1: Hypothetical outcomes and treatment effects for 5 units.

A special case – though one that's often implicitly (or unknowingly) assumed – is one in which δ_i is *homogenous* (or invariant or constant) across units. This contrasts with *heterogeneous* treatment effects that vary with i. Assuming homogeneity is a strong assumption, since it means that the treatment has the same effect for all units in the population. Yet as we will see, it creates problems when estimating various quantities of interest.

i	Y^0	Y^1	δ_i	D
1	4	8	4	0
2	4	8	4	0
3	5	9	4	0
4	5	9	4	1
5	4	8	4	1

Note that there are two types of heterogeneity: *systematic* and *truly random*. For instance, we could predict that women respond systematically differently than men to a treatment, in which case we could interact the two (more on this later). Yet even aside from this, there could still be residual heterogeneity. That is often referred to as the **random effects** model.

i	Y^0	Y^1	δ_i	D

Table 2.2: Hypothetical outcomes for 5 units with homogenous treatment effects.

This is not the only function we might be interested in. People interested in relative risks could study the ratio of outcomes or of probability of outcomes

$$\frac{Y_i^1}{Y_i^0}$$

We could also focus on transformations, such as the likelihood ratio of Y:

$$\frac{\mathbb{P}(Y_i^1 = 1)}{\mathbb{P}(Y_i^0 = 1)}$$

But the most common case is that we are interested in δ or a variant thereof.

2.3 The problem

The problem is that we never observe the UTE because we don't simultaneously observe the two outcomes jointly. The data we have looks like Table 2.3.

This is the **fundamental problem of causal inference** (Rubin 1974, 690; Holland 1986). Instead we see a partial story. As a result, we cannot recover the UTE without generating a counterfactual outcome.

i	Y^0	Y^1	Y^{observed}	δ_i
1	4		4	?
2	2		2	?
3	5		5	?
4		7	7	?
5		9	9	?

i	Y^0	Y^1	Y^{observed}	δ_i

Table 2.3: Observed data if units 1 to 3 were untreated and 4-5 are treated.

2.4 Average treatment effects (ATE)

One thing that stands out from the previous table is: while we don't have the counterfactual for individual units, we do have data for both Y^0 and Y^1. One could wonder: couldn't we get some kind of *average* treatment effect (or *expected* causal effect)?

> **Note**: moving from *unit* to *average* causal effect feels like moving the goal posts. Yet average effects might be more interesting. UTE tell us what happened to a specific unit. But if we are interested in policy recommendations, we will be interested in saying what will happen *on average* if the treatment D is implemented.

Definition 2.2 (Average treatmente effect)**.** Define the **average treatment effect** (ATE) as

$$\begin{aligned}\mathbb{E}[\delta] &= \mathbb{E}[Y^1 - Y^0] \\ &= \mathbb{E}[Y^1] - \mathbb{E}[Y^0]\end{aligned} \tag{2.2}$$

Five things to note.

1. Virtually all the problems in causal inference come from finding a way to generate reliable predictions for the missing counterfactual observations (or at least their expected value). The mathematical expectation $\mathbb{E}[\cdot]$ is an (unobserved) population parameter. It will thus have to be **estimated**.

2. Estimands that are about moments of the population are only defined if the population has these moments. If Y is Cauchy-distributed, for instance, it doesn't make sense to talk about $\mathbb{E}[Y]$ since it's undefined.

3. There are many ways within and outside quantitative methods to generate counterfactuals. Some historians try to spell out counterfactuals ("what if the UK had lost the Battle of Britain?"). Same with writers (Philip K. Dick's *Man in the High Castle*). War games are attempts to generate counterfactuals in strategic settings. And nothing prevents people to do this with qualitative methods.

4. We want to capture the *average* treatment effect. This raises an ambiguity: is it a constant that's identical across all units, or is it a variable? If it's the latter, then we talk of **heterogeneous treatment effects**.

5. There are many things we could be interested in aside from $\mathbb{E}[\delta]$. We could be interested in the change in distribution, in the change of quintile, etc. For instance, if I look at the effect of a drug, I might be interested in finding out whether the spread of welness outcome increases or not. In addition, we can also be interested in other transformations (Colnet et al. 2023).

$$\begin{aligned}
\text{Risk ratio} &= \frac{\mathbb{E}[Y^1]}{\mathbb{E}[Y^0]} \\
&= \frac{\mathbb{P}[Y^1 = 1]}{\mathbb{P}[Y^0 = 1]} \\
\text{Excess Risk Ratio} &= \frac{\mathbb{E}[Y^1] - \mathbb{E}[Y^0]}{\mathbb{E}[Y^0]} \\
\text{Survival ratio} &= \frac{\mathbb{P}[Y^1 = 0]}{\mathbb{P}[Y^0 = 0]} \\
\text{Odds ratio} &= \frac{\left[\frac{\mathbb{P}[Y^1=1]}{\mathbb{P}[Y^1=0]}\right]}{\left[\frac{\mathbb{P}[Y^0=1]}{\mathbb{P}[Y^0=0]}\right]} \\
&= \frac{\mathbb{P}[Y^1 = 1]}{\mathbb{P}[Y^1 = 0]} \frac{\mathbb{P}[Y^0 = 0]}{\mathbb{P}[Y^0 = 1]}
\end{aligned}$$

2.5 Conditional average treatment effects (CATE)

There are other estimand we might be interested in. In particular, we might be interested in the average treatment effect for a specific subgroup. This is a **conditional average treatment effect** (CATE).

Definition 2.3 (Conditional ATE)**.** The conditional average treatment effect (CATE) identifies the ATE by an arbitrary subgroup $g \in \mathcal{G}$:

$$\text{CATE} = \mathbb{E}[Y^1|\mathbb{I}(g)] - \mathbb{E}[Y^0|\mathbb{I}(g)]$$

Among an infinite set of CATEs, there are two that are especially of interest. We can define the **average treatment effect on the treated** (ATT) and **average treatment on the control** (ATC) as (Equation 2.3):

$$\begin{aligned} \mathbb{E}[Y^1|D=1] - \mathbb{E}[Y^0|D=1] &\equiv \text{ATT} \\ \mathbb{E}[Y^1|D=0] - \mathbb{E}[Y^0|D=0] &\equiv \text{ATC} \end{aligned} \tag{2.3}$$

The ATT is the effect of the treatment among those in the population who get the treatment (and likewise for the ATC).

> **Example**: suppose that the treatment is having a laptop at school and the outcome is test scores. ATT would be the effect of the laptop among students are typically have a laptop. ATC is the effect of a laptop among those what typically don't have a laptop. The two might be different if, on average, δ differs between the two groups (i.e. $\mathbb{E}[\delta|D=1] \neq \mathbb{E}[\delta|D=0]$). For instance, students who buy laptops are also those who benefit more from having a computer and may have a larger average δ.

To see the difference between the three estimands (ATE, ATT, ATC), we can use our earlier sample (Table 2.4). There are differences, though they are quite small.

i	Y^0	Y^1	δ_i	D
1	4	8	4	0
2	2	8	6	0
3	5	6	1	0
4	5	7	2	1
5	4	9	5	1
ATE	= 3.6	(18/5)		
ATT	= 3.5	(7/2)		
ATC	= 3.7	(11/3)		

Table 2.4: (True) ATE, ATT, and ATC for this population.

At this stage, we haven't really any reason to say why the three estimands would differ or not. Suffice to say that there is not a unique and identical δ_i for all i. Instead, the expected value of δ might vary depending on the subset of the population we are looking at.

2.6 Estimation: bias and inconsistency

The problem to be solved is: how can we estimate population estimands (ATE, ATT, ATC) from a random sample? Given that we always miss half the data on the outcome, this will generally be a challenge in observational settings.

> **Note**: we talk about *observational* data when the treatment assignment isn't controled by the researcher. In contrast, *experimental* data is generated under a design controled by the researcher. A *randomized* experiment refers to the type of design used by the researcher (here: random assignment of the treatment).

Intuitively, one way to estimate the ATE is to compute the estimate $\tilde{\delta}$

$$\tilde{\delta} = \frac{1}{|N\mathbb{I}(D=1)|}\sum D_i Y_i - \frac{1}{|N\mathbb{I}(D=0)|}\sum Y_i(1-D_i)$$

This is simply the difference between the mean of the treatment group and the mean of the control group.

The question is: will $\mathbb{E}[\tilde{\delta}] = \text{ATE}$? This will often not be the case in observational setting.

Realized Y	$D = 0$	$D = 1$	**Overall**
Y^0	a	b	$\mathbb{E}[Y^0] = a(1-\pi) + \pi b$
Y^1	c	d	$\mathbb{E}[Y^1] = c(1-\pi) + \pi d$

Table 2.5: Decomposition of the average treatment effect. c and b cannot be observed. Note for instance that d should be read as: $d = \mathbb{E}[Y^1|D = 1]$ (same for the other entries). π is the share of the population that receives the treatment.

The reason is that the naive estimate misses three key components, as shown in Table 2.5.

- a: untreated outcome for units that aren't typically treated (observable).
- d: treated outcome for units that are typically treated (observable).
- b and c: counterfactual cases (unobservable).
- π: population share of treated units.

We can visualize the problem in Figure Figure 2.1.

We can then revisit our definition of ATE:

$$
\begin{aligned}
\text{ATE} \equiv \mathbb{E}[\delta] &= \mathbb{E}[Y^1] - \mathbb{E}[Y^0] \\
&= \underbrace{\left[\pi \mathbb{E}[Y^1|D=1] + (1-\pi)\overbrace{\mathbb{E}[Y^1|D=0]}^{\text{unobservable}}\right]}_{\text{weighted average of } Y^1} \\
&- \underbrace{\left[\pi \overbrace{\mathbb{E}[Y^0|D=1]}^{\text{unobservable}} + (1-\pi)\mathbb{E}[Y^0|D=0]\right]}_{\text{weighted average of } Y^0} \\
&= [\pi d + c(1-\pi)] - [a(1-\pi) + \pi b] \qquad \text{above notation}
\end{aligned}
$$

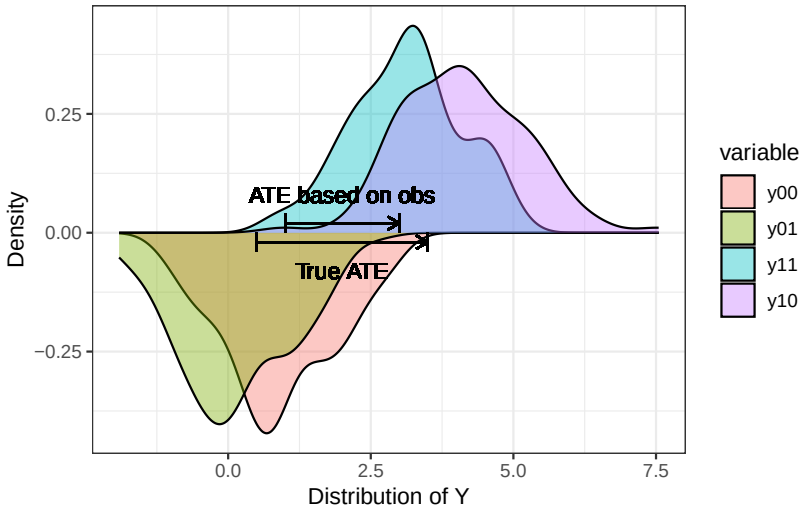

Figure 2.1: Example of bias generated by unobserved counterfactuals. $E[Y^0|D = 0] = 1, E[Y^0|D = 1] = 0, E[Y^1|D = 0] = 4, E[Y^1|D = 1] = 3$. Assume all group to be of equal size. Naive ATE: 2. True ATE: 3

where π is the proportion of units that are typically treated.

"Typically" since we could have problems such as units accidentally not taking the treatment, non-compliers, etc.

What this decomposition tells us is that $\mathbb{E}[Y^1]$ isn't just the outcome of the treated units. It is the (weighted) mean of the (mean) outcomes of the treated units (which can be observed) and the (mean) outcomes of the control units *had they been treated* (which cannot be observed). The weight is π. And the same applies to $\mathbb{E}[Y^0]$.

Note the difference between the ATE and $\tilde{\delta}$: the equation of $\tilde{\delta}$ only includes 2 out of the 5 terms in the ATE. It is missing b, c, and π. Putting the two side by side makes this clear:

$$\begin{aligned}
\text{ATE} &= \mathbb{E}[Y^1] - \mathbb{E}[Y^0] \\
&= [\pi d + c(1-\pi)] - [a(1-\pi) + \pi b] \\
&\text{versus} \\
\tilde{\delta} &= \mathbb{E}[Y^1|D=1] - \mathbb{E}[Y^0|D=0] \\
&= d - a
\end{aligned}$$

We can already intuitively guess that $\tilde{\delta}$ = ATE if $d = c$ and $a = b$.

Later, we will see how panel data allows us to deal with systematic differences in baseline via the use of difference-in-difference models.

But let's dig deeper. We can characterize the biases that ignoring these unobservable values creates. To do so, we can rearrange the earlier decomposition:

$$
\begin{aligned}
\pi d + (1-\pi)c - \pi b - (1-\pi)a &= \mathbb{E}[\delta] && \text{by def} \\
\pi d + c - \pi c - \pi b - a + \pi a &= \mathbb{E}[\delta] && \text{distributing} \\
-a &= \mathbb{E}[\delta] - c - \pi d + \pi c + \pi b - \pi a && \text{moving around} \\
d - a &= \mathbb{E}[\delta] + d - c - \pi d + \pi c + \pi b - \pi a && \text{add } d \text{ both sides} \\
d - a &= \mathbb{E}[\delta] + (b-a) + [(1-\pi)[(d-b) - (c-a)]] && \text{magic}
\end{aligned}
$$

Note that we can rewrite these last quantities as follows (see Table 5):

$$
\begin{aligned}
d - a &= \tilde{\delta} \\
d - b &= \mathbb{E}[Y^1 - Y^0 | D = 1] \\
&= \mathbb{E}[\delta | D = 1] \\
c - a &= \mathbb{E}[Y^1 - Y^0 | D = 0] \\
&= \mathbb{E}[\delta | D = 0]
\end{aligned}
$$

Thus we can now rewrite this more explicitly (Expression 2.4):

$$
\begin{aligned}
\underbrace{\mathbb{E}[Y^1|D=1] - \mathbb{E}[Y^0|D=0]}_{\text{naive difference } \tilde{\delta}} = \underbrace{\mathbb{E}[\delta]}_{\text{ATE}} & \\
&+ \underbrace{[\mathbb{E}[Y^0|D=1] - \mathbb{E}[Y^0|D=0]]}_{\text{baseline bias}} \\
&+ (1-\pi) \underbrace{[\mathbb{E}[\delta]D=1] - \mathbb{E}[\delta|D=0]]}_{\text{differential average treatment effect bias}}
\end{aligned}
\tag{2.4}
$$

There are thus two reasons why a difference between a control group and a treatment group could lead to an inaccurate estimate of ATE:

Note that this is almost exactly what we will resdiscover below with the Heckman selection problem (Heckman and Vytlacil 2005). The difference is that π doesn't seem to be relevant.

1. **Baseline bias**. The two groups are different at baseline. The potential Y^0 for those who are treated isn't the same as the (possibly observed) one for the control. This can be caused by confounding and leads to **endogeneity** and cause a **selection** effect (Heckman, Urzua, and Vytlacil 2006).

2. **Differential treatment effect bias**. The average treatment effect isn't the same for the treated and the control. On average, δ could be different for the two groups. This is a problem of **essential heterogeneity** (Heckman, Urzua, and Vytlacil 2006).

   Note: when we talk about **identification**, we generally talk about a research design in which the biases (such as baseline and differential treatment effects, but also others connected to measurement error, etc.) are plausibly eliminated in order to recover the estimand of interest (Joshua D. Angrist and Krueger 1999).

To see this, define $Y_i^0 = \mathbb{E}[Y^0] + \varepsilon_i^0$. We sometimes talk about *selection* because units that have (say) a high ε_i could self-select into the treatment arm, thus creating correlation between ε and D. In regression analysis, this leads to biased estimates.

Does living near the sea increase life expectancy?

A recent study by Cao, Martins, and Wu (2025) tests the effect of living close to the ocean (and other water bodies) on life expectancy. The estimate a linear model at the individual level:

$$\text{Life expectancy}_i = \alpha + \beta \mathbb{I}(\text{Living within 50km of the ocean}) + \ldots + \varepsilon_i,$$

where $\mathbb{I}$ is an indicator that equals 1 if an individual lives in a neighborhood that is within 50km of the ocean, and 0 otherwise. That is their treatment.
Their estimate of β is positive ($\hat{\beta} = 0.3$, 95% confidence interval: 0.26 to 0.39). They conclude that living near the ocean causes an increases life expectancy. But can you think of reasons why this may be misleading?

This is why we need to make two assumptions (and implement a design that makes them likely to hold) to recover the ATE (or only one to recover the ATT or ATC):

1. Same potential treated Y: $\mathbb{E}[Y^1|D=1] = \mathbb{E}[Y^1|D=0]$
2. Same potential control Y: $\mathbb{E}[Y^0|D=1] = \mathbb{E}[Y^0|D=0]$

We then have three cases.

1. If both assumptions hold: we can recover the ATE (which turns out to be equal to the ATT and ATC).
2. If only the first assumption holds, then we can recover:

$$\begin{aligned}\tilde{\delta} &= \mathbb{E}[Y^1|D=1] - \mathbb{E}[Y^0|D=0] \\ &= \mathbb{E}[Y^1|D=0] - \mathbb{E}[Y^0|D=0] \qquad \text{by assumption} \\ &\equiv \text{ATC}\end{aligned}$$

3. If only the second assumption holds, then we can recover:

$$\begin{aligned}\tilde{\delta} &= \mathbb{E}[Y^1|D=1] - \mathbb{E}[Y^0|D=0] \\ &= \mathbb{E}[Y^1|D=1] - \mathbb{E}[Y^0|D=1] \qquad \text{by assumption} \\ &\equiv \text{ATT}\end{aligned}$$

Example: Consider the effect of getting a solar panel on household income. Can we assume that the expected income of those who got the panel would have been the same *had they not obtained a panel* than the expected income of those who haven't bought one? If so, the naive $\tilde{\delta}$ recovers the ATT.

2.7 Linear model

Let us set biases aside and see how the causal model discussed above can be connected to the workhorse of econometrics: the **linear model** (i.e., a model that's linear in parameters). Compactly, we write (see Notation box for more details):

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon$$

This notation can be traced back (at least) to Aitken (1936).[1]

[1] Replace $\mathbf{X}$ with $\mathbf{P}$ and β with a (Aitken uses y for $\hat{y}$).

By "linear," we mean linear *in parameters*. The linear model admits nonlinearity in *variables* (e.g., by adding a squared term of X, taking the log of X, or taking its square root).

The model links a set of variables (commonly referred to as *independent variables*, *treatments*, *interventions*, *explanatory variables*, *right-hand side variables*, *regressors*, etc.) to another variable (*dependent variable*, *outcome*, *left-hand side variable*, etc.). To account for stochasticity in any kind of **data-generating process** , we include an *error term* ε whose exact nature will be defined below.

By convention, the independent variable(s) are generally referred to as $\mathbf{X}$ and the dependent variable as $\mathbf{y}$. Sometimes, when we are interested in a particular (generally dichotomous) treatment, we will use D.

Typically, we will either (a) be interested in measuring the **causal effect** of a subset of the variables in $\mathbf{X}$, or (b) be interested in **forecasting y**. In general, in this class, we will focus on (a).

2.8 Foundations of the linear model

What are the foundations of the linear model? Suppose you came up with a theory that states that a treatment D increases Y. In the counterfactual language, this means that Y would have had another value had D had itself another value. Let us first begin with the case in which $D \in \{0, 1\}$ is a dichotomous variable. Then:

1. Let's say we are interested in the expected value of Y, i.e. $\mathbb{E}[Y]$.
2. We are asking whether $\mathbb{E}[Y]$ is affected by D; we are then therefore interested in the conditional expectation $\mathbb{E}[Y|D]$.
3. If $D = 0$, we write:

$$\mathbb{E}[Y|D] = \mathbb{E}[Y|D = 0]$$

Likewise, if $D = 1$:

$$\mathbb{E}[Y|D] = \mathbb{E}[Y|D = 1]$$

4. Thus:

$$\mathbb{E}[Y|D] = \mathbb{E}[Y|D = 0] + D(\mathbb{E}[Y|D = 1] - \mathbb{E}[Y|D = 0])$$

5. Let us define

$$\alpha := \mathbb{E}[Y|D = 0]$$

and

$$\beta := (\mathbb{E}[Y|D = 1] - \mathbb{E}[Y|D = 0])$$

We can rewrite:

$$\mathbb{E}[Y|D] = \alpha + \beta D$$

6. Lastly, let us add Y on both sides (because $\mathbb{E}[Y|D]$ is not directly observable, whereas Y is).

$$Y = \alpha + \beta D + (Y - \mathbb{E}[Y|D])$$

7. Let us define $\varepsilon := (Y - \mathbb{E}[Y|D])$ as the difference between a given value of Y (y_i) and the expected value of Y at D.

We find the **linear model** :

$$\begin{aligned} \mathbb{E}[Y|D] &= \alpha + \beta D && \text{from step 5} \\ Y &= \alpha + \beta D + \varepsilon && \text{from step 6} \\ Y &= \mathbb{E}[Y|D] + \varepsilon && \text{by plugging the 1st eq in the 2nd} \end{aligned}$$

The last equation corresponds to the **conditional expectation function** (CEF) decomposition theorem (see (J. Angrist and Pischke 2008), p.32).

Generalizing this to a matrix of variables $\mathbf{X}$ (which may include D), we obtain the expression we started with, namely:

$$\begin{aligned} \mathbb{E}[\mathbf{y}|\mathbf{X}] &= \mathbf{X}\beta \\ \mathbf{y} &= \mathbf{X}\beta + \varepsilon \\ \mathbf{y} &= \mathbb{E}[\mathbf{y}|\mathbf{X}] + \varepsilon \end{aligned}$$

Note that the first expression gives us a natural interpretation of β_k: the partial derivative of $\mathbb{E}[\mathbf{y}]$ with respect to variable X_k is β_k, which is thus the marginal effect of X_k on $\mathbb{E}[\mathbf{y}]$:

$$\frac{\partial \mathbb{E}[\mathbf{y}]}{\partial X_k} = \beta_k.$$

The parameter β_k, then, measures the effect of a small change in X_k on the expected value of Y.

Here, we note two things:

1. The (population) CEF may or may not be linear. In what follows, we will generally assume it is. If it's not exactly linear, then $\mathbf{X}\beta$ will *approximate* it. How well it does so is case-specific.
2. A model that is **saturated** is linear by construction. The model is saturated if $\mathbf{X}$ is entirely composed of dichotomous variables and their interactions. In such a case, each conditional expectation $\mathbb{E}[Y|D = d, X = x, Z = z, ...]$ has its own unique parameter or combination of parameters.

Here's an example of a saturated model. Suppose we have a model in which we want to test whether the average price of a cars sold is higher for consumers with more/less formal education education, where education is defined as a categorical variable (with 4 values):

$$\text{Education}_i \in \{\text{Less than High School, High School, College, PhD}\}.$$

We could then estimate:

$$\text{Price of car sold}_i = \alpha + \beta \text{High School}_i + \gamma \text{College}_i + \lambda \text{PhD}_i + \varepsilon_i,$$

where each variable is a dichotomous indicator. Then, each value of education has its own estimate:

- $\mathbb{E}[\text{Price}|\text{Less than HS}] = \alpha$
- $\mathbb{E}[\text{Price}|\text{High School}] = \alpha + \beta$
- $\mathbb{E}[\text{Price}|\text{College}] = \alpha + \gamma$
- $\mathbb{E}[\text{Price}|\text{PhD}] = \alpha + \lambda$

You can also construct saturated models with more than one variable. You would then need to include a number of interaction effects. For instance, if we included gender in addition to education, we would need to interact it with every dummy above. See more later on interaction effects.

When the treatment is **continuous**, we cannot discretize it in a way that guarantees linearity. We might then want to start with a flexible formulation such as:

$$\mathbf{y} = f(\mathbf{X}) + \varepsilon.$$

We would then impose restrictions on $f()$ ranging from a fully parametric approach (in which we replace $f()$ with any desired function) to semi- and non-parametric specifications. More on this later.

3 Ordinary least squares: point estimation

This version: December 16, 2025

Caveat

These notes are offered in support of *Introduction to Econometrics* (MGT 581). They were written to help students understand the logic behind important topics in econometrics. But they don't (always) replace the insights and depth that textbooks can provide. This is why we connect each session to chapters of our core textbooks. See the syllabus for more details.

3.1 Readings

People interested in least squares and point estimation may refer to: B. Hansen (2022a) (ch.4), Stachurski (2016) (ch.11-12), Verbeek (2018) (ch.2.1-2.3, 2.8-2.9, and 3), Hayashi (2000) (ch1.1-1.2) Stock and Watson (2011) (ch.4 and 6), Wooldridge (2009) (ch.2-3). See Morgan and Winship (2014) (ch.6) for the connection between least squares and causal inference.

3.2 Estimators

The vector β is a set of unknown and unobservable parameters. These parameters capture the relation between $\mathbf{X}$ and $\mathbf{y}$. At this stage, this relation could be correlational or it could be causal. We don't know just yet. There are two problems here:

1. Can we map from β to some causal estimand of interest?
2. Since we don't know β, we will have to estimate it from data, i.e., from a sample. How?

One way to think of what econometrics is to see this as a problem of drawing inferences from the data about these parameters, $\mathbb{P}(\beta|\mathbf{X})$. To make inferences, we need first an **estimator**, namely a rule that takes data as input and produces estimates as output. There exists several such methods, including various forms of **least squares** (ordinary least squares, weighted least squares, etc.), **methods of moments**, **maximum likelihood**, **Bayesian estimators**, and so forth. We will see a few of them, starting with ordinarily least squares.

3.3 Ordinary least squares

We start with the linear model (see box for notation):

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon$$

Notation

I generally try to use the following notation. For a sample that contains n units (e.g., consumers, firms, countries, etc.) and k right-hand side parameters, we write:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n\times 1} \qquad \mathbf{X} = \begin{bmatrix} 1 & x_{1,2} & z_{1,3} & \cdots & w_{1,k} \\ 1 & x_{2,2} & z_{2,3} & \cdots & w_{2,k} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n,2} & z_{n,3} & \cdots & w_{n,k} \end{bmatrix}_{n\times k}$$

$$\beta = \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \vdots \\ \tau_k \end{bmatrix}_{k\times 1} \qquad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}_{n\times 1}$$

Here, the letters x, z, etc. refer to random variables (I number them here to help count them, but I'll stop doing

it from now on). For instance, in a dataset on firms, x could be the number of employees, z annual spending on R&D, etc. The column of 1's is included to serve as a constant.
Note that $\mathbf{X}$ is a stacked matrix of vectors $\mathbf{x}_i'$, where each vector represents the data for unit i. In that sense, $\mathbf{X}$ is like a spreadsheet with each unit being a separate row, and each variable being a column. We sometimes call it the **design matrix**.
We then write (dropping the number for the variables):

$$\begin{aligned} y_i &= \alpha + \beta x_i + \gamma z_i + ... + \tau w_i + \varepsilon_i && \text{unit } i \\ y_i &= \mathbf{x}_i'\beta + \varepsilon_i && \text{matrix notation} \\ \mathbf{y} &= \mathbf{X}\beta + \varepsilon && \text{matrix notation} \end{aligned}$$

When we move to data and estimated quantities, we write:

$$\begin{aligned} y_i &= \hat{\alpha} + \hat{\beta} x_i + \hat{\gamma} z_i + ... + \hat{\tau} w_i + u_i && \text{observed } y \\ \hat{y}_i &= \hat{\alpha} + \hat{\beta} x_i + \hat{\gamma} z_i + ... + \hat{\tau} w_i && \text{predicted } y \\ \mathbf{y} &= \mathbf{X}\hat{\beta} + \mathbf{u} && \text{same in matrix} \\ \hat{\mathbf{y}} &= \mathbf{X}\hat{\beta} \end{aligned}$$

where u_i is called the **residual** and represents the difference between the observed value of y for unit i and its predicted value $\hat{y}$. Note that the residuals are not exactly the same as the error term. To see this (B. Hansen 2022a, 173):

$$u_i = y_i - \mathbf{X}\hat{\beta} = (\mathbf{X}\beta + \varepsilon_i) - \mathbf{X}\hat{\beta} = \varepsilon_i - \mathbf{X}(\hat{\beta} - \beta).$$

Thus the residual is the error term plus some additional randomness created by the fact that, for a given sample, $\hat{\beta} \neq \beta$ almost certainly.

The least squares estimator is a rule to generate estimates of the vector β. Most textbooks call these estimates $\hat{\beta}$. Some call them $\mathbf{b}$. When needed, we will add the subscript $\hat{\beta}_{ols}$ to emphasize that these estimates were obtained from ordinary

least squares. We write:

$$\mathbf{y} = \mathbf{X}\hat{\beta} + \mathbf{u}.$$

Least squares, as the name suggests is a minimization problem, in which we *minimize the sum of squared residuals* (aka SSR).[1] When we weight all observations equally, we call this ***ordinary* least squares**.

In contrast to *weighted* least squares, which we will discuss later.

The minimization problem is:

$$\arg\min_{\hat{\beta}} \text{SSR} := \arg\min_{\hat{\beta}} \mathbf{u}'\mathbf{u},$$

where $\mathbf{u}$ is the vector of residuals defined as the difference between *observed* values of y_i and *predicted* values $\hat{y}_i$ of y for unit i.

We can equivalently write:

$$\arg\min_{\hat{\beta}} \text{SSR} = \arg\min_{\hat{\beta}} (\mathbf{y} - \mathbf{X}\beta)^2$$

You may ask here: why *squared* total residuals? Why not just total residuals, or cubic residuals, or something else? Two answers:

1. We will see that minimizing squared residuals (i.e. using OLS) yields estimates with very desirable properties when necessary assumptions are met: OLS is unbiased, consistent, and (more rarely) efficient.
2. As a minimization problem, OLS will yield a unique solution, which other methods will often fail to do so.

[1] The history of least squares is more than 200 years old. Its first formulation was made by Legendre in 1806 (Legendre 1806, viii): "la méthode qui me paroît la plus simple et la plus générale, consiste à rendre minimum la somme des quarrés des erreurs (…). La méthode dont je viens de parler (…) que j'appelle *Méthode des moindres quarrés* peut être d'une grande utilité." A few years later, Gauss proved that the ordinary least square solution corresponds to the maximum likelihood estimator (Gauss 1857). See Merriman (1877). See also Aitken (1936).

We can solve this minimization problem by using first and second order conditions.

$$
\begin{aligned}
\text{SSR} &= \mathbf{u}'\mathbf{u} \\
&= (\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta}) && \text{by definition of residuals} \\
&= (\mathbf{y}' - \hat{\beta}'\mathbf{X}')(\mathbf{y} - \mathbf{X}\hat{\beta}) && \text{by properties of transpose} \\
&= \mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{X}\hat{\beta} - \hat{\beta}'\mathbf{X}'\mathbf{y} + \hat{\beta}'\mathbf{X}'\mathbf{X}\hat{\beta} && \text{distributing} \\
&= \mathbf{y}'\mathbf{y} - 2\mathbf{y}'\mathbf{X}\hat{\beta} + \hat{\beta}'\mathbf{X}'\mathbf{X}\hat{\beta} && \text{reorganizing}
\end{aligned}
$$

At this stage, we can take the first order condition with respect to the vector $\hat{\beta}$:

$$
\begin{aligned}
\frac{\partial \text{SSR}}{\partial \hat{\beta}} &= 0 - 2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta} = 0 \\
2\mathbf{X}'\mathbf{X}\hat{\beta} &= 2\mathbf{X}'\mathbf{y} && \text{moving things around} \\
\mathbf{X}'\mathbf{X}\hat{\beta} &= \mathbf{X}'\mathbf{y} && \text{dividing by 2} \\
(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} && \text{pre-multiplying both sides; A2} \\
\mathbf{I}\hat{\beta} &= \underbrace{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}}_{k\times 1} && \text{where } \mathbf{I} \text{ is the identity matrix}
\end{aligned}
$$

Note that $\mathbf{X}$ is of dimension $n \times k$; thus $\mathbf{X}'$ is of dimension $k \times n$ and $(\mathbf{X}'\mathbf{X})$ is of dimensions $k \times k$. Finally, $\mathbf{X}'\mathbf{y}$ is of dimension $k \times 1$, thus the whole thing is $k \times 1$. It all works out.

We reach our first important result: $\hat{\beta}_{ols} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$. Given that $\hat{\beta}$ is a vector of scalars, we refer to $\hat{\beta}_k$ as the **point estimate** of β_k for variable k.[2]

The only assumption we needed (**A2**) is for the inverse of $(\mathbf{X}'\mathbf{X})$ to exist.

Assumption: full rank/no perfect multicollinearity

Assumption 2 (Full rank). The design matrix $\mathbf{X}$ (of dimension $n \times k$) is of full rank with probability 1.

[2] We focus on point identification, that is, the estimation of a scalar or a vector. There is a rich literature on **set identification** which derives upper and lower bounds on what the treatment effect could be. See (Manski 1990; Tamer 2010).

The rank of a matrix is the number of linearly independent columns (here: variables).
When this assumption is met, there is no perfect multicollinerity across variables in $\mathbf{X}$, and the inverse matrix $(\mathbf{X}'\mathbf{X})^{-1}$ exists.
Violations typically occur when the matrix $\mathbf{X}$ includes variables that perfectly predict each other. See below for a discussion of causes.

Note. Some textbooks take a slightly different path to arrive at the same conclusion. Instead of minimizing the *total* sum of squared residuals, we can equivalently minimize the *average* of the sum of squared residuals. We can then write the minimization problem in terms of *expectations*:

$$\min_{\hat{\beta}} = \mathbb{E}[(\mathbf{y} - \mathbf{X}\hat{\beta})^2]$$
$$\hat{\beta} = \mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}]\mathbb{E}[\mathbf{X}'\mathbf{y}]$$

Note. We can think of SSR as the **loss function**, often labeled as $\mathcal{L}$. It is the penalty we are facing when making prediction errors.

Note. We can connect OLS to other estimators, such as **LASSO** or **ridge regression**, by extending the loss function to include constraints. In the case of LASSO, we have:

$$\mathcal{L}_{\text{LASSO}} = \|\mathbf{y} - \mathbf{X}\beta\|^2 - \lambda\|\beta\|_1$$

Note. So far, we have no way to interpret $\hat{\beta}$ causally – it is strictly a solution to an optimization problem. In fact, we can't even interpret population β causally. To do this, we will need additional assumptions to be discussed later.

Additional matrices

Note that there are a few useful concept to keep in mind.

The **projection matrix** or **hat matrix P** is defined as:

$$\mathbf{P} := \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

We then obtain:

$$\mathbf{P}\mathbf{y} = \mathbf{X}\hat{\beta},$$

which is the vector of $\hat{y}$. Further, if we define $\mathbf{M}$ as the **residual maker matrix**:

$$\mathbf{M} := \mathbf{I} - \mathbf{P},$$

we obtain:

$$\mathbf{M}\mathbf{y} = (\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1})\mathbf{y} = \mathbf{y} - \mathbf{P}\mathbf{y} = \mathbf{y} - \mathbf{X}\hat{\beta} = \varepsilon$$

3.4 Special case: OLS with single regressor

It is useful to write down the special case of a single regressor to capture the intuition behind least square regression.

To begin with, we are now minimizing:

$$\min \sum u_i^2 = \min \sum (y_i - \hat{\alpha} - \hat{\beta}x_i)^2$$

We can solve this by calculus. Now, we could directly take the partial derivatives of this last expression using the chain rule. Or we can do it by hand. By the chain rule, we get:

$$\begin{aligned}
\frac{\partial \text{SSR}}{\partial \hat{\alpha}} &= -2\sum (y_i - \hat{\alpha} - \hat{\beta}x_i) && \text{extracting constant} \\
0 &= 2\left(\sum y_i - \sum \hat{\alpha} - \sum \hat{\beta}x_i\right) && \text{distributing} \\
&= \sum y_i - n\hat{\alpha} - \hat{\beta}\sum x_i && \text{dividing by 2}
\end{aligned}$$

and:

$$\begin{aligned}
\frac{\partial \text{SSR}}{\partial \hat{\beta}} &= \sum -2\left(x_i y_i - \hat{\alpha} x_i - \hat{\beta} x_i^2\right) \\
0 &= -2\left(\sum x_i y_i - \hat{\alpha}\sum x_i - \hat{\beta}\sum x_i^2\right) && \text{distributing} \\
&= \sum x_i y_i - \hat{\alpha}\sum x_i - \hat{\beta}\sum x_i^2 && \text{dividing by 2}
\end{aligned}$$

This gives us two equations with two unknowns. We can reshape them as the **normal equations** (I also drop the subscripts for readability):

$$\begin{aligned}
n\hat{\alpha} + \hat{\beta}\sum x &= \sum y \\
\hat{\alpha}\sum x + \hat{\beta}\sum x^2 &= \sum xy
\end{aligned}$$

There are various ways to solve these equations. One way is to use some matrix algebra. Specifically:

$$\begin{bmatrix} n & \sum x \\ \sum x & \sum x^2 \end{bmatrix} \begin{bmatrix} \hat{\alpha} \\ \hat{\beta} \end{bmatrix} = \begin{bmatrix} \sum y \\ \sum xy \end{bmatrix}$$

We can then isolate the parameters, assuming that the inverse of the first matrix exists. In practice, this requires the absence of perfect multicollinearity. In the bivariate case, it means that the variance of X should be greater than zero.

$$\begin{bmatrix} \hat{\alpha} \\ \hat{\beta} \end{bmatrix} = \begin{bmatrix} n & \sum x \\ \sum x & \sum x^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum y \\ \sum xy \end{bmatrix} \qquad \text{by A2}$$

Taking the inverse, we have:

$$\begin{bmatrix} \hat{\alpha} \\ \hat{\beta} \end{bmatrix} = \frac{1}{n\sum x^2 - (\sum x)^2} \begin{bmatrix} \sum x^2 & -\sum x \\ -\sum x & n \end{bmatrix} \begin{bmatrix} \sum y \\ \sum xy \end{bmatrix}$$

But we can simplify this a bit: note that $\sum x = n\bar{y}$ and $\sum x = n\bar{y}$. Define $s_{xx} := \sum(x_i - \bar{x})^2 = \sum x^2 - n\bar{x}^2$, $s_{yy} := \sum(y_i -$

$\bar{y})^2 = \sum y^2 - n\bar{y}^2$, $s_{xy} := \sum(x_i - \bar{x})(y_i - \bar{y}) = \sum xy - n\bar{x}\bar{y}$. Therefore:

$$\begin{aligned}
\hat{\beta} &= \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2} && \text{from above} \\
&= \frac{n\sum xy - n^2\bar{x}\bar{y}}{n\sum x^2 - (n\bar{x})^2} \\
&= \frac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n\bar{x}^2} && \text{cancelling n} \\
&= \frac{s_{xy}}{s_{xx}} \\
&= \frac{\widehat{\mathrm{Cov}}(X,Y)}{\hat{\mathbb{V}}(X)} && \text{dividing both terms by n}
\end{aligned}$$

Therefore, $\hat{\beta}$ is the covariance of X and Y, scaled by the variance of X.

As for $\hat{\alpha}$:

$$\begin{aligned}
n\hat{\alpha} + \hat{\beta}\sum x &= \sum y && \text{from the normal equations} \\
n\hat{\alpha} + \hat{\beta}n\bar{x} &= n\bar{y} \\
\hat{\alpha} + \hat{\beta}\bar{x} &= \bar{y} && \text{removing n} \\
\hat{\alpha} &= \bar{y} - \hat{\beta}\bar{x}
\end{aligned}$$

The intercept $\hat{\alpha}$ is therefore the average value of y minus $\hat{\beta}$ times the average value of x.

3.5 Model specifications and their interpretation

3.5.1 "Vanilla" OLS

The linear model expresses the conditional expectation of y given $\mathbf{X}$ (for k independent variables, including a constant):

$$\mathbb{E}[Y|X, Z, ...K] = \mathbf{X}\beta = \alpha + \beta X + \gamma Z + ... + \lambda_k K$$

And for a given sample, using OLS estimates, we can plug in:

$$\hat{\mathbb{E}}[Y|X, Z, ...K] = \hat{y} = \hat{\alpha} + \hat{\beta}X + \hat{\gamma}Z + ... + \hat{\lambda}_k K.$$

Thus, the predicted values $\hat{y}$ are conditional expectation (or conditional mean) set at specific values of $X, Z, ..., K$.

Taking the partial derivative of $\mathbb{E}[Y]$ with respect to an arbitrary variable X yields:

$$\frac{\partial \mathbb{E}[Y|Z, ..., K]}{\partial X} = \beta$$

Thus, β is the effect of a small change in X on the *expected value* of Y. We sometimes refer to β as the *marginal effect* of X on Y (it is clear from the previous equation why it is the "marginal" effect), or just the *effect* of X on Y. A few observations:

- For a dummy variable ($X \in \{0, 1\}$), β is akin of "turning on" the variable from 0 to 1 (since the "slope" from 0 to 1 is constant and thus equals the derivative).
- We write the model under the assumption that X is a *cause* of Y, by which we mean that if we could modify the value of X, $\mathbb{E}[Y]$ would change as well (Pearl 2009). However, $\hat{\beta}$ is not automatically an estimate of that causal effect. For the OLS estimate $\hat{\beta}$ to have a causal interpretation, we will need to understand how to draw inferences with OLS (more in the next chapter).

"Vanilla" OLS: illustration

Example 3.1 (Quantifying the effect of working from home)**.** Does working from home increase or reduce employee performance? This is what N. Bloom et al. (2015) wanted to study. They partnered with Ctrip, a Chinese travel agency. Ctrip wanted to see if letting people at their call center work from home would affect performance. If working from home doesn't hurt, then it would also be an opportunity for Ctrip to reduce the rental space they need for their call centers.
The experiment took place during a period of 9 months. Volunteers were either assigned to working from home

(treatment group) or continued to work at the office (control group). The team of economists working with Ctrip recorded a bunch of performance metrics, which would serve as dependent variables. This includes an overall performance score (standardized to have mean zero and standard deviation of 1), the (log of) the number of calls per minute as well as the (log of the) time spent on the phone. Using logs is something we will discuss later. For now, suffice to say that in a log-linear model (logged DV, untransformed treatment), we can interpret it as follows: the treatment changes the expected value of the outcome by $100 \cdot \widehat{\beta}$ percent.

The models N. Bloom et al. (2015) estimate are:

$$\text{Employee performance}_{i,t} = \beta \text{Treatment} + \gamma_i + \tau_t + \varepsilon_{i,t},$$

where γ and τ are fixed effects (which we will discuss later) and β is our treatment effect. The models are estimated with OLS. (We include the subscripts i for employees and t for weeks, since the researchers recorded performance over 9 months.)

The results are reported in Table 3.1. The numbers on the *Treatment* row are $\widehat{\beta}$ for the effect of the treatment (letting people work from home) on overall performance (column 1), calls per minute (log) (column 2), and minutes on the phone (log) (column 3). We see that the intervention increased overall performance by 0.23 units on the z-scale. More concretely, it increased calls per minute by about 12 percent and minutes spent on the phone by about 9 percent. Thus, working from home increased performance across the board.

The result? Ctrip ended up giving the option of working from home to the entire firm.

	Overall	Calls/min (log)	Min on phone (log)
Treatment	0.23	0.12	0.09
S.E.	(0.06)	(0.03)	(0.03)
N emp.	249	134	134
Obs	17,806	9,426	9,426

	Overall	Calls/min (log)	Min on phone (log)

Table 3.1: Extract from Table 3.1, Table II. Dependent variables listed at the top of the column. The treatment equals 1 if a worker was allowed to work from home and zero otherwise.

3.5.2 Log transformation

Natural logarithmic transformations are popular in economics (in what follows, *log*() always refers to the *natural* log, except otherwise stated). Part of the reason is that there are theoretical quantities of interest that lend themselves well to being expressed as logs. For instance, a Cobb-Douglas model in which a firm i's production is a function of labor L and capital K:[3]

$$y_i = L_i^{\alpha} K_i^{\beta}.$$

This model isn't linear in parameters (α and β are in the exponent). However, we can transform it:

$$log(y_i) = \alpha \text{log(L)}_i + \beta \text{log(K)}_i,$$

which can be estimated with OLS.

Another reason is that some important economic variables are very (right-)skewed. For instance, some variables have outliers with very large values (e.g., GDP). Logging these variables "tames" them.

The challenge lies in the interpretation of the parameters when a variable is log-transformed. We can distinguish four cases:

1. Linear-linear. This is what we saw so far.
2. Linear-log
3. Log-linear
4. Log-log

[3]See Kenneth Benoit's excellent set of notes: here.

Linear-log. This corresponds to:

$$y = \alpha + \beta \log(X) + \varepsilon$$

Recall that β is the slope, which here corresponds to the effect of increase $log(X)$ by 1. But note:

$$1 = log(e),$$

where e is Euler's number (about 2.72). Thus, by the properties of log:

$$log(X) + 1 = log(X) + log(e) = log(eX).$$

Thus, β is the same as the change in $\mathbb{E}[Y]$ caused by multiplying X by 2.72, or, equivalently, increasing it by 172%.

Now, this is not typically the kind of change in X we are interested in. More commonly, we are interested in an increase of X by, say, 1% or 10% or 100%. It entirely depends on the context of your treatment. You should always pick a Δ (change in the treatment) that is economically meaningful, even for the linear-linear model.

The solution is straightforward: instead of adding $log(e)$ (which multiplies X by e), we can add $log(1 + p)$ for whatever p we are interested in (p is the percentage change we want to study, e.g. 0.3 for 30%). This will then multiply X. For instance, if we want to estimate the increase of X by 20%, we would use $log(1 + 0.2) = log(1.2)$:

$$\text{Effect of increasing } X \text{ by } 20\% = \beta \cdot log(1.2).$$

Here, there's a small trick that is useful to calculate this, since we typically don't compute logs in our heads. For small values of p, we can use the following approximation:

$$log(1 + p) \approx p.$$

Thus, we can straightforwardly capture the effect of a p percentage increase in X:

$$\text{Effect of increasing } X \text{ by } p \approx p \cdot \beta.$$

For instance, if $p = 0.01$ (i.e., an increase of X by 1%), then:

$$\text{Effect of increasing } X \text{ by } 1\% \approx \tfrac{\beta}{100}.$$

Log-linear. This corresponds to:

$$\log(y_i) = \alpha + \beta X_i + \varepsilon_i$$

By definition:

$$\beta = \log(y|X+1) - \log(y|X),$$

which we can rewrite (for readibility) as y being a function of X ($y(X)$ reads: the expected value of Y at X):

$$\beta = \log(y(X+1)) - \log(y(X)),$$

By the properties of logs (difference of logs is the log of their ratio), we can rewrite:

$$\beta = log\left(\frac{y(X+1)}{y(X)}\right).$$

Let's exponentiate:

$$exp(\beta) = \frac{y(X+1)}{y(X)}.$$

Note that the right-hand side is the ratio of the "new" over the "old" value of Y ("new" meaning the expected value of y when X has been increased by one unit). For instance, if it equals 1.2, then the new value is 1.2 times the old value. If we substract 1, we can interpret this as a percentage change. For a ratio of, say, 1.2, it means that Y increases on average by 20% ($1.2-1 = 0.2$) when X increases by one unit. Thus, β will tell us something about the percentage change in Y.

Again, there is a nifty trick here, which is that:

$$exp(\beta) \approx 1 + \beta.$$

Thus, a one unit change in X corresponds to multiplying Y by $1+\beta$, which corresponds to a change of Y by $100 \cdot \beta$ percent. For instance, if $\beta = 0.2$, then a one-unit increase of X multiplies Y by 1.2, that is, Y increases by 20% $(0.2 \cdot 100)$.

If we want to use a different change in X (instead of 1 unit), we simply multiply the exponential. For instance: $exp(2\beta)$ tells us the effect of increasing X by 2 units.

Log-log. This corresponds to:

$$\log(y_i) = \alpha + \beta \log(X_i) + \varepsilon_i.$$

We can combine the logic of the log-linear and linear-log models. If we increase X by p percent, we write:

$$\beta \cdot log(1 + p).$$

Define $k = 1+p$ (that is, k is what you multiply X by). Then:

$$\frac{y(kX)}{y(X)} = exp(\beta log(k)) = k^{\beta}.$$

For instance, taking $p = 0.01$ (and thus $k = 1.01$), we obtain:

$$\frac{y(1.01 \cdot X)}{y(X)} = 1.01^{\beta}$$

So for an increase of X by 1%, Y is multiplied by 1.01^{β}. Equivalently, we can say that Y increases by $k^{\beta} - 1$ percent. But note that for small enough k:

$$\beta \approx k^{\beta} - 1.$$

Let's take $k = 1.01$ (thus $p = 0.01$ or 1 percent): β is the expected change (in percent) of Y when X increases by 1 percent.

This is why β is often referred to as an **elasticity**. In economics, the elasticity price of demand is the change in quantity demanded (in percent) when the price increases by 1 percent. The log-log model captures precisely this.

To sum up (keeping in mind that some of these results are approximations valid for small changes only):

- Linear: when X increases by one *unit*, Y changes by β *units* (on average).
- Linear-log: when X increases by one *percent*, Y changes by $\beta/100$ *units* (on average).
- Log-linear: when X increases by one *unit*, Y changes by $100 \cdot \beta$ *percent* (on average).
- Log-log: when X increases by one *percent*, Y changes by β *percent* (on average).

Log model: illustration

Example 3.2 (Log-log model)**.** In a highly cited paper, Mankiw, Romer, and Weil (1992) try to explain why some countries have higher GDP per capita than others. Building on a famous model by Solow (1956), they test the conjecture that the key driver for economic growth is investments (i.e., capital accumulation).
Using subscript c for country, they estimate a model of the following kind:

$$\log\left(\frac{\text{GDP}}{\text{\# working-age people}}\right)_c = \alpha + \beta \log\left(\frac{\text{Investment}}{GDP}\right)_c$$

This is a log-log model. The OLS estimates are reported in Figure 3.1. Looking at non-oil producing countries (column 1), they found that:

- $\hat{\alpha} = 5.48$ (that's *CONSTANT*)
- $\hat{\beta} = 1.42$ (that's *ln(I/GDP)*)

The latter is what we were looking for: it says that for every additional percent of investments (normalized by GDP), GDP per worker increases by 1.4 percent on average.

TABLE I
ESTIMATION OF THE TEXTBOOK SOLOW MODEL

Dependent variable: log GDP per working-age person in 1985			
Sample:	Non-oil	Intermediate	OECD
Observations:	98	75	22
CONSTANT	5.48	5.36	7.97
	(1.59)	(1.55)	(2.48)
ln(I/GDP)	1.42	1.31	0.50
	(0.14)	(0.17)	(0.43)
$\ln(n + g + \delta)$	−1.97	−2.01	−0.76
	(0.56)	(0.53)	(0.84)
$\bar{R}^2$	0.59	0.59	0.01
s.e.e.	0.69	0.61	0.38

Figure 3.1: Source: Mankiw, Romer, and Weil (1992, 414).

3.5.3 Nonlinear treatment

Suppose you have a continuous treatment, or one that takes more than two values. A linear regression model $y = \alpha + \beta X$ offers a linear approximation of the effect of X on Y. But this might not be likely. Maybe we have theoretical reasons to expect nonlinearities that we cannot easily model. Or maybe we just don't want to assume linearity. There are two approaches to handle this. First, it makes sense to **dichotomize** or **bin** the treatment. Second, we can make the model nonlinear by including polynomials of the variable in question.

Regarding the binning approach, we move from:

$$Y_i = \alpha + \beta X_i$$

to:

$$Y_i = \alpha + \beta \mathbb{I}[X_0 \leq X_i < X_1] + \gamma \mathbb{I}[X_1 \leq X_i < X_2] ...$$

where $\mathbb{I}$ is an indicator that equals 1 if the condition inside the brackets is met. This allows the effect of X to vary based on the range in which it is located. Note that this is akin to making dummies for each group. For instance, if X is years of schooling, we would have a dummy for 10 years, another for 11 years, etc. (with one category omitted to serve as baseline).

3.5.4 Polynomial models

The interpretation of point estimates can become more complicated with nonlinear models and must be adapted accordingly. By "nonlinear," we mean nonlinearity in *variables*. The linear model remains linear in *parameters*.

One type of nonlinear model is the **quadratic model**. This captures situations in which the marginal effect of a variable varies according to its starting value.

Concretely, write:

$$\mathbb{E}[Y|X] = \alpha + \beta X + \gamma X^2.$$

This yields a derivative that is a function of X:

$$\frac{d\mathbb{E}[Y|X]}{dX} = \beta + 2\gamma X.$$

In such a case, the effect of X is not constant and depends on 'where' we are with respect to X.

Something that might be interesting is to estimate the max or min, which is:

$$-\frac{\beta}{2\gamma}.$$

Of course, nothing (aside from data) prevents you from adding higher terms. You could have a cubic polynomial, for instance, if you want:

$$\mathbb{E}[Y|X] = \alpha + \beta X + \gamma X^2 + \lambda X^3.$$

Polynomial model: illustration

Example 3.3 (Cubic treatment effect)**.** Fang, Keane, and Silverman (2008) examine how a healthcare insurance called Medigap affects average medical spending. Their model is:

$$\begin{aligned}\text{Medical spending (USD)}_i = \alpha + \beta\text{Medigap}_i + \gamma\text{Age}_i \\ + \lambda\text{Age}_i^2 + \phi\text{Age}_i^3 + \varepsilon_i,\end{aligned}$$

where age is a person's age *minus* 65 (so that the polynomial is centered on zero when people become eligible for this healthcare program). The main treatment effect is to reduce average medical spending by USD 4,393. But what is of interest to us here is the cubic polynomial of age. The results are reported in Table 3.2.
As we see (column 1), we have: $\hat{\gamma} = 388$, $\hat{\lambda} = 1.9$, and $\hat{\phi} = 0.12$. The marginal effect of one year is:

$$\frac{\partial \text{Spending (USD)}_i}{\partial \text{Age}_i} = \gamma + 2\lambda \text{Age}_i + 3\phi \text{Age}_i^2$$

Thus, we can compute that for someone who is 67 years old (two years after 65), the marginal effect of getting older is:

$$\frac{\partial \widehat{\text{Spending}} \text{ (USD)}_i}{\partial \text{Age}_i}|_{\text{Age}=2} = 388 + (2 \cdot 1.9 \cdot 2) + (3 \cdot 0.12 \cdot (2^2)) \approx 397,$$

or an increase in medical spending by USD 397.
In contrast, the marginal effect for someone who is 85:

$$\frac{\partial \widehat{\text{Spending}} \text{ (USD)}_i}{\partial \text{Age}_i}|_{\text{Age}=20} = 388 + (2 \cdot 1.9 \cdot 20) + (3 \cdot 0.12 \cdot (20^2)) \approx 608,$$

or an increase in medical spending by USD 608 on average.

TABLE 2
ORDINARY LEAST SQUARES REGRESSION RESULTS OF TOTAL MEDICAL EXPENDITURE ON "MEDIGAP" COVERAGE IN THE MCBS

	A. Without Health Controls			B. With Direct Health Controls		
Variable	All (1)	Female (2)	Male (3)	All (4)	Female (5)	Male (6)
Medigap	−4,392.7***	−6,037.4***	−1,863.4***	1,937.0***	1,677.3***	2,420.9***
	(346.5)	(455.5)	(538.8)	(257.2)	(348.0)	(395.8)
Female	270.0	. . .	. . .	−751.6***	. . .	. . .
	(356.2)			(283.3)		
Age − 65	387.5***	460.6***	292.9	394.5***	417.5***	355.4*
	(138.0)	(175.5)	(228.5)	(117.2)	(144.6)	(196.8)
$(\text{Age} - 65)^2$	1.9	−1.8	5.6	−27.5***	−32.0***	−22.8
	(10.6)	(13.2)	(18.8)	(9.2)	(11.4)	(16.2)
$(\text{Age} - 65)^3$	.12	.17	.07	.47**	.55**	.47
	(.22)	(.27)	(.43)	(.21)	(.25)	(.38)
State dummy	Yes	Yes	Yes	Yes	Yes	Yes
Year dummy	Yes	Yes	Yes	Yes	Yes	Yes
Observations	15,945	9,725	6,220	14,129	8,371	5,758
Adjusted R^2	.073	.092	.060	.211	.196	.252

NOTE.—The dependent variable is total medical expenditure. All regressions are weighted by the cross-section sample weights. Health controls included in panel B are described in detail in the Data Appendix under the category Health. A total of 71 health indicators are included. Robust standard errors clustered at the individual level are in parentheses.
* Significant at 10 percent.
** Significant at 5 percent.
*** Significant at 1 percent.

Table 3.2: Source: Fang, Keane, and Silverman (2008), Table 2.

3.5.5 Interaction models

In the linear model $y = \alpha + \beta X$, we are making an implicit assumption: the average treatment effect β is constant. But this seems often implausible. For instance, consider these examples:

- The effect of an ad on a product's sales might differ for young adults versus older customers.
- The effect of a new technology on a country's output (GDP) might differ for a democracy versus a dictatorship.
- The effect of a policy announcement on a politician's level of support might differ for women versus men.
- The effect of a Work-from-Home policy on a firm's performance might differ for a manufacturing firm versus a financial firm.

For these reasons, we may sometimes want to investigate whether the average treatment effect varies by group, which we call **heterogeneous treatment effects**. Here, there are two approaches:

1. You know the groups you want to study. For instance, you want to see whether your map app creates more engagement for car owners versus people who use bikes. This is what we will discuss below.
2. You don't know the groups and want the data to reveal them. For instance, it could be that your app works well for men aged 27-35 who live in suburbs, but you don't know this. To reveal heterogeneity, recent developments in machine learning have been suggested (Athey, Tibshirani, and Wager 2019; Chernozhukov et al. 2018).

Aside from revealing heterogeneity, interaction models are also useful because they allow the construction of models that avoid assumptions about linearity of the conditional expectation function (i.e., $\mathbb{E}[Y|\mathbf{X}]$). We talk here of **saturated models** and we will return to this later.

The starting point to allow heterogeneity is to build an interactive model in which independent variables enter the model independently (we call these "constituent terms") and as their joint product ("interaction term") (Brambor, Clark, and Golder 2006):

$$\mathbb{E}[Y|X, Z] = \alpha + \beta X + \gamma Z + \tau XZ.$$

We are interested in the effect of X, which we obtain by taking the partial derivative with respect to X:

$$\frac{\partial \mathbb{E}[Y|X, Z]}{\partial X} = \beta + \tau Z.$$

Here's what's key: the effect of an increase of X on $\mathbb{E}[Y]$ will depend on the value of τZ. This differs from the simpler linear model, in which the effect of X was β and did not depend on anything else. Now, the effect of an increase of X will depend not only on β, but also on τZ. Since τ is a constant (and presumably non-zero), the effect of X will vary depending on the value of Z. If τ is large enough, you can end up in situations in which the effect of X could be, say, positive for some values of Z and negative for others. Using the imaginary examples listed above, you could imagine:

- An ad increases sales among old customers but reduces it for young customers.
- A new technology has zero effect on GDP in dictatorships but a positive effect in democracies.
- etc.

What analysts typically do is to select a few meaningful values of Z and compute the marginal effect of X at these values, using $\hat{\beta}$ and $\hat{\tau}$ estimated via OLS. If Z is a dummy (dichotomous) variable, then this simply boils down to computing the effect of X when $Z = 0$ and then again when $Z = 1$. You can also augment this by computing the value of the marginal effect at a range of values of Z (say, $[Z_0, Z_1]$) and then plotting the marginal effect on the y-axis as a function of Z (x-axis).

Note that you can increase the complexity of these models by interacting more than two variables. You could have:

$$y = \alpha + \beta X + \gamma Z + \lambda W + \tau XZ + \rho ZW + \phi XZW + \varepsilon.$$

The thing to keep in mind is that, once you introduce interaction effects, you want to include all possible combinations. For instance, you would need very good reasons to estimate this model:

$$y = \alpha + \beta X + \tau XZ + \rho ZW + \phi XZW + \varepsilon,$$

where Z and W have been omitted. This imposes weird assumptions that are hard to justify (Brambor, Clark, and Golder 2006).

Interaction model: illustration

Example 3.4 (Interaction effect)**.** In the 1990s, macroeconomist started observing a decoupling between US and European economies. In the US, productivity (measured as output per worker) increased, whereas it didn't in Europe. In the long term, this creates considerable diverging GDP paths.
Is this divergence caused by differing management styles? Or was it caused by the rise of internet and related technologies? N. Bloom, Sadun, and Reenen (2012) looked at

the performance of European firms. The asked whether the effect of IT spending was larger for US-owned than for non-US owned firms. One hypothesis is that US managers knew better how to benefit from investments in IT, so that spending one dollar on IT would have a larger effect if done under US than under European (or non-US more generally) management.
After collecting a bunch of data, they estimate the following model:

$$\begin{aligned}\text{Output/worker (log)}_i = \beta\text{US owner}_i + \gamma\text{IT/worker (log)}_i \\ + \lambda\beta\text{US owner}_i \cdot \text{IT/worker (log)}_i \\ + ... + \varepsilon_i,\end{aligned}$$

where i denotes a firm.
The effect of IT spending then is as follows. For a non-US owned firm based in Europe:

$$\frac{\partial\text{Output/worker (log)}_i}{\partial\text{IT/worker (log)}} = \gamma;$$

and for a US-owned, Europe-based firm:

$$\frac{\partial\text{Output/worker (log)}_i}{\partial\text{IT/worker (log)}} = \gamma + \lambda$$

The results are reported in Table 3.3 in which we add variables one after the other. Given the log, the interpretation is a bit tricky (see logs elsewhere in this chapter). But here, we can say the following:

- A firm that is US-owned is 6-7 percent more productive than a Europe-owned one (models 1 and 2).
- Roughly doubling the amount of IT per worker increases productivity by 5 percent, regardless of the owner's origin (model 2)
- Roughly doubling the amount of IT per worker for a European owner increases productivity by 4 percent for a European-owned firm and by 6 (2+4) percent for a US-owner firm.

From this and from other data, the authors conclude that "Americans do it better", that is, are more efficient in turning IT into output (at least in the 2000s).

	M1	M2	M3
US owner	0.07	0.06	0.07
IT capital (log)		0.05	0.04
IT (log) * US owner			0.02
Obs	21,746	21,746	21,746

Table 3.3: Extract from N. Bloom, Sadun, and Reenen (2012), Table 2 (only reporting the parameters of three variables). Analysis of Europe-based firm performance. The dependent variable is a firm's output per worker (logged). *IT capital* is IT spending per worker (logged). *US owner* indicates whether a firm is owned by a US firm (yes=1; 0 if the firm is owned by Europeans).

There is another reason to use interaction effect models. Under some conditions, interaction models allow us to relax the assumption of linearity of the *conditional expectation function*. Consider the case in which D is a dummy treatment. If our model is:

$$y_i = \alpha + \beta D_i + \varepsilon_i,$$

then there is no problem with linearity. We can compute $\mathbb{E}[Y|D = 0]$ and $\mathbb{E}[Y|D = 1]$. There is no other values of D (say, $D = 3$) that could introduce nonlinearity.

But now, suppose you want to include an additional independent variable $Z \in \{0, 1\}$:

$$\mathbb{E}[Y|D] = \alpha + \beta D + \gamma Z$$

Is the conditional expectation function still linear? Not necessarily, though we are here assuming it is. The problem is the following. We can compute the following expected values for

Y:

$$\begin{aligned}
\mathbb{E}[Y|D=0, Z=0] &= \alpha \\
\mathbb{E}[Y|D=1, Z=0] &= \alpha + \beta \\
\mathbb{E}[Y|D=0, Z=1] &= \alpha + \gamma \\
\mathbb{E}[Y|D=1, Z=1] &= \alpha + \beta + \gamma
\end{aligned}$$

Here's the key: the treatment effect is β regardless of the value of Z. To see this, take the difference between the 4th and the 3rd expression (the treatment effect of X conditional on $Z = 1$): it's β. Then take the 2nd minus the 1st expression (where we condition on $Z = 0$): it's β again. Thus, this *additive* model leaves no room for variation in treatment effect. But this is the same as *assuming* linearity of the conditional expectation function.

To relax this assumption, we would like to allow a unique estimate for each combination of independent variables. We can achieve this by including interaction effects:

$$\mathbb{E}[Y|D] = \alpha + \beta D + \gamma Z + \lambda(D \cdot Z)$$

This allows us to generate distinct treatment effects depending on Z:

$$\begin{aligned}
\mathbb{E}[Y|D=0, Z=0] &= \alpha \\
\mathbb{E}[Y|D=1, Z=0] &= \alpha + \beta \\
\mathbb{E}[Y|D=0, Z=1] &= \alpha + \gamma \\
\mathbb{E}[Y|D=1, Z=1] &= \alpha + \gamma + \beta + \lambda
\end{aligned}$$

Note now that the ATE when $Z = 1$ is $\beta + \lambda$, whereas it is β when $Z = 0$. This allows the treatment effect to vary. We call this a **saturated model**. In such models, there is no problem of linearity *by construction*. Note of course that this can only be done in models with discrete and finite set of categories (like dummy variables). Linearity problems return when the covariates are continuous.

3.6 Frisch–Waugh–Lovell Theorem

The OLS estimates of the parameters of a regression with more than one independent variable are thus quite straightforward: $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$. While this is nicely compact, it doesn't tell us all that much about how we should think of $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$.

To help us, we can use the **Frisch–Waugh–Lovell theorem** (or the **regression anatomy theorem**) (Frisch and Waugh 1933; Lovell 1963) to get a better insight into what is going on.[4] The theorem states the following.

Theorem 3.1 (Frisch-Waugh-Lovell)**.** *Suppose we have:*

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon.$$

Partition $\mathbf{X}$ *into* $\mathbf{X}_2$ *and* $\mathbf{X}_2$*, and likewise* β *into* β_1 *and* β_2*.*

1. *Estimate the following model with OLS:*

$$\mathbf{y} = \mathbf{X}_2\beta_2$$

   and estimate the residuals $\hat{\varepsilon}_y$

2. *Estimate the following model with OLS:*

$$\mathbf{X}_1 = \mathbf{X}_2\gamma$$

   and estimate the residuals $\hat{\varepsilon}_{\beta_2}$

3. *You can then regress* $\hat{\varepsilon}_y$ *on* $\hat{\varepsilon}_{\beta_2}$ *and estimate the model's parameter with OLS:*

$$\hat{\varepsilon}_y = \hat{\varepsilon}_{\beta_2}\psi$$

The Frisch-Waugh-Lovell Theorem shows that:

$$\hat{\psi} = \hat{\beta}_1$$

We can also get Frisch-Waugh-Lovell in a slightly different manner:

[4] See J. Angrist and Pischke (2008) and Filoso (2013) for a discussion. The presentation here follows B. Hansen (2022a, 82).

1. Suppose you have a model such that:

$$y_i = \alpha + \beta x_{i,1} + \delta w_{i,2} + ... + \gamma z_{i,k} + \varepsilon_i$$

with k independent variables.

2. Define the residual $\tilde{x}_i = x_i - \hat{x}$ from the regression of x on all other independent variables, such that:

$$x_i = \rho + \lambda w_{i,2} + ... + \tau z_{i,k} + \mu_i$$

Then, we can express $\hat{\beta}$ as:

$$\hat{\beta} = \frac{Cov(Y, \tilde{x})}{Var(\tilde{x})}$$

In other words, $\hat{\beta}_x$ will equal the covariance of Y and the residuals from the regression of x on all other independent variables, scaled by the variance of these residuals. But note that this is the same as the following three steps:

1. Regress x on all other independent variables
2. Obtain the residuals from this regression ($\tilde{x}$)
3. Regress y on $\tilde{x}$, because $\hat{\beta}_x = Cov(Y, \tilde{x})/Var(\tilde{x})$ in the bivariate case.

One way to think about this is to note that $\tilde{x}$ is *whatever variation is left of x once we explain it with all other confounding variables*. So in a sense we purge from x everything that could be attributed to the other confounders. And then we finally scale it by the residuals' variance.

This is a very useful result for a number of reasons. First, it tells us that $\hat{\beta}_x$ will depend on all other independent variables *as long as they are correlated with x*. We can see this from 2: suppose that w is correlated with x, then $\lambda \neq 0$. But then, $\tilde{x}$ will change depending on whether w is included in the regression or not. And therefore, so will $\hat{\beta}$.

We can therefore see the following key implication: if a variable Z is (causally) correlated with X and Y, then we need to

include it in our model. Otherwise, we suffer from **omitted variable bias**. See below for more on this.

A related implication concerns the design of our research. If we can **randomize** our independent variable, then by construction *nothing* is correlated (in expectation) with X (since X is randomly assigned), and therefore we don't *need* to adjust for anything (there are other reasons to adjust for third factors, which we'll discuss later). This is why **randomized controlled trials** (RCTs) are so popular.

It seems that the first such experiment was conducted by James Lind, a Scottish physician, who was trying to find a remedy to scurvy.

3.7 Multicollinearity

3.7.1 Perfect multicollinearity

So far, the only assumption we made to derive OLS estimates is that $\mathbf{X}'\mathbf{X}$ is invertible. This, as we saw, is the same as saying that $\mathbf{X}$ must be of full rank, which in turn is the same as saying that no variable is a linear combination of others. The assumption we make here is the absence of **perfect multicollinearity**. Without it, we cannot compute a minimum to the least squares problem and we are stuck.

There are two reasons why this assumption may not be fulfilled:

1. Incorrectly specified model.
2. Insufficient data and bad (or unlucky) sampling.

The former is a problem of *theory*. The latter is a data problem. Let's clarify. First, we may face the problem of building up a model that contains variables that, by construction, are perfectly collinear. For instance, assuming a world with two genders for simplicity, we may try to estimate:

$$\text{Wage}_i = \alpha + \beta\text{Female}_i + \gamma\text{Male}_i + \varepsilon_i.$$

Here, $\mathbf{X}$ is not of full rank: if a unit is a female (=1), then we know by construction that male=0, and vice versa. Thus, the two variables perfectly predict each other. Likewise, if we do

an analysis of financial flows and want to control for seasonality (and using daily data indexed by t):

$$\text{CAPEX}_t = \alpha + \beta T_t + \gamma \text{Winter}_t + \rho \text{Spring}_t + \phi \text{Summer}_t + \lambda \text{Fall}_i + \varepsilon_i$$

If the observation is measured in winter, you know by construction that the other variables will be zero.

The solution here is easy: modify your model to remove one of the variable. This has no consequence. For instance:

$$\text{Wage}_i = \alpha + \beta \text{Female}_i + \varepsilon_i.$$

Then, β is the difference between $\mathbb{E}[Y|\text{Male}]$ and $\mathbb{E}[Y|\text{Female}]$. Furthermore, note that $\mathbb{E}[Y|\text{Male}] = \alpha$. We call *male* the **omitted category**, but note that while it's omitted, we still recover the expected value of Y for that category.

The second reason for perfect multicollinearity is an "unlucky" sample. For instance, if we tried to get data on wages and gender, we might randomly end up with a sample that only contains women. Then, we couldn't estimate the model, since there is no variance on that variable. The solution here, typically, is to collect more data.

A version of this story that is less about lack of luck is a poor sampling design. For instance, if you want to study the effect of being above 65 years old (=1) on public opinion about a product (adjusting for having a college degree), then do not run your survey on the grounds of a university, since you might only end up with young people with a college degree.

3.7.2 High multicollinearity

High multicollinearity – a large but not perfect correlation between independent variable(s) – is *not* a violation of the Gauss-Markov or of any assumption in the linear model and OLS. As such, we don't require anything aside from a lack of *perfect* multicollinearity when we work with these models and estimation methods.

This being said, high multicollinearity can still be annoying. The issue is *not* that it will create bias or inconsistency. Recall that nothing about these results require low multicollinearity. No, the problem is that high multicollinearity will make $(\mathbf{X}'\mathbf{X})^{-1}$ larger. And since this is part of the expression of the variance of $\widehat{\beta}$, this means that high multicollinearity will reduce t statistics and make it more difficult to detect a statistical effect. More on this later.

To illustrate the problem, consider the following example. Suppose that we want to measure the effect of people's height on their wages, and we want to adjust for gender. We would build the following model:

$$\text{Wage}_i = \alpha + \beta\text{Height}_i + \gamma\text{Male}_i + \varepsilon_i$$

But suppose that our data only contains tall men and short women. The two variables will be highly, though not perfectly, correlated. Then it will be very hard to tease out the effect of height versus the effect of gender.

You can check whether your model suffers from high multicollinearity. The most common test is the **Variance Inflation Factor** (VIF). Consider a model such as:

$$y = \alpha + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_k x_k + \varepsilon.$$

Then, variable x_i's VIF is:

$$\text{VIF}_{x_i} = \frac{1}{1 - R_i^2},$$

where R_i^2 is the R-squared of a regression of x_i on all other independent variables x. Typically, people start to worry when VIF is greater than 10. But there's no hard rule about this.

What's the solution to high multicollinearity? To the degree that the problem is a lack of independent variation across variables, the solution will be to add more data.

What you should *not* do is to drop the variable(s) that is (are) causing the problem. A variable's inclusion in a model is based on theoretical grounds. Either it is a confounder or it isn't. The

fact that it's "annoying" to have a confounder that's highly correlated with your treatment isn't a good justification to remove it.

Review questions

Exercise 3.1. Derive the ordinary least *cubes* estimator. To simplify, consider the model $y = \beta x$ (no intercept). What is it? Is there a closed-form solution? Bonus: implement this estimator in R or Python.

Exercise 3.2. When we bin variables into dummy indicators (see nonlinear models), why do we omit one category?

Appendix

Matrix algebra

Matrix algebra

A quick refresher on matrices and matrix algebra. Matrices are indexed by two dimensions, row $\times$ columns, and are often referred to in capitalized letters in bold fonts. An $n \cdot k$ matrix $\mathbf{X}$ corresponds to:

$$\mathbf{X} = \begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,3} & \cdots & x_{1,k} \\ x_{2,1} & x_{2,2} & x_{2,4} & \cdots & x_{2,k} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{n,1} & x_{n,2} & x_{n,3} & \cdots & x_{n,k} \end{bmatrix}_{n \times k}$$

A matrix in which one of the dimensions equals 1 is a vector (row vector if it is of length $1 \times k$, column vector if $n \times 1$). Since column vectors are more common, it is typically assumed that a "vector" refers to a the latter. Vectors are sometimes labeled in lower-case bold fonts (e.g., $\mathbf{x}$).
The transpose of a matrix flips the rows and columns, such that entry $x_{i,j}$ becomes $x_{j,i}$ for all i, j. It is denoted $\mathbf{X}^\top$ or $\mathbf{X}'$.

$$\mathbf{X} = \begin{bmatrix} A_{1,1} & B_{1,2} & C_{1,3} \\ D_{2,1} & E_{2,2} & F_{2,4} \end{bmatrix}_{2 \times 3}$$

$$\mathbf{X}' = \begin{bmatrix} A_{1,1} & D_{1,2} \\ B_{2,1} & E_{2,2} \\ C_{3,1} & F_{3,2} \end{bmatrix}_{3 \times 2}$$

Addition. Matrices can be added as long as they are **conformable**, i.e., have the same dimensions.
Multiplication. To multiply matrices, we need the concept of **inner product** (or **scalar product**).[5] Consider two vectors, $\mathbf{x}$ and $\mathbf{z}$, the inner product is:

$$\mathbf{x}'\mathbf{z} = \sum_{i}^{n} x_i z_i = \mathbf{z}'\mathbf{x} = c,$$

where c is a scalar.
The product of two matrices yields a new matrix, whose entry i, j will be the inner product of the ith row of the first matrix and the jth column of the second matrix. (For this to work, we need the first matrix to have the same number of columns as the second matrix's number of rows.) We then end up with:

$$\mathbf{X}_{n\times k}\mathbf{Z}_{k\times l} = \mathbf{W}_{n\times l}$$

Properties of addition, products, and transposes:

- $\mathbf{X}(\mathbf{W}+\mathbf{Z}) = \mathbf{XW}+\mathbf{XZ}$
- $(\mathbf{XZ})' = \mathbf{Z}'\mathbf{X}'$

Inverse. The inverse of a matrix, written $\mathbf{X}^{-1}$, is a matrix that satisfies the following property:

$$\mathbf{XX}^{-1} = \mathbf{X}^{-1}\mathbf{X} = \mathbf{I},$$

where $\mathbf{I}$ is the **identity** matrix, a square $(l \times l)$ matrix, with diagonal entries $\mathbf{I}_{i,i} = 1$ and off-diagonal entries $\mathbf{I}_{i,j} = 0$. The logic is the following: if you multiply a scalar by its inverse, you get 1 (e.g., $8 \cdot 1/8 = 1$). Likewise, the inverse is what you meed to multiply a matrix $\mathbf{X}$ to obtain the identity matrix. Note that the inverse may not exist.

[5] Not to be confused with **direct product** in which x_{ij} is multiplied by $z_{i,j}$ for all i, j.

4 Inference with OLS

This version: December 16, 2025

> **Caveat**
>
> These notes are offered in support of *Introduction to Econometrics* (MGT 581). They were written to help students understand the logic behind important topics in econometrics. But they don't (always) replace the insights and depth that textbooks can provide. This is why we connect each session to chapters of our core textbooks. See the syllabus for more details.

4.1 Readings

Here are references on least squares and (causal) inference: (B. Hansen 2022a) (ch. 6-7, and 9), (Stachurski 2016) (ch. 13.1), (Wooldridge 2009) (ch. 4-5), (Verbeek 2018) (ch. 2.4-2.6 and 4), (Hayashi 2000) (ch2.3-2.4), (Stock and Watson 2011) (ch. 5-7).

4.2 Performance of estimators

Our main result ($\hat{\beta}_{ols} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$) did not require any assumption aside from the matrix $\mathbf{X}$ being of full rank. Yet this only guarantees the *existance* of a unique vector $\hat{\beta}_{ols}$ that solves the minimization problem in a given **sample**. It doesn't tell us anything about whether this rule has desirable properties and allows us to say anything about the true, unobservable, **population** vector β.

The properties we may want to achieve are:

- **Unbiasedness**. Note that $\hat{\beta}_{ols}$ is a random variable, since it is the product of random variables. Unbiasedness refers to the property that the expected value of $\hat{\beta}$ obtained via some estimator equals the true population value of β, namely:
$$\mathbb{E}[\hat{\beta}] = \beta$$

- **Consistency**. We may want the distribution of our estimate $\hat{\beta}$ to get increasingly more precise around the true value β as our sample size goes toward infinity. A consistent estimate is one that converges in probability to β. That is:
$$\text{plim}\hat{\beta} = \beta$$

- **Efficiency**. Due to sampling variation, estimators will generate estimates that vary from sample to sample. What we would like is for the estimator to have the smallest possible variance compared to any alternative estimator $\tilde{\beta}$. Typically, we restrict the competition to *unbiased* alternative estimators.
$$\mathbb{V}[\hat{\beta}] \leq \mathbb{V}[\tilde{\beta}], \forall \text{ unbiased } \tilde{\beta}$$

  Note. An estimator can be biased yet consistent. For instance, the estimator for the population mean: $\hat{\mu}_x = \frac{1}{n+1}\sum x$ is biased but consistent: as n increases, the bias disappears.

  Note. An estimator can be unbiased but inconsistent. For instance, if you estimate the population mean by picking the first value from your sample, your estimate would be unbiased, but the distribution of the error this estimator isn't converging to the true population mean.

What we want to figure out is: under what conditions is OLS unbiased, consistent, and/or efficient?

Unbiasedness: example

Let us establish that the sample average $\bar{x}$ is an unbiased

estimator of the population expectation $\mathbb{E}[X]$. We define:

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

What we need to establish is that $\mathbb{E}[\bar{x}] = \mathbb{E}[X]$. To do so, we take expectations of the previous expression:

$$\begin{aligned}
\mathbb{E}[\bar{x}] &= \mathbb{E}\left[\frac{1}{N}\sum_{i=1}^{N} x_i\right] \\
&= \frac{1}{N}\mathbb{E}\left[\sum_{i=1}^{N} x_i\right] && \text{extracting constant} \\
&= \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}[X] \\
&= \frac{1}{N}N\mathbb{E}[X] && \text{n times } \mathbb{E}[X] \\
&= \mathbb{E}[X]
\end{aligned}$$

which completes the proof.

4.3 Assumptions so far

So far, the only assumptions we made to derive the OLS estimator is to state that the design matrix $\mathbf{X}$ is of full rank (no perfect multicollinearity) with probability 1 (**A2**).

Let us add a few additional assumptions, some of which we will relax later.

Throughout, except if otherwise stated, we assume that $\mathbf{y}$ and $\mathbf{X}$ are **random variables**. See earlier notes on measure theory and see **A3**. We will generally also assume that units in our sample are independently drawn from identical distributions (often referred to as the **IID** assumption). We also generally assume that these distributions have finite first moments and finite and positive second moments.

Assumption: IID with finite moments

Assumption 3 (IID). The random variables X, Y are randomly drawn from independent and identical distributions (IID).
In addition, X, Y have finite first moment and finite nonzero second moments.

This assumption combines three ideas: random sampling, IID, and properties of the distribution of X and Y. Finite moments excludes certain types of distributions, such as Cauchy, but it's otherwise not overly restrictive in practice.
Later, we will discuss what happens when the variables are not IID (with independence being a focus in particular).

Aside from this, we also start by assuming that $\mathbf{y}$ can be expressed as a linear function of $\mathbf{X}$, where linearity is about the parameters connecting X to Y (**A4**). For instance:

$$Y = X^{\beta} \qquad \text{not linear}$$
$$Y = \beta X + \gamma X^2 \qquad \text{linear in parameters}$$

Later, we will see under what conditions this assumption is plausible.

Assumption: Linearity

Assumption 4 (Linearity). Y can be expressed as a linear function of X, such that:

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon,$$

where $\mathbf{X}$ can include any transformation (linear or not) of an arbitrary set of random variables.
Note: this assumption does not mean that the relation between X and Y is linear. Rather, this relation is linear *in parameters*.
This assumption can hold by construction with discrete treatments in saturated models.

4.4 Unbiasedness

We want to show that $\mathbb{E}[\hat{\beta}] = \beta$. To do so, we must find a way to connect $\hat{\beta}$ to the true parameter β. Starting with the estimator, we can write:

$$\begin{aligned}
\hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} && \text{from FOC/SOC of OLS, incl. A2} \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \varepsilon) && \text{by A4} \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon && \text{properties of matrices} \\
&= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon && \text{properties of matrices}
\end{aligned}$$

We can now take the expectations:

$$\begin{aligned}
\mathbb{E}[\hat{\beta}] &= \mathbb{E}[\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon] \\
&= \mathbb{E}[\beta] + \mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon] && \text{distributing expectations} \\
&= \beta + \mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon] && \text{since } E(c) = c \\
&= \beta + \mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbb{E}[\varepsilon|\mathbf{X}]] && \text{by law of iterated expectations} \\
&= \beta + \mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{0}] && \text{by A5} \\
&= \beta
\end{aligned}$$

which concludes the proof. We needed here to make a new assumption: the conditional expected value of ε given $\mathbf{X}$ equals zero (**A5**) (see Davidson and MacKinnon (2004, 90) or Stock and Watson (2011, 143)).

The expectation could also have gone right through the second expression if we are willing to claim that $\mathbf{X}$ are fixed and not random variables. Then we could just have assumed that $\mathbb{E}[\varepsilon] = 0$. See Davidson and MacKinnon (2004, 90).

In sum, under **A4** to **A5**, OLS will yield **unbiased** estimates of the population parameter β.

⚠ Assumption: exogeneity/zero conditional expectation

Assumption 5 (Exogeneity). Conditional on predictors, the error term equals zero in expectation, that is:

$$\mathbb{E}[\varepsilon|\mathbf{X}] = 0$$

This assumption can be built from two ideas. To see this, note that this equivalent to writing:

$$\mathbb{E}[\varepsilon_i|\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_i, ..., \mathbf{x}_n] = 0.$$

That is, the expected error term for unit i is zero conditional on the values of all independent variables for all

units $1, 2, ...i, ..., n$. Now, we can simplify this to the much weaker assumption:

$$\mathbb{E}[\varepsilon_i|\mathbf{x}_i] = 0,$$

but only if we bring in our earlier assumption of independence (**A3**). Recall that we started by assuming that our observations are IID – one of the "I" stands for independence. Thus, our exogeneity assumption, as stated here (i.e., $\mathbb{E}[\varepsilon|\mathbf{X}] = 0$) can be met by combining $\mathbb{E}[\varepsilon_i|\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_i, ..., \mathbf{x}_n] = 0$ with independence.
Note that this assumption implies that $Cor(\varepsilon, \mathbf{X}) = 0$ and that $\mathbb{E}[\mathbf{X}'\varepsilon] = 0$.

4.5 Asymptotic distribution of $\hat{\beta}_{ols}$

To do inference from a sample to the population, we must understand the nature of $\hat{\beta}$. The starting point is that it is a **random variable**, unlike β which is a constant (at least for now). Why does it vary? Because of sampling! From one sample s to another, we will obtain $\hat{\beta}_s$ that are different from each other. See for instance Figure 4.1, which shows how three different samples yield somewhat different values of $\hat{\beta}$ (0.58, 0.87, and 0.9). We call this **sampling variance**.

We are interested in understanding the distribution of $\hat{\beta}$ as N tends to infinity, that is, its **asymptotic distribution**. Take again Figure 4.1: if we could sample over and over again (say, an infinite number of times, or at least a very large number of timess), we could build the full distribution of $\hat{\beta}$; this is what the light blue distribution does in the second row of the figure (here: for 100 samples).

The challenge is that we don't know, exactly, what this distribution looks like. We generally cannot re-sample to generate new $\hat{\beta}$.

The workaround will be to see whether we can borrow results from asympototic statistics to make claims. Concretely (and we

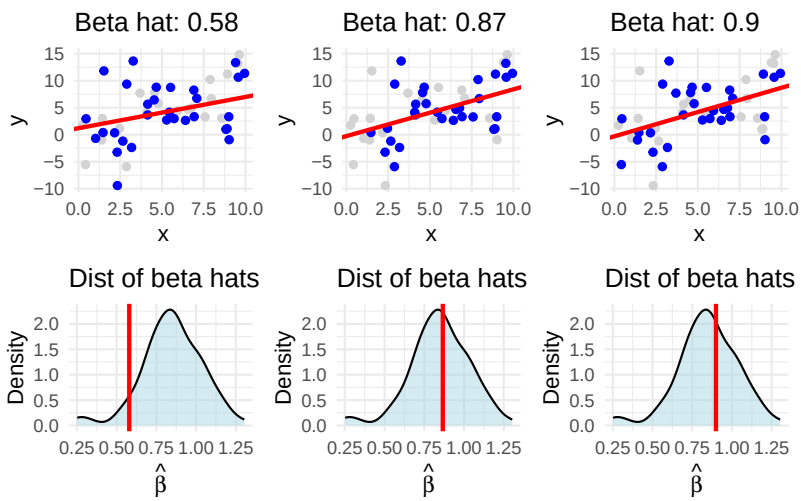

Figure 4.1: **Sampling variance**. Three samples from a common population. 30 observations (blue) are randomly sampled from a population of 50 observations (gray), generating slightly different beta.hat. True model: y = 0.5 + 0.7x + N(0,5). Bottom row: distribution of beta hats from 100 Monte Carlo simulations. The red line reports the beta hat from the figure above.

will work through this next), we will borrow the **Lindeberg–Lévy Central Limit Theorem** (CLT) (Theorem 1.2). If you recall, it states that, for a sequence of random IID variables $X_1, X_2, ..., X_n$, the asymptotic distribution of the centered and scaled sample mean $\bar{X}$ is normal (Gaussian):

$$\frac{\sqrt{N}(\bar{X} - \mu)}{\sigma} \sim \mathcal{N}(0, 1).$$

If we could use this, somehow, for estimates using OLS, then we would know how $\hat{\beta}$ is asymptotically distributed. To top it off, it would be a very nice and well-known distribution, which would allow us to make all kinds of statements regarding the probability of observing ranges of values of $\hat{\beta}$.

The way we will do this is the following: we will try to reformulate $\hat{\beta}$ in a form approaching Lindeberg-Lévy. Then we will see whether we can apply it. In doing so, we will use three theorems from statistics: (1) the Lindeberg-Lévy Central Limit Theory, (2) the Law of Large Numbers, and (3) Slutsky's Theorem.

Step 1. Let's reformulate $\hat{\beta}$ in a way that mimics Lindeberg-Lévy. We start with:

$$\begin{aligned}
\hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} && \text{by definition} \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}'\beta + \varepsilon) && \text{by A4} \\
&= \beta + (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\varepsilon) && \text{canceling terms}
\end{aligned}$$

define the expression $\sqrt{N}(\hat{\beta} - \beta)$ using that last expression:

$$\begin{aligned}
\hat{\beta} - \beta &= (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\varepsilon) && \text{shifting terms} \\
\sqrt{N}(\hat{\beta} - \beta) &= \sqrt{N}(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\varepsilon) && \text{multiplying by sqrt N} \\
&= \frac{N}{\sqrt{N}}(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\varepsilon) && \text{sqrt N is same as N/sqrt N} \\
&= N(\mathbf{X}'\mathbf{X})^{-1}\frac{1}{\sqrt{N}}(\mathbf{X}'\varepsilon) && \text{distributing} \\
&= \left(\frac{\mathbf{X}'\mathbf{X}}{N}\right)^{-1}\frac{1}{\sqrt{N}}(\mathbf{X}'\varepsilon) && \text{putting N inside the inverse}
\end{aligned}$$

Step 2. Since we want to know what happens in asymptotia, we must study this expression as $N \to \infty$.

$$\text{plim}[\sqrt{N}(\hat{\beta} - \beta)] = \text{plim}\left[\left(\frac{\mathbf{X}'\mathbf{X}}{N}\right)^{-1} \frac{1}{\sqrt{N}}(\mathbf{X}'\varepsilon)\right]$$

Let us divide this expression in two parts:

$$\left(\frac{\mathbf{X}'\mathbf{X}}{N}\right)^{-1} \quad \text{and} \quad \frac{1}{\sqrt{N}}(\mathbf{X}'\varepsilon)$$

First expression ($\left(\frac{\mathbf{X}'\mathbf{X}}{N}\right)^{-1}$). What will happen to the first expression as N goes to infinity? By the **Law of Large Numbers**, we can write:

$$\frac{\mathbf{X}'\mathbf{X}}{N} \xrightarrow{p} \mathbb{E}[\mathbf{X}'\mathbf{X}] := \mathbf{Q},$$

which is a positive-definite matrix that *converges in probability* to $\mathbf{Q}$.[1] Then, by A2, we know that the inverse of this quantity is defined, that is, $\mathbf{Q}^{-1}$ exists. Informally, this tells us that (a) the firm term converges to something, and (b) it converges in probability.

Second expression ($\frac{1}{\sqrt{N}}(\mathbf{X}'\varepsilon)$). Note that it is a sum of products $\mathbf{X}'\varepsilon$ (scaled by $1/\sqrt{N}$):

$$\mathbf{X}'\varepsilon = \begin{bmatrix} 1_1 & 1_2 & \dots & 1_n \\ x_1 & x_2 & \dots & x_n \\ z_1 & z_2 & \dots & z_n \\ \dots & & & \\ k_1 & k_2 & \dots & k_n \end{bmatrix} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \dots \\ \varepsilon_n \end{bmatrix}$$

At this stage, we can invoke the Lindeberg-Lévy CLT, assuming that ε is IID. The reason is a bit technical, but we can do a short detour to explain it. You may wonder why we can apply Lindeberg-Lévy, since it is generally presented as a result about the sample mean ($\bar{X}$) and not, say, as $\mathbf{X}'\varepsilon$. There are two steps to clarify this:

Technically, Lindeberg-Lévy requires ε to be IID. But this assumption can be relaxed. The CLT may also hold if ε is not identically distributed and exhibits dependence. You would need something like the Lindeberg–Feller CLT (with independence across ε) or Billingsley's CLT (with weak dependence)

[1] We sometimes write *plim* instead of $\xrightarrow{p}$.

1. The CLT is actually about the *sum* of random variables, not their (sample) mean. To see this, note that we can rewrite the CLT as:

$$\begin{aligned}\frac{\sqrt{N}(\bar{X}-\mathbb{E}[X_i])}{\sigma} &= \frac{N(\bar{X}-\mathbb{E}[X_i])}{\sqrt{N}\sigma} \\ &= \frac{N\bar{X}-N\mathbb{E}[X_i]}{\sqrt{N}\sigma} \\ &= \frac{\sum(X_i-\mathbb{E}[X_i])}{\sqrt{N}\sigma} \\ &= \frac{\sum X_i - N\mathbb{E}[X_i]}{\sqrt{N}\sigma}\end{aligned}$$

   The last expression is equivalent to the one used by Billingsley (1995, 357) in his definition of the Lindeberg-Lévy CLT.

2. From there, take the expression $\sum(X_i - \mathbb{E}[X_i])$ and plug in $\mathbf{X}'\varepsilon$ instead of X. Note that $\mathbf{X}'\varepsilon$ is the same as $\sum^N x_i \cdot \varepsilon_i$ (once you exit matrix notation). Thus, we can write:

$$\sum^N (x_i \cdot \varepsilon_i - \mathbb{E}[x_i \cdot \varepsilon_i])$$

   This is a sum of the product of IID variables, centered by the second term! Furthermore, note that $\mathbb{E}[x_i \cdot \varepsilon_i] = 0$ by the exogeneity assumption. This therefore simplifies to:

$$\sum^N (x_i \cdot \varepsilon_i - \mathbb{E}[x_i \cdot \varepsilon_i]) = \sum^N (x_i \cdot \varepsilon_i) \equiv \mathbf{X}'\varepsilon$$

   Thus, assuming that the data are IID and that the exogeneity assumption holds, then we can use the CLT on this quantity.

Note here that we can do two extra steps that will be useful. First, we can confirm that, indeed, this second term has a mean of zero (as we would expect based on the CLT). Second, we can express the actual variance of this quantity (to replace the placeholder σ^2 used in the general statement of the CLT). Let's quickly do this and then let's put all pieces back together.

We can obtain the *first moment* of this distribution immediately:

$$\begin{aligned}\mathbb{E}\left[\frac{1}{\sqrt{N}}(\mathbf{X}'\varepsilon)\right] &= \frac{1}{\sqrt{N}}\mathbb{E}[\mathbf{X}'\varepsilon] \\ &= 0 && \text{by A5}\end{aligned}$$

This is thus a mean-zero vector.

Let's get to the *second moment*. Ultimately, we will want to find the variance of the whole thing:

$$\mathbb{V}\left[\sqrt{N}(\hat{\beta}-\beta)\right] = \mathbb{V}\left[\mathbf{Q}^{-1}\frac{1}{\sqrt{N}}(\mathbf{X}'\varepsilon)\right]$$

We already noted that $\mathbf{Q}$ is finite and invertible. We can extract it from the variance like a constant by pre- and post-multiplying. Let's do this at the end. Instead, to ensure that this expression is finite (and thus this CLT assumption is met), let us derive the second part.

$$\begin{aligned}\mathbb{V}\left[\frac{1}{\sqrt{N}}\mathbf{X}'\varepsilon\right] &= \frac{1}{N}\mathbb{V}\left[\mathbf{X}'\varepsilon\right] && \text{property of var} \\ &= \frac{1}{N}\mathbb{V}\left[\sum_{i=1}^{N} x_i\varepsilon_i\right] && \text{by def of matrix prod} \\ &= \frac{1}{N}\sum_{i=1}^{N}\mathbb{V}[x_i\varepsilon_i] && \text{by A5} \\ &= \frac{1}{N}N\mathbb{E}[\mathbf{X}'\varepsilon\varepsilon'\mathbf{X}] && \text{back to matrix} \\ &= \mathbb{E}[\mathbf{X}'\varepsilon\varepsilon'\mathbf{X}]\end{aligned}$$

Here, we need to assume that:

$$\mathbb{E}[\mathbf{X}'\varepsilon\varepsilon'\mathbf{X}] < \infty$$

If so, we can state that:

$$\frac{1}{\sqrt{N}}\mathbf{X}'\varepsilon \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathbf{X}'\varepsilon\varepsilon'\mathbf{X}])$$

Putting it all together. From here, let's put all the pieces back together. Using the Law of Large numbers, we were able to characterize the first expression. Using the CLT, we were able to characterize the second expression. We have:

$$\text{By LLN: } (\mathbf{X}'\mathbf{X}/N)^{-1} \xrightarrow{p} \mathbf{Q}^{-1},$$

and

$$\text{By CLT: } \frac{1}{\sqrt{N}}\mathbf{X}'\varepsilon \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathbf{X}'\varepsilon\varepsilon'\mathbf{X}]).$$

We can combine the two statements using **Slutsky's Theorem** Theorem 4.1 (B. Hansen 2022b, 171).

Theorem 4.1 (Slutsky)**.** *If $A_n \xrightarrow{p} A$ and $B_n \xrightarrow{d} B$ as $n \to \infty$, then:*

1. $A_n + B_n \xrightarrow{d} A + B$.
2. $A_n B_n \xrightarrow{d} AB$.
3. $\frac{B_n}{A_n} \xrightarrow{d} \frac{B}{A}$ *if* $A \neq 0$.

Using the second result from Slutsky, this means that:

$$\begin{aligned}
\sqrt{n}(\hat{\beta} - \beta) &= \underbrace{\left(\frac{\mathbf{X}'\mathbf{X}}{N}\right)^{-1}}_{A_n} \underbrace{\frac{1}{\sqrt{N}}(\mathbf{X}'\varepsilon)}_{B_n} \\
&\xrightarrow{d} \mathbf{Q}^{-1}\mathcal{N}(0, \mathbb{E}[\mathbf{X}'\varepsilon\varepsilon'\mathbf{X}]) && \text{by def of variance} \\
&\xrightarrow{d} \mathcal{N}(0, \mathbf{Q}^{-1}\mathbb{E}[\mathbf{X}'\varepsilon\varepsilon'\mathbf{X}]\mathbf{Q}^{-1}) && \text{by properties of variance} \\
&:= \mathcal{N}(0, \Omega)
\end{aligned}$$

where Ω is defined as $\mathbf{Q}^{-1}\mathbb{E}[\mathbf{X}'\varepsilon\varepsilon'\mathbf{X}]\mathbf{Q}^{-1}$ and is referred to as the **variance-covariance matrix**.

We can conclude that under the following assumptions:

- **A2** (full rank): needed to compute $\hat{\beta}$
- **A3** (IID and finite second moment): needed for Lindeberg-Lévy CLT

- **A4** (linearity): needed to link $\hat{\beta}$ to β.
- **A5** (exogeneity): needed for $\hat{\beta}$ to be centered on β (so that $\mathbb{E}[\hat{\beta}] - \beta = 0$)...

that:

$$\sqrt{N}(\hat{\beta} - \beta) \sim \mathcal{N}(0, \Omega)$$

> **Note**. $\varepsilon\varepsilon'$ is an $n \times n$ matrix, which is then multiplied by $\mathbf{X}$ (an $n \times k$ matrix). The result is an $n \times k$ matrix. This is then pre-multiplied by $\mathbf{X}'$ (an $k \times n$ matrix), leaving us with a $k \times k$ matrix. Since $(\mathbf{X}'\mathbf{X})$ is a $k \times k$ matrix as well, Ω as a whole is $k \times k$.
>
> **Note**. We can relax the IID assumption and instead invoke other versions of the CLT, such as the Lindeberg–Feller CLT (independent but not identically distributed ε) or the Billingsley CLT (dependent ε_i, weak dependence, finite variance).
>
> **Note**. Sometimes, people will equivalently write $\frac{\sqrt{N}(\hat{\beta}-\beta)}{\sqrt{\Omega}} \sim \mathcal{N}(0, 1)$. We can also write: $\hat{\beta} \sim \mathcal{N}(\beta, \frac{\Omega}{N})$.
>
> **Note**. Sometimes, textbooks use Ω to refer solely to $\mathbb{E}[\mathbf{X}'\varepsilon\varepsilon'\mathbf{X}]$ and thus write $\mathcal{N}(0, \mathbf{Q^{-1}}\Omega\mathbf{Q^{-1}})$.

From there, we can define the **standard error** of $\hat{\beta}_k$ as the square root of the diagonal entry of the kth row of Ω/N.

4.6 Estimating standard errors

From the asymptotic distribution of $\hat{\beta}$ (or, more precisely, the asymptotic distribution of $\sqrt{N}(\hat{\beta} - \beta)$), we know that:

$$\Omega = \mathbf{Q}^{-1}\mathbb{E}[\mathbf{X}'\varepsilon\varepsilon'\mathbf{X}]\mathbf{Q}^{-1}$$

The complication comes from ε, which isn't observed, and thus makes the expression $\varepsilon\varepsilon'$ challenging. First, note that this is a

$n \times n$ matrix. Taking expectations (and assuming it exists; see **A3**):

$$\mathbb{E}[\varepsilon\varepsilon'] = \begin{bmatrix} \mathbb{E}[\varepsilon_1^2] & \mathbb{E}[\varepsilon_1\varepsilon_2] & ... & \mathbb{E}[\varepsilon_1\varepsilon_n] \\ \mathbb{E}[\varepsilon_2\varepsilon_1] & \mathbb{E}[\varepsilon_2^2] & ... & \mathbb{E}[\varepsilon_2\varepsilon_n] \\ ... & ... & ... & ... \\ \mathbb{E}[\varepsilon_n\varepsilon_1] & ... & ... & \mathbb{E}[\varepsilon_n^2] \end{bmatrix}$$

Note that we can simplify this a tiny bit by noting that $\mathbb{E}[\varepsilon_i^2] = \mathbb{V}[\varepsilon_i]$ (or equivalently σ_i^2):

Thanks to the fact that the unconditional error term $\mathbb{E}[\varepsilon] = 0$ by construction.

$$\mathbb{E}[\varepsilon\varepsilon'] = \begin{bmatrix} \mathbb{V}[\varepsilon_1] & \mathbb{E}[\varepsilon_1\varepsilon_2] & ... & \mathbb{E}[\varepsilon_1\varepsilon_n] \\ \mathbb{E}[\varepsilon_2\varepsilon_1] & \mathbb{V}[\varepsilon_2] & ... & \mathbb{E}[\varepsilon_2\varepsilon_n] \\ ... & ... & ... & ... \\ \mathbb{E}[\varepsilon_n\varepsilon_1] & ... & ... & \mathbb{V}[\varepsilon_n] \end{bmatrix}$$

Note as well that we can similarly replace the off-diagonal entries by using the fact that $\mathbb{E}[\varepsilon_i\varepsilon_j] = \text{Cov}[\varepsilon_i, \varepsilon_j]$:

$$\mathbb{E}[\varepsilon\varepsilon'] = \begin{bmatrix} \mathbb{V}[\varepsilon_1] & \text{Cov}[\varepsilon_1, \varepsilon_2] & ... & \text{Cov}[\varepsilon_1, \varepsilon_n] \\ \text{Cov}[\varepsilon_2, \varepsilon_1] & \mathbb{V}[\varepsilon_2] & ... & \text{Cov}[\varepsilon_2, \varepsilon_n] \\ ... & ... & ... & ... \\ \text{Cov}[\varepsilon_n, \varepsilon_1] & ... & ... & \mathbb{V}[\varepsilon_n] \end{bmatrix}$$

This is why $\mathbb{E}[\varepsilon\varepsilon']$ is also called the **variance-covariance matrix of error terms** (it's a matrix of the variances of ε_i and of covariances of ε_i and ε_j).

> **Note**. Ω is also a variance-covariance matrix. Be careful to specify whether you are talking about the variance-covariance matrix of error terms (like) or of parameters.

But now we are stuck. We have $n(n+1)/2$ unknown parameters to figure out. Here, we may want to simplify this expression via two additional assumptions. The first one is the lack of autocorrelation across units or across time (no autocorrelation); the second is that the error term has a constant and identical variance for all units (homoskedasticity).

Because it's a symmetrical matrix ($Cov(a, b) = Cov(b, a)$).

Assumption: No autocorrelation

Assumption 6 (Autocorrelation). For any two unit $i, j \in N$, where $i \neq j$, we have:

$$Cor[\varepsilon_i, \varepsilon_j] = 0$$

This assumption is typically violated in the following cases:

1. Serial correlation. Error term of a unit measured at time t is correlated with the error term at time $t-1$. For instance, in finance, a stochastic shock that increases returns of a firm in the first quarter of 2025 may also affect it in the second quarter of 2025.
2. Spatial correlation. Suppose you are doing a study on Swiss municipalities. A high error term in Lausanne may also generate a high error term in Geneva.

We then simplify the variance-covariance matrix of errors as:

$$\mathbb{E}[\varepsilon\varepsilon'] = \begin{bmatrix} \sigma_1^2 & 0 & ... & 0 \\ 0 & \sigma_2^2 & ... & 0 \\ ... & ... & ... & ... \\ 0 & ... & ... & \sigma_n^2 \end{bmatrix} \qquad \text{via A6}$$

where σ_i^2 is the variance of the error term of unit i.

If we are willing to assume that the variance is identical for all units i (the assumption of **homoskedasticity** (**A7**), then this simplifies further to:

$$\begin{aligned} \mathbb{E}[\varepsilon\varepsilon'] &= \begin{bmatrix} \sigma^2 & 0 & ... & 0 \\ 0 & \sigma^2 & ... & 0 \\ ... & ... & ... & ... \\ 0 & ... & ... & \sigma^2 \end{bmatrix} && \text{by A7} \\ &= \sigma^2 \mathbf{I} && \text{where } \mathbf{I} \text{ is the identity matrix} \end{aligned}$$

We sometimes refer to error terms ε that are independent (no autocorrelation) and homoskedastic as a **spherical** error terms.

Assumption: homoskedasticity

Assumption 7 (Homoskedasticity). The conditional variance of the error term is a constant:

$$\mathbb{V}[\varepsilon|\mathbf{X}] = \sigma^2.$$

Note that the key here is that σ^2 is a constant and doesn't vary by values of $\mathbf{X}$ (i.e., $\sigma^2 \neq \sigma_i^2$).

Under **A6** and **A7**, we are left with a single unknown, σ^2. This allows us to simplify the variance:

$$\begin{aligned}
\Omega &= \mathbf{Q}^{-1}\mathbb{E}[\mathbf{X}'\varepsilon\varepsilon'\mathbf{X}]\mathbf{Q}^{-1} && \text{from above} \\
&= \mathbf{Q}^{-1}\mathbb{E}[\mathbf{X}'\sigma^2\mathbf{I}\mathbf{X}]\mathbf{Q}^{-1} && \text{by A7} \\
&= \sigma^2\mathbf{Q}^{-1}\mathbb{E}[\mathbf{X}'\mathbf{X}]\mathbf{Q}^{-1} \\
&= \sigma^2\left(\frac{\mathbf{X}'\mathbf{X}}{N}\right)^{-1}\left(\frac{\mathbf{X}'\mathbf{X}}{N}\right)\left(\frac{\mathbf{X}'\mathbf{X}}{N}\right)^{-1} && \text{by definition} \\
&= \sigma^2 N\left(\mathbf{X}'\mathbf{X}\right)^{-1} && \text{simplifying} \\
&= \sigma^2\mathbf{Q}^{-1} && \text{equivalently}
\end{aligned}$$

Then, since $\mathbb{V}[\hat{\beta}] = \Omega/N$:

$$\begin{aligned}
\mathbb{V}[\hat{\beta}] &= \frac{\Omega}{N} \\
&= \frac{1}{N}\sigma^2\mathbf{Q}^{-1} && \text{from above} \\
&= \frac{1}{N}\sigma^2\left(\frac{\mathbf{X}'\mathbf{X}}{N}\right)^{-1} && \text{by def of } \mathbf{Q} \\
&= \frac{1}{N}\sigma^2 N(\mathbf{X}'\mathbf{X})^{-1} && \text{extracting } N \\
&= \sigma^2(\mathbf{X}'\mathbf{X})^{-1} && \text{simplifying}
\end{aligned}$$

We can consistently estimate σ^2 with s^2 (some textbooks use $\hat{\sigma}^2$), the (estimated) **standard error of the regression** :

$$s^2 = \frac{\sum_i^N u_i^2}{N-k}$$

where N is the sample size, k the number of independent variables (including a constant), and u the residual.

This yields:

$$\hat{\mathbb{V}}[\hat{\beta}]_{\text{homoskedastic}} = \frac{\sum_i^N u_i^2}{N-k}(\mathbf{X}'\mathbf{X})^{-1}$$

Now, in general, it is unlikely for error terms to be homoskedastic. Thus we can just skip this assumption. However, we could also test for it. One such test is called **Breusch-Pagan test** and goes as follows (Breusch and Pagan 1979):

We follow the presentation of the test by Wooldridge (2002, 126).

1. Estimate the linear model

   $$y = \mathbf{X}\beta + \varepsilon$$

   with OLS. Derive the residuals:

   $$u_i = y_i - \hat{y}$$

   and square them (u_i^2).
2. Estimate the linear model:

   $$u^2 = \mathbf{X}\gamma + \mu,$$

   and evaluate the F statistic that all parameters $\gamma_1, \gamma_2, ...\gamma_k$ are jointly equal to zero (we will see F statistics a bit later). This corresponds to the null hypothesis that we have *homoskedasticity*. If we reject this null hypothesis (at a pre-set critical threshold α, e.g., 0.05), then we face the problem of *heteroskedasticity*.

If we are not willing to make the homoskedasticity assumption, meaning that we expect **heteroskedasticity**, then we are left with the earlier matrix, which we can estimate using the residuals $\mathbf{u}$ instead of the errors. Doing so, and replacing sample

averages for expected values yields:

$$
\begin{aligned}
\frac{\Omega}{N} &= \frac{1}{N} \underbrace{\left(\frac{\mathbf{X}'\mathbf{X}}{N}\right)^{-1}}_{k\times k} \underbrace{\left(\frac{\mathbf{X}'\mathbf{u}\mathbf{u}'\mathbf{X}}{N}\right)}_{k\times k} \underbrace{\left(\frac{\mathbf{X}'\mathbf{X}}{N}\right)^{-1}}_{k\times k} \\
&= \frac{1}{N}\left(\frac{1}{N}\sum_i^N X_i X_i'\right)^{-1}\left(\frac{1}{N}\sum_i^N X_i X_i' u_i^2\right)\left(\frac{1}{N}\sum_i^N X_i X_i'\right)^{-1} && \text{NB: } X_i \text{ is a row vector} \\
&= \left(\sum_i^N X_i X_i'\right)\left(\sum_i^N X_i X_i' u_i^2\right)\left(\sum_i^N X_i X_i')\right) && \text{simplifying} \\
&= (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{u}\mathbf{u}'\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1} && \text{in matrix notation}
\end{aligned}
$$

These are so-called "**robust**" or **Huber-Eicker-White** estimates of the (co)variance (P. J. Huber 1967; White 1980; Freedman 2006) . As before, the standard error of parameter $\widehat{\beta}_k$ is the square root of the diagonal value of row k. Note that this a *biased* but *consistant* estimator of the variance Ω/N (we won't prove it, but see White (1980)).

In practice, we generally avoid assuming homoskedasticity for two reasons. First, we generally don't have good reasons to assume it. Second, if the relation between X and Y isn't linear and thus we are approximating $\mathbb{E}[Y]$, then we will have heteroskedasticity by construction. Whenever our treatment is continuous, it is hard to assume that its effect is linear over its entire support, and thus it seems unlikely that homoskedasticity will hold.

4.7 Standard errors for interaction effects

As discussed in the point estimation discussion, we sometimes believe that the effect of X varies depending on the value of some third variable Z. This requires the inclusion of an interaction effect:

$$y = \alpha + \beta X + \gamma Z + \lambda XZ + \varepsilon,$$

with estimated marginal effect:

$$\frac{\partial \widehat{\mathbb{E}}[Y]}{\partial X} = \hat{\beta} + \hat{\lambda} Z.$$

The standard error of this marginal effect is slightly more complicated than that of a single parameter:

$$se(\hat{\beta} + \hat{\lambda}Z) = \sqrt{\mathbb{V}[\hat{\beta}] + Z^2\mathbb{V}[\hat{\lambda}] + 2Z \cdot Cov[\hat{\beta}, \hat{\lambda}]},$$

where $\mathbb{V}[\hat{\beta}]$, $\mathbb{V}[\hat{\lambda}]$, and $Cov[\hat{\beta}, \hat{\lambda}]$ can be obtained from Ω.

4.8 Efficiency

4.8.1 Gauss-Markov theorem

If our assumptions A3-A5 are met (which guarantee unbiasedness), and if the error terms are independent (no autocorrelation, A6) and have constant variance (homoskedasticity, A7) (both setting the result that $\mathbb{V}[\hat{\beta}] = \sigma^2(\mathbf{X}'\mathbf{X})$), then we obtain the following result (see (B. Hansen 2022a), p.104).

Theorem 4.2 (Gauss-Markov)**.** *Take any unbiased linear estimator $\tilde{\beta}$ of β. Then:*

$$\mathbb{V}[\tilde{\beta}] \geq \sigma^2(\mathbf{X}'\mathbf{X}).$$

In other words, the variance of any non-OLS (unbiased) estimator can at best equal the variance obtained by OLS, as long as Assumptions **A2** to **A7** are met. Some people refer to this as OLS being **BLUE**: best linear unbiased estimator (i.e., the most efficient estimator among unbiased estimators).

This result establishes that, under these assumptions, OLS is **efficient** – another desirable property in addition to unbiasedness .

⚠ ● Proof of Gauss-Markov

Let me sketch a proof of the Gauss-Markov theorem. What we want to show is that any non-OLS linear unbiased estimator of β has a higher variance than $\hat{\beta}_{ols}$. I'll refer to an arbitrary competing estimator $\tilde{\beta}$.

We start with the model:

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon$$

Since we are limiting ourselves to *linear* estimators, we can write such an estimator $\tilde{\beta}$ as a function of y:

$$\tilde{\beta} = m + M\mathbf{y}$$

where m and M are a vector and a matrix of constants, respectively. In OLS, for instance, $M = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.
Next, we assume that $\tilde{\beta}$ is unbiased (since we are only considering the category of *unbiased* linear estimators). This implies that:

$$\begin{aligned}
E(\tilde{\beta}|\mathbf{X}) &= E(m + M\mathbf{y}|\mathbf{X}) \\
&= m + M\mathbb{E}[\mathbf{y}|\mathbf{X}] && \text{(distributing the expect.)} \\
&= m + M\mathbb{E}[\mathbf{X}\beta + \varepsilon|\mathbf{X}] && \text{by A4} \\
&= m + M\mathbf{X}\beta + M\mathbb{E}[\varepsilon|\mathbf{X}] && \text{(distr. expect.)} \\
&= m + M\mathbf{X}\beta && \text{by A5}
\end{aligned}$$

The requirement for $\tilde{\beta}$ to be unbiased, thus, is:

$$\begin{aligned}
m &= 0 \\
M\mathbf{X} &= \mathbf{I}
\end{aligned}$$

Note that OLS satisfies this, since for least squares we have $M = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and thus $M\mathbf{X} = \mathbf{I}$. The implication is that a linear unbiased estimator will look like:

$$\tilde{\beta} = M\mathbf{y},$$

with M satisfying the above requirement. So, a typical non-OLS estimator will be (wlog):

$$M = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + C$$

where C is any matrix. Since $\tilde{\beta}$ is unbiased, we know that $M\mathbf{X} = \mathbf{I}$ must hold. At the same time:

$$\begin{aligned}
M\mathbf{X} &= [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + C]\mathbf{X} \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} + C\mathbf{X} \\
&= \mathbf{I} + C\mathbf{X}
\end{aligned}$$

The implication is that $C\mathbf{X} = 0$.
So, we can now compute the variance of $\tilde{\beta}$. Let's write:

$$\begin{aligned}
\tilde{\beta} &= My \\
&= M(\mathbf{X}\beta + \varepsilon) && \text{(Assumption ??)} \\
&= M\mathbf{X}\beta + M\varepsilon && \text{(distr. M)} \\
&= \mathbf{I}\beta + M\varepsilon && \text{(to be unbiased)} \\
\tilde{\beta} - \beta &= M\varepsilon
\end{aligned}$$

The variance-covariance matrix is thus:

$$\begin{aligned}
\mathbb{E}[(\tilde{\beta} - \beta)(\tilde{\beta} - \beta)'] &= \mathbb{E}[M\varepsilon(M\varepsilon)'] \\
&= \mathbb{E}[M\varepsilon\varepsilon' M'] && \text{(properties of matrices)} \\
&= M(\mathbb{E}[\varepsilon\varepsilon'])M' && \text{(distr. expect.)} \\
&= M\sigma^2\mathbf{I}M' && \text{by A3 (independence) and A7} \\
&= \sigma^2 MM'
\end{aligned}$$

So what is MM' again? It is:

$$\begin{aligned}
MM' &= [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + C][(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + C]' \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\
&+ (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'C' + C\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + CC' \\
&= (\mathbf{X}'\mathbf{X})^{-1} + CC' && \text{(since } C\mathbf{X} = 0\text{)}
\end{aligned}$$

Thus, the variance of $\tilde{\beta}$ and $\hat{\beta}_{ols}$ are:

$$\begin{aligned}
\sigma^2 MM' &= \sigma^2[(\mathbf{X}'\mathbf{X})^{-1} + CC'] && \text{(for alternative estimator)} \\
\sigma^2 MM' &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1} && \text{(for OLS)}
\end{aligned}$$

Since CC' is a quadratic term, it is positive. Thus, the variance of the rival estimator $\tilde{\beta}$ is larger than the variance of the OLS estimator, except if CC' happens to be zero. But if it is, then the estimator is actually OLS.

4.8.2 Cramér-Rao lower bound

One may sometimes wonder: can't we find an estimator with a variance of zero or almost zero? The answer is no: since we are

looking at stochastic data-generating processes with random samples, estimators will always have a positive variance. This intuition is captured in the **Cramér-Rao Lower Bound theorem**, which characterizes the lowest variance of an unbiased estimator (see (Gailmard 2014), p.307).

The general statement of the Cramér-Rao theorem is the following. Let $X = (X_1, X_2, \dots, X_n)$ be a random sample from a probability distribution with PDF $f(X|\beta)$, where β is our parameter of interest, and let $\hat{\beta}$ be an unbiased estimator of β. The variance of $\hat{\beta}$ is bounded below by the inverse of the Fisher information of the data-generation process, namely:

$$\mathbb{V}[\hat{\beta}] \geq \frac{1}{\mathcal{I}(\beta)}$$

where $\mathcal{I}(\beta)$ is the **Fisher information** given by:

$$\mathcal{I}(\beta) = -\mathbb{E}\left[\frac{\partial^2}{\partial\beta^2}\log f(X|\beta)\right] = \mathbb{E}\left[\left(\frac{\partial}{\partial\beta}\log f(X|\beta)\right)^2\right]$$

To see how this applies, note that the linear model can be rewritten as a conditional normal distribution centered on $\hat{y}$ (either by assuming normally distributed error terms or by using our earlier asymptotic results):

$$y \sim \mathcal{N}(\mathbf{X}\beta, \sigma^2\mathbf{I})$$

The **log-likelihood** function for this model is (see Appendix):

$$\log L(\beta, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)$$

The *score* (first derivative of the log-likelihood with respect to β) is:

$$\frac{\partial \log L}{\partial\beta} = \frac{1}{\sigma^2}\mathbf{X}'(\mathbf{y} - \mathbf{X}\beta) = -\frac{1}{\sigma^2}(\mathbf{X}'\mathbf{X}\beta)$$

The **Fisher information** $\mathcal{I}(\beta)$ is the *negative expected value* of the second derivative of the log-likelihood with respect to β. First, we compute the second derivative:

$$\frac{\partial^2 \log L}{\partial\beta\partial\beta'} = -\frac{1}{\sigma^2}\mathbf{X}'\mathbf{X}$$

Next, take the expected value of this quantity:

$$\mathcal{J}(\beta) = -E\left[\frac{\partial^2 \log L}{\partial\beta\partial\beta'}\right] = \frac{1}{\sigma^2}\mathbf{X}'\mathbf{X}$$

Taking the inverse, we find that:

$$\frac{1}{\mathcal{J}(\beta)} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} = \mathbb{V}[\hat{\beta}]$$

The variance of the OLS estimator is exactly equal to the inverse of the Fisher information, and thus it achieves the Cramér-Rao lower bound. This is the best (smallest) that is feasible, another way to say that OLS – under all assumptions listed above, including homoskedasticity – is efficient.

4.9 Consistency

The idea of consistency is that, as $N \to \infty$, the probability that an estimate is farther from its true value than some arbitrary constant ε goes to zero. Formally, with say that $\hat{\beta}$ converges in probability to β iff:

$$\lim_{N\to\infty} Pr(|\beta - \hat{\beta}| \geq \varepsilon) = 0, \forall\varepsilon > 0.$$

More compactly, this is written as:

$$plim\hat{\beta} = \beta$$

or

$$\hat{\beta} \xrightarrow{p} \beta$$

Informally, constitency is the idea that, as the sample size increases, the probability of getting a value of $\hat{\beta}$ beyond some threshold ε goes toward zero. Put another way, we want to obtain increasingly more precise estimates as the sample size increases.

We can use our earlier results on the asymptotic distribution of $\hat{\beta}$ to help us out. The reason is that the logic is the same:

we wanted to know what happens to $\hat{\beta}$ as N went to infinity. We want to do the same here. Thus, many of the results below repeat what we had seen earlier.

To establish this result, we can write:

$$\begin{aligned}
plim\hat{\beta} &= plim(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \\
&= plim(\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon) && \text{see unbiasedness above} \\
&= plim\beta + plim(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon \\
&= \beta + plim(\mathbf{X}'\mathbf{X})^{-1}plim(\mathbf{X}'\varepsilon) && \text{since } \beta \text{ is a constant}
\end{aligned}$$

Now, as N increases, so will $\mathbf{X}'\mathbf{X}$. To avoid blowing up, we normalize by N/N:

$$plim\hat{\beta} = \beta + plim\left(\frac{1}{N}(\mathbf{X}'\mathbf{X})\right)^{-1} plim\left(\frac{1}{N}\mathbf{X}'\varepsilon\right)$$

What we want is for the last term to disappear. Let's take its components one by one. By the **Law of Large Numbers**, we can claim that:

$$plim\left(\frac{1}{N}\mathbf{X}'\mathbf{X}\right) = \mathbb{E}[\mathbf{X}'\mathbf{X}] := \mathbf{Q},$$

that is, the first term converges to its expected (finite) value, which is a postitive definite (and inversible) matrix $\mathbf{Q}$. We now have:

$$plim\hat{\beta} = \beta + \mathbf{Q}^{-1}plim\left(\frac{1}{N}\mathbf{X}'\varepsilon\right)$$

We only have the last term to tackle. To ensure that the probability limit of $\hat{\beta}$ is β, we must show that:

$$plim\left(\frac{1}{N}\mathbf{X}'\varepsilon\right) = 0$$

If we take expectations, we see that:

$$plim\left(\frac{1}{N}\mathbb{E}[\mathbf{X}'\varepsilon]\right) = 0 \qquad \text{by A5}$$

Thus, under assumptions **A3** to **A5**, and thanks to the Law of Large Numbers, **OLS is consistant** .

4.10 Taking a deep breath

Here are some of the main results we have obtained so far:

- **Unbiasedness** of OLS : we need a random sample of IID variables and a linear model (**A3** and **A4**), a full rank matrix of regressors (**A2**), and exogeneity (**A5**).
- **Consistency** of OLS : same assumptions (with an assist from the Law of Large Numbers).
- **Efficiency** of OLS : same assumptions plus the assumption of homoskedasticity (**A7**), which together gives us the Gauss-Markov theorem .

We are not making many assumptions, but we are making strong ones:

- Exogeneity is very hard to meet with observational (i.e., non-experimental) data.
- Homoskedasticity is very rarely justified.
- Independence depends on the issue that you are studying.

4.11 Omitted variable

What are the consequences of specifying our model incorrectly by omitting a variable z? Without loss of generality, let us go over the case in which we have two regressors, X and Z. We are interested in the effect of X but we can't include Z (e.g., it's not possible to measure it).

Suppose that the true model is:

$$y_i = \lambda + \tau x_i + \gamma z_i + \mu_i,$$

and assume that all assumptions listed so far are met (including $E(\mu|x, z) = 0$). τ is the true treatment effect. Yet we erroneously estimate:

$$y_i = \alpha + \beta x_i + \varepsilon_i.$$

When we do this, we move γz into ε. What are the consequences of such a decision? Since we started with the bivariate case, we can write:

$$\begin{aligned}
\hat{\beta} &= \frac{Cov(X,Y)}{\mathbb{V}[X]} \\
&= \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2} && \text{canceling the denominators} \\
&= \frac{\sum y_i(x_i-\bar{x})}{\sum x_i(x_i-\bar{x})} && \text{algebra trick}
\end{aligned}$$

We can plug in the true model for y:

$$\begin{aligned}
\frac{\sum y_i(x_i-\bar{x})}{\sum x_i(x_i-\bar{x})} &= \frac{\sum(x_i-\bar{x})(\lambda+\tau x_i+\gamma z_i+\mu_i)}{\sum x(x-\bar{x})} \\
&= \frac{\lambda\sum(x_i-\bar{x})+\tau\sum x_i(x_i-\bar{x})+\gamma\sum(x_i-\bar{x})z_i+\sum(x_i-\bar{x})\mu_i}{\sum x_i(x_i-\bar{x})} \\
&= \tau+\gamma\frac{\sum(x_i-\bar{x})z_i}{\sum x_i(x_i-\bar{x})}+\frac{\sum(x_i-\bar{x})\mu_i}{\sum x_i(x_i-\bar{x})}
\end{aligned}$$

Note that we can't simplify $\sum(x_i-\bar{x})z_i$ nor $\sum(x_i-\bar{x})\mu_i$ because neither z nor μ are constant.

Taking expectations, we get:

$$\begin{aligned}
\mathbb{E}[\hat{\beta}] &= \mathbb{E}[\tau]+\mathbb{E}\left[\gamma\frac{\sum(x_i-\bar{x})z}{\sum x_i(x_i-\bar{x})}\right]+\mathbb{E}\left[\frac{\sum(x_i-\bar{x})\mu}{\sum x(x_i-\bar{x})}\right] \\
&= \tau+\gamma\mathbb{E}\left[\frac{\sum(x_i-\bar{x})z_i}{\sum x_i(x_i-\bar{x})}\right]+\mathbb{E}\left[\frac{\sum(x_i-\bar{x})\mu_i}{\sum x_i(x_i-\bar{x})}\right] && \text{taking constants out} \\
&= \tau+\gamma\mathbb{E}\left[\frac{\sum(x_i-\bar{x})z_i}{\sum x_i(x_i-\bar{x})}\right] && \text{if A5 is met}
\end{aligned}$$

Instead of our findings to establish unbiasedness, we now have a new additional term on top of our true value τ. But note that the term in the expectation is simply:

$$\mathbb{E}\left[\frac{\sum(x_i-\bar{x})z_i}{\sum x_i(x_i-\bar{x})}\right]=\frac{\mathrm{Cov}[X,Z]}{\mathbb{V}[X]}:=\omega.$$

This is the same as the parameter of a regression of Z on X and thus captures the effect of X on Z. Define this effect ω. We

can write that the estimate of the 'wrong' regression combines the following terms (where $i \to j|k$ means the effect of i on j, conditional on k):

$$\mathbb{E}[\widehat{\beta}] = \overset{x \to y|z}{\widehat{\tau}} + \underbrace{\gamma}_{z \to y|x} \cdot \overset{x \to z}{\widehat{\omega}}$$

The bias itself equals the difference between the average of the (badly) estimated effect ($\widehat{\beta}$) and the true effect (τ):

$$\text{Bias} := \mathbb{E}[\widehat{\beta}] - \tau = \gamma\omega$$

The bias will depend on γ (the effect of Z on Y adjusting for X) and on the ω (the effect of X on Z). If these two are *both* nonzero, then **OLS will be biased** .

Things get worse. Not only is OLS biased, but it is also **inconsistent** . Since γ and ω are constants, nothing would make them go away as $n \to \infty$.

Therefore, a variable Z that is both causing X and Y will make $\widehat{\beta}_{\textbf{ols}}$ biased and inconsistent if it is omitted from the estimated model.

- $\widehat{\beta}$ will be **biased** $\mathbb{E}[\widehat{\beta}] \neq \beta$.
- $\widehat{\beta}$ will be **inconsistent** (increasing the sample size won't help).

4.12 Null Hypothesis testing

4.12.1 Simple hypothesis

We move on to the world on (social) scientific theories. Recall that OLS yields an estimate $\widehat{\beta}$ from a sample. Yet, setting aside asymptotic properties of OLS, we don't know what the true value of β is.

Hypothesis testing is the process of answering the question: *given a hypothesis that we may have about β, how confident are we about it after seeing $\widehat{\beta}$?* For instance, if we hypothesize that

$\beta = -5$, and our sample yields $\hat{\beta} = 45.5$, will be start doubting our hypothesis?

In practice, hypothesis testing is conducted as follows.

Step 1. Set your **null hypothesis**, typically called H_0. If we are interested in a single parameter β_D (for the effect of D on Y), we would then set it as:

$$H_0 : \beta_D = c.$$

We could also use **composite** values (e.g., an interval), but this is rare. In practice, the most common null hypothesis is that $c = 0$, which means that the treatment D has no effect on Y. But other hypotheses exist. For instance, in macroeconomics, you may want to test whether the fiscal multiplier is 1.

Step 2. Set your **rival** or **alternative hypothesis** (H_a or H_1). Commonly, we will either be interested in whether β_D is *different* from c or whether it is larger (or smaller) than c. In the first case, we say that we are conducting a **two-tailed test** and in the latter case we are conducting a **one-tailed test**.

$$H_a : \beta_D \neq c$$
$$H_a : \beta_D > c$$
$$H_a : \beta_D < c$$

Step 3. We then estimate our model and obtain some value $\hat{\beta}_D$. Now, the probability that $\hat{\beta}_D = \beta_D$ exactly is zero (there is zero probability mass at a single point). We will always get an estimate that differs from its true value. The question we are asking is: how 'different' must $\hat{\beta}$ be from c for us to doubt $\beta_D = c$?

To answer this, we can rely on our results on the asymptotic distribution of $\hat{\beta}_D$: as $N \to \infty$, it will be distributed $\hat{\beta}_D \sim \mathcal{N}(\beta, \Omega/N)$. We can replace β with our null hypothesis, yielding:

$$\hat{\beta}_D \sim \mathcal{N}(c, \Omega/N)$$

In other words, *if our null hypothesis were true*, then $\hat{\beta}_D$ would be distributed normal around c. If $c = 0$, then this distribution is centered on zero.

Note that since we need to estimate σ^2 with s^2, we replace the normal distribution with the Student's t distribution, whose sole parameter is the number of **degrees of freedom**, i.e., $N - k$. The difference between the two is small; as the degrees of freedom tend to infinity, the t distribution converges to the standard normal distribution $\mathcal{N}(0, 1)$.

Given that we know the distribution, we can then also form rigorous statements regarding the probability of obtaining a value like $\hat{\beta}_D$ or more extreme.[2] Concretely, we can standardize our parameter to a value that we shall call t:

$$\tilde{t} = \frac{\hat{\beta}_D - c}{se(\hat{\beta})}$$

I write $\tilde{t}$ to denote the *ex post* t statistic obtained from a given sample.

For instance, consider $c = 0$, $\hat{\beta}_D = 3$, and $se(\hat{\beta}) = 1.5$, we obtain:

$$\tilde{t} = \frac{3}{1.5} = 2$$

The t statistic tells us how many standard errors away $\hat{\beta}$ is from the constant c.

The question is: *is being two standard errors away from c a lot?* To answer this, we build on the fact that we know the exact distribution of t and that we can thus infer the probability of obtaining a given t value or more extreme. Refer back to Student t's distribution table and see how much mass there is left and right of that t value.

Concretely, we can:

1. Compute the probability of obtaining a t equal to or greater than $\tilde{t}$. This would be done in a one-tailed test that $\beta > c$. (Likewise, we could compute the probability of obtaining a negative t value that is smaller than $\tilde{t}$ if we tested $\beta < c$.)

[2] The probability of $\hat{\beta}$ being a particular value is zero. See the chapter on probability and measure theory.

2. Compute the probability that $|t| > |T|$. This would be done in a two-tailed test.

The probability in question is a **p-value** (or just p). To reiterate, it tells us: *what is the probability of obtaining a t value greater than T* (one-tailed test) or *what is the probability of obtaining a t value greater than the absolute value of T* (two-tailed test). As stated above, we can compute it by comparing the t statistic we obtained against a t_{N-k} distribution.

Step 4. We started by saying that we wanted to evaluate the plausibility of our null hypothesis. More formally, what we meant is to say: *given some criterion of plausibility, do we have enough evidence to reject the null hypothesis?*

The criterion is going to be as follows. For an α selected by the analyst ($0 < \alpha < 1$), we define a **Type I** error as:

$$\begin{aligned}\alpha &:= \mathbb{P}[\text{Type I error}] \\ &= \mathbb{P}[\text{reject } H_0 | H_0 \text{ is true}]\end{aligned}$$

You will often see analysts using $\alpha = 0.05$ (i.e., setting the probability of a Type I error at 5%) or smaller.

The last step then is to assess whether the probability of obtaining a given t statistic or more extreme is smaller or larger than α. We reject the null hypothesis if $p - value < \alpha$: in such a case, our estimate $\hat{\beta}_D$ is very unlikely if the null hypothesis is true, leading us to cast doubt over H_0's truthfulness. If this condition is met, we say that we *reject the null hypothesis*. Equivalently, we say that $\hat{\beta}$ *is statistically significant*.

If, however, $p > \alpha$, then we say that *we fail to reject the null hypothesis*: we don't have enough evidence to cast doubt over it.

Note that instead of comparing our p-value to α, we can equivalently compare our t statistic to a critical value of t^c that corresponds to α. For instance, since we know the distribution of the t statistic, we can identify the critical t^c value beyond which there is less than α of mass under the curve (or equivalently, the set of values t_i that lie in this part of the distribution; we call this the **rejection** or **critical region** (Casella and Berger

2021), p.374). For large N, for instance, we know that there is 2.5% of the area under the t distribution to the right of 1.96. Likewise, there is 2.5% to the left of -1.96. Thus, we know that t values that are either larger than 1.96 or smaller than -1.96 have a less than 5% of occurring if our null hypothesis is true.

There has been considerable criticism of hypothesis testing in recent years (Gelman and Stern 2006; Gelman 2013; Gelman and Loken 2013; Altman and Krzywinski 2017). Some of the concerns include:

- Hypothesis testing tests hypotheses that are never true: $\beta = 0$ is a strawman that never occurs. Given enough data, we will then always end up rejecting the null hypothesis.
- A correlary to the previous point: a point estimate can be significant even if its effect is tiny and unimportant. We must not confuse *statistical significance* with *practical significance* (or *economic significance*). Likewise, a point estimate could be very large and meaningful, yet if its standard error is large as well, it will be insignificant. But this doesn't mean its effect is zero!
- Hypothesis testing leads to a binary outcome (significant/insignificant) that is too coarse. We might prefer to make statements about the range of plausible values for β.
- Hypothesis testing leads to biases in terms of research that is published: if there is a preference for "positive" findings (i.e., significance), then research will overproduce studies that show effects even when there are none.

4.12.2 Multiple restrictions

In econometrics and statistics, when testing a hypothesis involving multiple restrictions, the null hypothesis (H_0) typically takes the form:

$$H_0 : \mathbf{r}'\beta = \tilde{\mathbf{r}}$$

where R is a vector of coefficients corresponding to the restrictions, β represents the parameter vector, and $\mathbf{r}$ is a vector of constants. Typically, we will be interested in the null vector $\mathbf{r}' = [0 \quad 0 \quad ... \quad 0]$. The alternative hypothesis (H_1) suggests that the restrictions don't hold:

$$H_1 : \mathbf{r}'\beta \neq \tilde{\mathbf{r}}$$

For instance, we could compare the following two models:

$$\begin{aligned} y &= \tau + \beta x + \varepsilon \\ y &= \tau + \beta x + \gamma z + \lambda w + \mu \end{aligned}$$

where the restrictions to be tested would be:

$$\mathbf{r}'\beta = [1 \quad 1 \quad 0 \quad 0] \begin{bmatrix} \tau \\ \beta \\ \gamma \\ \lambda \end{bmatrix}$$

In simpler terms, this corresponds to:

$$H_0 : \gamma = \lambda = 0.$$

To test this, we cannot rely on the t test (which tests a single constraint). Instead, we may use the F test, which allows us to determine if a set of linear restrictions on the parameters of a model is valid. The test statistic compares the restricted model (H_0) to the unrestricted model (under H_1).

The F-statistic is calculated as:

$$\begin{aligned} F &= \left(\frac{\sum (y_i - \hat{y}_r)^2 - \sum (y_i - \hat{y}_u)^2}{(n - k_r) - (n - k_u)} \right) \Big/ \left(\frac{\sum (y_i - \hat{y}_u)^2}{n - k_u} \right) \\ &= \frac{\left(\frac{\sum u_r^2 - \sum u_u^2}{k_u - k_r} \right)}{\left(\frac{\sum u_u^2}{n - k_u} \right)} \end{aligned}$$

where k_u and k_r are the number of parameters in the unrestricted and the restricted model (and thus $k_u - k_r = q$ equals

the number of restrictions). The residuals u_u and u_r refer to the unrestricted and restricted model, respectively.

The F statistic is distributed with two parameters $F(q, n-k_u)$ distribution. Since we know its distribution, we can compute p values that are evaluated as the mass under the F distribution at or beyond a given F statistic. If F statistic is "large" (using the F distribution), it means that there is little mass beyond this point, indicating that it is implausible to obtain such an F under the null hypothesis. In that case, we reject the null hypothesis. As in hypothesis testing, the threshold of Type I error (α) is up to the analyst's judgement, but typical values are $\alpha = 0.05$ or smaller.

4.13 Confidence intervals

Confidence interval flips the approach used in hypothesis testing. In hypothesis testing, we try to see if our estimate is plausible compared to a value that is dictated by theory, not by the data. In most cases, we say something like: "we believe that $\beta = 0$, and let's see if the data tell us otherwise." In confidence interval, we center our analysis on $\widehat{\beta}$ and build uncertainty around it (a **confidence interval**).

A confidence interval is an interval that has, *ex ante*, an $1-\alpha$ probability of containing the true value β. The value $1-\alpha$ is referred to as the **coverage (probability)** of a confidence interval.

I insist on "*ex ante*": the statement is about the coverage *before* we draw a random sample, not about a given confidence interval obtained *ex post* once the data is obtained. The interval computed by your software either contains β or it doesn't, so the probability *ex post* is 1 or 0.

We compute confidence intervals as:

$$CI_{1-\alpha}[\beta] = [\widehat{\beta} - t^c_{N-k} \cdot se(\widehat{\beta}); \widehat{\beta} + t^c \cdot se(\widehat{\beta})]$$

where t^c is the critical t value at α with $N-k$ degrees of freedom. For instance, if $\alpha = 0.05$ and the sample is large, the critical t value is 1.96.

What happens this is the following: over repeated samples, this interval will capture the true value of β $1-\alpha$ percent of the time.

This confidence interval can straightforwardly be obtained from two pieces:

1. Our unbiasedness result, which allows us to replace β with $\mathbb{E}[\hat{\beta}]$. Thus, over repeated samples (i.e., in expectation), the distribution of $\hat{\beta}$ is centered on β.
2. Our sampling distribution of $\hat{\beta}$.

To show this, recall:

$$t = \frac{\hat{\beta} - \beta}{se(\hat{\beta})} \sim t_{N-k}.$$

We can then write:

$$Pr\left[-t^c_{N-k} \leq t \leq t^c_{N-k}\right] = 1 - \alpha$$

where t^c_{N-k} is the critical t value for $\alpha/2$. This result comes from the t distribution's definition. We can then replace:

$$\begin{aligned}
1-\alpha &= \mathbb{P}\left[-t^c_{N-k} \leq \frac{\hat{\beta}-\beta}{se(\hat{\beta})} \leq t^c_{N-k}\right] && \text{by definition} \\
&= \mathbb{P}\left[-t^c_{N-k} se(\hat{\beta}) \leq \hat{\beta} - \beta \leq t^c_{N-k} se(\hat{\beta})\right] && \text{moving standard errors} \\
&= \mathbb{P}\left[\hat{\beta} - t^c_{N-k} se(\hat{\beta}) \leq \beta \leq \hat{\beta} + t^c_{N-k} se(\hat{\beta})\right] && \text{moving } \hat{\beta}
\end{aligned}$$

Thus, the confidence interval is centered on β. If we repeatedly did this calculation over multiple samples, and took the expectation of this probability, we end up with a probability of 1-α.

Note that there is a link to hypothesis testing. Without loss of generality, let's say that we were earlier interested in testing the hypothesis that $\beta = 0$. Then, if our confidence interval contains 0, we automatically know that we will fail to reject the null hypothesis. Conversely, if our confidence interval doesn't contain 0, then we know that we would reject the null hypothesis. To see this, note that the null hypothesis assumes a normal

distribution around 0, whereas confidence intervals build a distribution around $\hat{\beta}$. If $\hat{\beta}$ is larger than, say, two standard errors from 0, then it must also that $\hat{\beta}$ minus two standard errors is larger than 0.

If: $\hat{\beta} > 0 + 1.96se(\hat{\beta})$	significant under hypothesis testing
Then: $\hat{\beta} - 1.96se(\hat{\beta}) > 0$	
and thus: $0 \notin [\hat{\beta} - 1.96 \cdot se(\hat{\beta}); \hat{\beta} + 1.96 \cdot se(\hat{\beta})]$	confidence intervals don't include 0

Thus, confidence intervals and hypothesis testing are two separate ways to do inference, but they are connected.

4.14 Model fit

We are often interested in knowing how our model performed. By "model," we mean how our independent variables and the parametric assumptions we place on them allow us to explain an outcome. This is distinct from OLS properties: we are now evaluating our model *ex post* here.

There are several ways to measure **model fit**.

4.14.1 R^2

The R^2 (aka **coefficient of determination**) of a model is the share of the variation in $\mathbf{y}$ that is explained by $\mathbf{X}\hat{\beta}$. Formally:

$$R^2 = 1 - \frac{\sum u_i^2}{\sum (y - \bar{y})^2}$$

The numerator of the second term is the sum of squared residuals (aka residual sum of squares). The denominator is the sum of square total (or total sum of squares).

R^2 will be located on the $[0, 1]$ interval, with larger values denoting a larger share of $\mathbf{y}$ explained by our model. Its advantage lies in (a) it being easy to interpret, and (b) telling us how much variation is left that could affect our results. This has been used by analysts to engage in robustness tests to see

the degree to which omitted variables could affect our results (Oster 2019).

The disadvantage of R^2 is that it is extremely sensitive to the sample you obtain. Holding the model constant, increasing the variance of X generally increases the R^2, making it very difficult to compare across datasets.

Furthermore, R^2 is weakly increasing in additional random variables: adding variables can only reduce $\sum u_i^2$ and thus increase R^2. As a result, you may sometimes encounter the **adjusted** R^2:

$$R_a^2 = 1 - (1 - R^2)\frac{n-1}{n-k},$$

for n observations and k parameters (including a constant). This quantity punishes R^2 for the inclusion of irrelevant variables (by increasing k). The downside is that it is not bounded on the [0,1] interval anymore.

4.14.2 σ^2

We already encountered the **standard error of the regression** (σ) or **residual variance** (σ^2). This is the unconditional variance of the error term and is estimated by s^2:

$$s^2 = \frac{\sum u_i^2}{n-k},$$

for n observations and k parameters (including a constant). Take the square root to estimate s, the estimated standard error of the regression (i.e., s is an estimate of σ).

The better the model fit, the smaller s^2 (and s). This indicates that the model has a low level of noise and thus performs well. You can interpret s^2 as follows: 95% of predicted/fitted values ($\hat{y}$) will be located within $+/-2s$ of observed values y. Informally and imprecisely, you can think of this as the average residual in your prediction.

What's nice is (a) that it is an estimate of a population parameter, and (b) is expressed in terms of the dependent variable. Thus, in a study on predicted income in millions of USD, $s = 3$

suggests that our predicted estimate will be off, on average, by USD 3 million. It is thus very interpretable.

4.14.3 F test

The F test, which we encountered above, tests several restrictions on parameters. We can expand this to the special case where we test the restriction that all parameters (minus the constant) are jointly equal to zero. This represents an assessment of the performance of the *overall* model (and not just restrictions on a small subset of parameters).

The null hypothesis is that (ignoring the intercept, since a model with a single parameter is the comparison point):

$$H_0 : \beta_1 = \beta_2 = ... = \beta_{k-1} = 0,$$

against the alternative hypothesis that some parameter(s) is different from zero. We call the estimated model the "unrestricted" model (i.e., where we don't restrict our parameters to equal zero) which we compare against the "restricted" model (where all parameters are zero, except the constant):

$$\begin{aligned}
&\text{Restricted: } \mathbf{y} = \alpha \\
&\text{Unrestricted: } \mathbf{y} = \alpha + \beta X + \gamma W + ... + \varepsilon
\end{aligned}$$

The F statistic is computed as (using r as subscript for restricted and u for unrestricted):

$$\begin{aligned}
F &= \left(\frac{\sum (y_i - \hat{y}_r)^2 - \sum (y_i - \hat{y}_u)^2}{(n - k_r) - (n - k_u)} \right) \Big/ \left(\frac{\sum (y_i - \hat{y}_u)^2}{n - k_u} \right) \\
&= \left(\frac{\sum (y_i - \bar{y})^2 - \sum (y_i - \hat{y}_u)^2}{(n - k_r) - (n - k_u)} \right) \Big/ \left(\frac{\sum (y_i - \hat{y}_u)^2}{n - k_u} \right) && \text{if testing sig of all parameters} \\
&= \left(\frac{\sum (y_i - \bar{y})^2 - \sum (y_i - \hat{y}_u)^2}{(k_u - k_r)} \right) \Big/ \left(\frac{\sum (y_i - \hat{y}_u)^2}{n - k_u} \right) \\
&= \left(\frac{\sum (y_i - \bar{y})^2 - \sum u_u^2}{(k_u - k_r)} \right) \Big/ \left(\frac{\sum u_u^2}{n - k_u} \right) && \text{by def} \\
&= \left(\frac{\sum (y_i - \bar{y})^2 - \sum u_u^2}{(k_u - 1)} \right) \Big/ \left(\frac{\sum u_u^2}{n - k_u} \right) && \text{restricted model has 1 parameter}
\end{aligned}$$

for k parameters.

Under the null hypothesis, the F statistic is distributed F. As with t tests, larger values are implausible under the null hypthesis.

4.15 Forecasts and predictions

We take a break here to discuss how to make predictions and forecasts. I said that econometrics is often about identifying causal effects. But it can also be about predictions and forecasts. This can be helpful for description ("here's a what a customer with a given income, gender, etc. would spend on our product") and for policy purposes ("here's how a 10 basis point decrease in interest rates would affect house sales").

We know that the predicted value of y, given a model $\mathbf{X}$ and parameters β, equals:

$$\hat{y} = \mathbf{P}\mathbf{y} = \mathbf{X}\hat{\beta},$$

where $\mathbf{P} := \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. We can think of predictions and forecasts in two ways.

1. We take an arbitrary vector $\mathbf{x}_f'$ that we are interested in and multiply it with β (where f denotes the design vector for which we want a prediction). For instance, if we are trying to identify how much a customer might spend in your shop as a function of their personal features (gender, age, income, etc.), we could create a synthetic customer who is female, 35 years old, earns CHF50k per year, etc. We would typically call this an **out-of-sample** prediction, at least if we don't already possess units with this profile in our data:

$$[\mathbf{x}_f'\hat{\beta}]_{1\times 1} = \begin{bmatrix} 1 & \text{Female} & \text{35yo} & \cdots & \text{CHF50k} \end{bmatrix}_{1\times k} \times \begin{bmatrix} \hat{\alpha} \\ \hat{\beta} \\ \vdots \\ \hat{\gamma}_k \end{bmatrix}_{k\times 1}$$

2. We can model $\mathbf{y}$ as a dynamic process, where past values of y predict present values of y. This is a **forecast** and involves dynamic modeling with time series (or panel data), which I will discuss later.

For now, let's focus on making predictions. The point prediction is straightforward: just compute $\mathbf{x}'\hat{\beta}$ for your vector of values of $\mathbf{X}$ that you are interested in. However, this doesn't give you any sense of uncertainty regarding this prediction. What we want is to add confidence intervals around this prediction. How do we do this?

The general idea will be to measure the standard error around $\hat{y}$ and then multiply it by a critical t value (e.g., $t^* = 1.96$ for a type I error rate (or α) of 0.05 and a large sample). Here, there is a subtlety to keep in mind:

1. If you want the confidence interval around the prediction/forecast of the *expected value* of Y, you will need to incorporate the uncertainty of $\hat{\beta}$, $\hat{\gamma}$, etc.
2. If you want the confidence interval around the prediction/forecast of a *new value* Y, you will need in addition the uncertainty around ε (the error term).

Let's start with the expected value. We can write:

$$\mathbb{E}[y|\mathbf{x}_f] = \mathbf{x}_f\beta,$$

where can be estimated with:

$$\hat{y}_f = \mathbf{x}_f\hat{\beta}$$

This quantity is a random variable for two reasons: (a) $\hat{\beta}$ is a random variable, and (b) there is uncertainty from the error term (which we ignore for now, since we focus on the mean value of y).

The variance of $\hat{y}_f$ can then be defined as:

$$\mathbb{V}[\hat{y}_f] = \mathbb{V}[\mathbf{x}_f\hat{\beta}] = \sigma^2 \left[\mathbf{x}_f'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_f\right],$$

and σ^2 (the variance of the regression) can be consistantly estimated by s^2 as we saw previously:

$$s^2 = \frac{\sum u_i^2}{N-k}.$$

If we want to measure the uncertainty of around Y, then we must add our error term. Note that the residual is (by construction):

$$u_f = y_f - \hat{y}_f = \underbrace{[\mathbf{x}_f'\beta + \varepsilon]}_{\text{true model at } f} - \hat{y}_f.$$

Since $\mathbf{x}_f'\beta$ is fixed, and since ε shouldn't covary with anything, the variance of u_f equals the sum of the variances of the two remaining terms:

$$\mathbb{V}[u_f] = \mathbb{V}[\varepsilon] + \mathbb{V}[\hat{y}_f].$$

Thus, we have the same variance as above plus the variance of the error term (again). We can write:

$$\mathbb{V}[\mathbf{x}_f\hat{\beta} + u_f] = \sigma^2 \left[\mathbf{x}_f'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_f\right] + \sigma^2$$

In either case, take the square root of $\mathbb{V}[\cdot]$ to obtain the standard error and multiply it by t^*.

4.16 Outliers and leverage

Sometimes, people worry about **outliers**, that is, values of variables that are *abnormal* in terms of their value compared to the overall distribution of the variable.

To figure out what the problems might be, we first define **leverage values** (B. Hansen 2022a, 83). The leverage of an observation i (i.e., a row of data) tells us how dependent OLS estimates *could* be on that observation (as we will see: it's a potential threat that may or may not be realized in practice). Each observation has its own leverage value, which is defined as the

diagonal of the projection matrix. Recall that the projection matrix (also known as the "hat" matrix) is:

$$\mathbf{P} := \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}',$$

where $\mathbf{Py} = \hat{y}$ gives us the predicted values of y (hence the label "hat" matrix: it 'puts' the hat on y).

For unit $i \in \{1, 2, ..., N\}$, we define the leverage value h_{ii} as:

$$h_{ii} := X_i^{'}(\mathbf{X}'\mathbf{X})^{-1}X_i,$$

where is the diagonal of the projection/hat matrix.

For instance, with three variables X, Z, W (we ignore the intercept here; add columns of 1 accordingly) and focusing on, say, unit 2, we would have the following leverage:

$$h_{2,2} = \begin{bmatrix} x_2 & z_2 & w_2 \end{bmatrix}_{1\times 3} \left(\begin{bmatrix} x_1 & x_2 & x_3 \\ z_1 & z_2 & z_3 \\ w_1 & w_2 & w_3 \end{bmatrix} \begin{bmatrix} x_1 & z_1 & w_1 \\ x_2 & z_2 & w_2 \\ x_3 & z_3 & w_3 \end{bmatrix} \right)^{-1}_{3\times 3} \begin{bmatrix} x_2 \\ z_2 \\ w_2 \end{bmatrix}_{3\times 1}$$

which yields a scalar, h_{ii}. This scalar will be relatively large when the vector $\mathbf{x}_i$ for unit i is different from the other vectors $\mathbf{x}_j$.

The leverage h_{ii} has several properties:

1. $h_{ii} \in [0, 1]$
2. $\sum_{i=1}^{N} h_{ii} = k$, where k is the number of independent variables.

An indicator that can be useful is the max of the leverage across all observations. If you then take several models, you can in principle compare what this max is and see how responsive it is to the inclusion of a given variable.

Now, here's what could be tricky. The leverage h_{ii} only tells us of i's *potential* influence. Whether it is realized depends on the *residual* u_i attached to it. What will generally happen is that observations that are outliers will have larger influence. To see this, we can use the special case of a single predictor and

an intercept, which we saw when we discussed OLS with one treatment. We have:

$$\text{Design matrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \dots & \dots \\ 1 & x_n \end{bmatrix}.$$

Then, you can write:

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} n & \sum x_j \\ \sum x_j & \sum x_j^2 \end{bmatrix},$$

whose inverse is:

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{n\sum x_j^2 - (\sum x_j)^2} \begin{bmatrix} \sum x_j^2 & -\sum x_j \\ -\sum x_j & n \end{bmatrix}.$$

We can then write the leverage as:

$$h_{ii} = \begin{bmatrix} 1 & x_i \end{bmatrix} \frac{1}{n\sum x_j^2 - (\sum x_j)^2} \begin{bmatrix} \sum x_j^2 & -\sum x_j \\ -\sum x_j & n \end{bmatrix} \begin{bmatrix} 1 \\ x_i \end{bmatrix}.$$

After some algebra, we obtain:

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum (x_j - \bar{x})^2}.$$

Here's where the problem might come from: the values with the largest h are those that are furthest from the mean (numerator of the second term). Values at the mean are not very informative and OLS is not going to "use" them as much as values that are very large (in absolute terms). Thus, values with very large x_i compared to the mean will have more influence on the regression line. If one observation has a particularly large h, then the concern is that the point estimate is not very 'robust' in the sense that another random sample might yield a very different result. Imagine, for instance, a regression of political views on income, and your sample accidentally includes Elon Musk: his leverage h_{Musk} will be very large and therefore affect $\hat{\beta}$. Next sample might then yield a very different $\hat{\beta}$ if you don't randomly get a billionaire in it.

Now, having units with large leverage doesn't invalidate our OLS estimates. We never made any assumption regarding the contribution of individual units. However, from a common-sense perspective, we might worry that our results could be affected by a handful of influent observations. They might represent values that are uncommon and that respond nonlinearly to the treatment. For instance, imagine a sample with 10 units, nine of which having an x between 0 and 10 and one having $x = 1,000,000$. It might well be that x has no effect on y at small values, but the response could be very large at high values. Then, including this observation would dramatically change the results, but do we believe them?

There is something neat that you can do here, in case you are worried about outliers (or more precisely, units with large leverage): the **leave-one-out** regression (LOOR) (B. Hansen 2022a, 84). This consists in estimating $\hat{\beta}$ based on taking out one observation out at a time (e.g., you drop one country or one firm from your sample and regress Y on $\mathbf{X}$). What's very neat is that you can do this without having to run N regressions. B. Hansen (2022a, 84) shows that you can estimate the LOOR estimate $\hat{\beta}_{-i}$ as follows:

$$\hat{\beta}_{-i} = \hat{\beta}_{\text{ols}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i\tilde{u}_i,$$

where:

$$\tilde{u}_i = \frac{u_i}{1 - h_{ii}},$$

where u_i is the residual of the OLS estimate of Y on $\mathbf{X}$. Here's the thing: if h_{ii} are small, then $\tilde{u}_i$ will be similar to u_i. And then, we generally will see that $\mathbf{x}_i\tilde{u}$ is small, and as a result, the LOOR estimate $\hat{\beta}_{-i}$ will be similar to the OLS estimate $\hat{\beta}$.

Review questions

Exercise 4.1. Are OLS estimates biased, consistent, and efficient with non-normal error terms? Run a Monte Carlo simulation to find out inductively the answer.

Exercise 4.2. Reproduce the proof of unbiasedness with OLS, flagging every assumption necessary along the way.

Exercise 4.3. Run a Monte Carlo to evaluate the information generated by competing measures of fit. Concretely, start with the model:

$$y = 1 + 2x + \mathcal{N}(0, 5)$$

and increase the range of X from $X \sim U[0, 1]$ to $X \sim U[0, 100]$. How does R^2 change? What about σ^2 and F?

Appendix

Log-likelihood of the linear model

Log-likelihoods are useful for the estimation of parameters under probabilistic models. We have used OLS for estimation so far, but other methods, such as maximum likelihood – which relies on maximizing the log of the likelihood – are powerful too. For now, the log-likelihood is useful as a step in establishing the result that OLS has the lowest feasible variance, i.e., it meets the **Cramér-Rao lower bound theorem**.

Consider the linear regression model:

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon$$

We can rewrite this as follows:

$$\mathbf{y} \sim \mathcal{N}(\mathbf{X}\beta, \sigma^2\mathbf{I}),$$

either by assuming that the error term is normally distributed with constant variance, or by using our asymptotic results (and assuming homoskedasity).

The probability density function of a (multivariate) normal distribution is (using the definition of the normal distribution):

$$f(\mathbf{y}|\beta, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)\right)$$

The **likelihood function** $L(\beta, \sigma^2)$ is the joint probability of observing $\mathbf{y}$ given the parameters β and σ^2. For the normal distribution, it is the product of the density functions for all observations. We then take the log of this likelihood function (i.e., the **log-likelihood**) because it's easier to differentiate.

$$\log L(\beta, \sigma^2) = \log\left(\frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)\right)\right)$$

We can simplify this expression in two steps.

1. The logarithm of the normalizing constant:

$$\log\left(\frac{1}{(2\pi\sigma^2)^{n/2}}\right) = -\frac{n}{2}\log(2\pi\sigma^2)$$

2. The logarithm of the exponential term:

$$\log\left(\exp\left(-\frac{1}{2\sigma^2}(\mathbf{y}-\mathbf{X}\beta)'(\mathbf{y}-\mathbf{X}\beta)\right)\right) = -\frac{1}{2\sigma^2}(\mathbf{y}-\mathbf{X}\beta)'(\mathbf{y}-\mathbf{X}\beta)$$

We can then write the **log-likelihood function** of the linear model:

$$\log L(\beta,\sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}(\mathbf{y}-\mathbf{X}\beta)'(\mathbf{y}-\mathbf{X}\beta)$$

From here, see the notes to obtain the Cramér-Rao lower bound.

5 Causal point identification

This version: December 16, 2025

> **Caveat**
>
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5.1 Readings

On the many meanings of identification, see Lewbel (2019). For limits to (point) identification, see Tamer (2010) and Manski (1990). For an introduction to DAGs in general, see Pearl (2009). A shorter introduction to DAG is Pearl, Glymour, and Jewell (2016). For DAGs with panel data, see Reisach et al. (2025).

5.2 Identification

Lower costs of computational power and the realization that econometric analysis has business value has triggered a rapid increase in empirical work in economics and other social sciences.

In parallel, the standards for what qualifies as trustworthy econometric analysis have also increased (Leamer 1983; Joshua

D. Angrist and Pischke 2010). This is often referred to as the **credibility revolution**.

This notion captures several ideas, but at its core is a desire for applied econometric analysis to generate (point) estimates that are well **identified**. Again, it's hard to pin down a single definition of what identification means, but in general, it will mean that *we are constructing a research design that credibly yields consistent estimates of a causal effect.* In the context of the linear model and OLS, this means that we want to make sure that our research design is at least consistent and, ideally, unbiased. By "credible," we mean that this research design depends as little as possible on implausible assumptions (or, at the very least, that makes these assumptions transparent).

We could also engage in identification of sets rather than points (Manski 1990).

Why is this so important? Well, imagine that you were tasked to generate the estimate of the effect of an advertisement campaign. How would you proceed? You will need to build a design that yields numbers you trust. And for this, "garbage can" regressions ('garbage in, garbage out') won't do.

Those involved in the credibility revolution have, fortunately, developed a set of tools and frameworks to improve the quality of empirical research. This ranges from the application of experimental methods to economic issues (next chapter) to the design of so-called **quasi-experimental** designs. But first, let us briefly investigate the sources of endogeneity (recall that this assumption was essential for OLS to yield consistent estimates). Then, we will do a quick introduction to directed acyclic graphs to have a good framework to engage in causal inference.

5.3 Endogeneity

5.3.1 What is endogeneity?

Suppose we have a linear model such as:

$$Y = \mathbf{X}\beta + \varepsilon,$$

where $\mathbf{X}$ is and $n \times k$ matrix (k parameters, including a constant) and β is a k vector of parameters. Then the **exogeneity**

assumption states that:

$$\mathbb{E}[\varepsilon|\mathbf{X}] = 0$$

When this condition is *not* met, the OLS estimates are **biased** and **inconsistent** (see chapter on inference with OLS). Here's one way to build your intuition on this: suppose we want to know the effect of some treatment D on $\mathbb{E}[Y]$ and we have:

$$y = \alpha + \beta_D D + \varepsilon$$

- If D is exogenous, then $\partial\mathbb{E}[Y|D]/\partial D = \beta_D$.
- If D is endogenous ($\mathbb{E}[\varepsilon|D] \neq 0$), then a change in D will operate via β_D (good) but also via ε (bad).

Since at best we can only know the direction of the bias (often we don't even know that!) but not its magnitude, and since more data won't help (why?), we won't know if the estimates we obtain from a given dataset are anywhere close to the true population value of β_D and we can't straightforwardly draw inferences from our sample.

Typically, endogeneity arises from these causes:

1. Omitted variable bias
2. Simultaneity bias
3. Measurement error

But that's not the only reasons why things might go wrong. A non-random sample can invalidate inference about a population, for instance. For instance, if you only survey rich people about their willingness to pay (WTP) for a new product, it's unlikely that you can make valid causal inferences about the population WTP.

5.3.2 Omitted variable bias

Suppose the true model is:

$$Y = \beta D + \gamma W + \varepsilon,$$

such that $\mathbb{E}[\varepsilon|D, W] = 0$. For simplicity, let's assume that all variables have been demeaned (we thus don't need to worry

about intercepts). Next, let's suppose that we estimate the "short" regression instead:

$$Y = D\rho + \mu$$

What we are wondering is: $\mathbb{E}[\mu|D] = 0$?

Let's start with the OLS definition of the slope of the "short" regression. Since all variables are demeaned, we can write:

$$\rho = \frac{\mathbb{E}[DY]}{\mathbb{E}[D^2]}.$$

Substitute the true model:

$$\mathbb{E}[DY] = \mathbb{E}\big[D(\beta D + \gamma W + \varepsilon)\big] = \beta\mathbb{E}[D^2] + \gamma\mathbb{E}[DW] + \mathbb{E}[D\varepsilon].$$

But by construction, the product of error terms and an independent variable are orthogonal:

$$\mathbb{E}[D\varepsilon] = 0$$

We can bring back the full expression for ρ (i.e., divide by $\mathbb{E}[D^2]$):

$$\rho = \beta + \gamma\frac{\mathbb{E}[DW]}{\mathbb{E}[D^2]},$$

which is the omitted variable bias formula: the parameter of the "short" regression equals the true causal effect plus the effect of the omitted variable on the outcome (γ) times the effect of treatment on the omitted variable ($\mathbb{E}[DW]/\mathbb{E}[D^2]$).

Next, we can return to μ and define it more precisely:

$$\begin{aligned}
\mu &= Y - D\rho \\
&= (\beta D + \gamma W + \varepsilon) - D\left(\beta + \gamma\frac{\mathbb{E}[DW]}{\mathbb{E}[D^2]}\right) \\
&= \gamma\left(W - D\frac{\mathbb{E}[DW]}{\mathbb{E}[D^2]}\right) + \varepsilon.
\end{aligned}$$

Use the fact that $\mathbb{E}[\varepsilon \mid D] = \mathbb{E}[\mathbb{E}[\varepsilon \mid D, W] \mid D] = 0$ (law of iterated expectations). We can then take the conditional

expectation of μ given D and find out when it equals zero (or not):

$$\mathbb{E}[\mu \mid D] = \gamma \left(\mathbb{E}[W \mid D] - D \frac{\mathbb{E}[DW]}{\mathbb{E}[D^2]} \right).$$

The right-hand side is not zero for a generic joint distribution of (D, W). Therefore, in general:

$$\mathbb{E}[\mu|D] \neq 0,$$

i.e. the residuals from the short regression are mean-dependent on D and we have endogeneity. The conditions for this *not* to happen are:

- $\gamma = 0$ (the omitted variable has no effect on the dependent variable).
- $\mathbb{E}[W \mid D] = D \cdot \mathbb{E}[DW]/\mathbb{E}[D^2]$ almost surely. This happens if $\mathbb{E}[W \mid D] = 0$, since both sides would then be equal to zero. In practice, this means that D is unrelated to W – our other strategy to obtain unbiased estimates.

If *either* happens, then $\mathbb{E}[\mu|D] = 0$ and we are 'saved.'

5.3.3 Measurement error in treatment

Suppose you have a treatment D and that the true model is:

$$Y = \beta D + \varepsilon$$

with $\mathbb{E}[\varepsilon|D] = 0$. As above, we demean D and thus skip the need for a y-intercept.

Unfortunately, we find out that D is imprecisely measured. Let's say you can only observe:

$$\tilde{D} = D + \mu,$$

We assume that $\mathbb{E}[\mu] = 0$ and that μ isn't correlated with D, ε, nor Y. It's a random shock and we call this the **classical errors-in-variables** problem. Such measurement error can happen, for instance, when you ask people questions and they

can't remember exactly the correct answer (eg, at what time did you wake up yesterday?). This happens fairly often when asking consumers about their past behavior.

Let's see what happens then:

$$\begin{aligned} Y &= \beta(\tilde{D} - \mu) + \varepsilon && \text{replacing } D \\ &= \beta\tilde{D} + (\varepsilon - \beta\mu) && \text{rearranging} \\ &= \beta\tilde{D} + \tilde{\varepsilon} && \text{redefining} \end{aligned}$$

where:

$$\tilde{\varepsilon} := \varepsilon - \beta\mu$$

We can now ask: does the following hold…

$$\mathbb{E}[\tilde{\varepsilon}|\tilde{D}] \stackrel{?}{=} 0$$

To show that it doesn't, it suffice to show that $\text{Cor}(\tilde{\varepsilon}, \tilde{D}) \neq 0$ (recall that a non-zero correlation implies dependence, but the reverse isn't true). And that's indeed the case since $\tilde{\varepsilon}$ contains μ, and since μ and $\tilde{D}$ are positively correlated (when the shock μ is positive and large, then $\tilde{D}$ will also be larger than average). Thus, we face an endogeneity problem if we can solely regress Y on $\tilde{D}$:

$$\mathbb{E}[\tilde{\varepsilon}|\tilde{D}] \neq 0.$$

OLS will yield biased and inconsistent estimates. We can spell

out the bias explicitly.

$$
\begin{aligned}
\hat{\beta} &= \frac{\text{Cov}(\tilde{D}, Y)}{\mathbb{V}(\tilde{D})} && \text{from def of OLS} \\
&= \frac{\text{Cov}(D+\mu, Y)}{\mathbb{V}(D+\mu)} && \text{from measurement error} \\
&= \frac{\text{Cov}(D, Y) + \text{Cov}(\mu, Y)}{\mathbb{V}(D) + \mathbb{V}(\mu) + Cov(D, \mu)} && \text{properties of var, cov} \\
&= \frac{\text{Cov}(D, Y)}{\mathbb{V}(D) + \mathbb{V}(\mu)} && \text{by assumption about } \mu \\
&= \frac{\mathbb{V}(D)\text{Cov}(D, Y)/\mathbb{V}(D)}{\mathbb{V}(D) + \mathbb{V}(\mu)} && \text{multiply and divide by var of } D \\
&= \beta \frac{\mathbb{V}(D)}{\mathbb{V}(D) + \mathbb{V}(\mu)} && \text{by def of } \beta_{\text{OLS}} \\
&= \beta \frac{\sigma_D^2}{\sigma_D^2 + \sigma_\mu^2} && \text{lighter notation}
\end{aligned}
$$

Thus, OLS will yield estimates that are biased toward zero, since $\sigma_D^2/(\sigma_D^2 + \sigma_\mu^2) \in (0, 1)$ if $\sigma_\mu^2 > 0$. We often call this **attenuation bias** (the estimate is an *attenuated* estimate of the true treatment effect).

5.3.4 Simultaneity bias

Some data-generating processes involve a system of equations. For instance, markets are typically characterized by two equations: one for demand and one for the supply side. We may therefore end up with situations in which two variables cause each other, such as price and quantity.

Consider the following two equations:

$$
\begin{aligned}
Y &= \beta D + \varepsilon \\
D &= \tau Y + \mu.
\end{aligned}
$$

As in the previous two cases, we demean D; we also demean now Y as well, so that we don't need any y-intercept. We will assume that $|\tau\beta| \neq 1$.

Suppose we are interested in the effect of D on Y and want to know if we can just estimate the first equation. Can we assume that $\mathbb{E}[\varepsilon|D] = 0$?

We can write:

$$\begin{aligned} D &= \tau Y + \mu \\ &= \tau(\beta D + \varepsilon) + \mu && \text{plugging in for } Y \\ D(1-\tau\beta) &= \tau\varepsilon + \mu && \text{rearranging} \\ D &= \varepsilon\frac{\tau}{1-\tau\beta} + \frac{\mu}{1-\tau\beta} && \text{defined since } |\tau\beta| \neq 1 \end{aligned}$$

But note that D will therefore be correlated with ε. As we stated above: a nonzero correlation implies dependence (though the reverse isn't true). As a result, $\mathbb{E}[\varepsilon|D] \neq 0$.

5.3.5 Endogeneity

To summarize: we have multiple threats that can generate endogeneity. And the exogeneity assumption is critical to obtain the most desirable properties of OLS, namely unbiasedness and consistency. Thus, a lot of research in econometrics has been about ways to handle this problem (Leamer 1983; Joshua D. Angrist and Pischke 2010). You may encounter this in references to concepts such as the **credibility** or the **identification revolution**. But before we look at this, we first take a detour via directed acyclic graphs as a method to make sense of data-generating processes.

5.4 Directed Acyclic Graphs

Once we move away from the world of RCTs and experimental data, we will have to work with **observational data**. The challenge here is the one discussed earlier: with observational data, we face the risk of confounding and erroneously attribute causal effects where there are only spurious correlations. That's because we don't control the assignment of treatment.

As a result, we are moving to the world of a single treatment linked to a given outcome. Instead, we must think hard about

the **data-generating process** (the process by which observational data is generated).

A useful tool to do so are **directed acyclic graphs** (DAGs). DAGs (Pearl 2009; Pearl, Glymour, and Jewell 2016; Pearl and Bareinboim 2024). DAGs allow us to construct models rigorously.

5.4.1 Graphs

A **graph** consists of two elements:

1. Vertices (or nodes). These will be variables.
2. Edges. These will be paths of dependence.

There exists different types of graphs:

- Directed or not. A directed graph establishes causal relations of dependence. Edges are then "arrows" from one node to another.
- Among directed graphs (DAG): cyclic or not. A directed graph in which a causal path can start and return to the same variable is *cyclic*.

Since we are interested in quantifying causal estimates, we will be working with directed graphs. And in most cases, but not all, we will be interested in directed acyclic graphs (DAGs).

In principle, a DAG can be expressed as a system of equations. For instance:

$$
\begin{aligned}
D &= 3Z + f_x(U_x) \qquad \text{with } f_x(U_x) \sim \mathcal{D}_x \\
Z &= f_z(U_z) \qquad \text{with } f_z(U_z) \sim \mathcal{D}_z \\
W &= 2Z + f_w(U_z) \qquad \text{with } f_w(U_w) \sim \mathcal{D}_w \\
Y &= 0.5D + 1.1Z - 0.4W + f_y(U_Y) \qquad \text{with } f_y(U_y) \sim \mathcal{D}_y
\end{aligned}
$$

This system contains endogenous and exogenous variables. Exogenous variables are not explained by the model. These are $f_i(U_i)$ which are drawn from some arbitrary distribution and are often called error terms. Endogenous variables are explained by the system.

Note that the definition of endogenous in a DAG is different from the way this concept is used in econometrics (and later this chapter).

However, in practice, we often don't know what the functional forms relating all variables are. Thus, we often use a visual representation of the graph, as in Figure 5.1. The system of equation above resembles the third DAG (minus the error terms that are removed to save space).

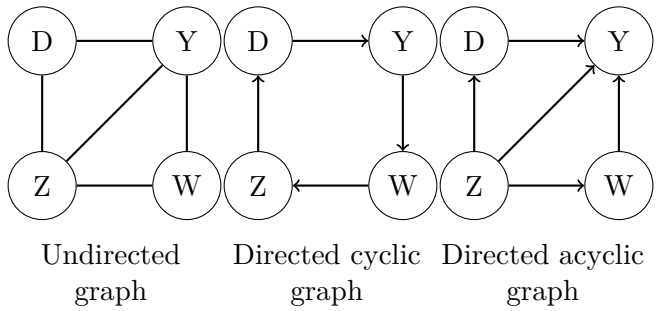

Figure 5.1: Three types of graphs.

5.4.2 Lessons from DAGs

We can use DAGs to figure out how to build a model to estimate the causal effect of D. We can procede in three steps.

First, there are three types of paths depending on relations between nodes:

1. **Forks**: a node has two causal arrows *departing* from it. For instance: $D \leftarrow Z \rightarrow Y$ has a fork (Z).
2. **Mediators**: a node has a causal arrow pointing at it and another pointing away. For instance: $D \rightarrow M \rightarrow Y$ has a mediator (M).
3. **Colliders**: a node has two causal arrows *pointing* at it. For instance: $D \rightarrow W \leftarrow Y$ has a collider (W).

Second, the key necessity to estimate causal effects of a treatment D on an outcome Y is that we must *block* **backdoors**. Backdoors are causal paths that go from the treatment to the outcome via a path that includes an arrow pointing at the treatment. For instance, suppose that you have the following two paths:

1. $D \rightarrow M \rightarrow Y$

2. $D \leftarrow Z \rightarrow Y$

Third, not all paths must be blocked. Only (a) *open* (b) *backdoors* must be blocked. This means that we must only worry about paths from D to Y where an arrow is pointing at D. And among these paths, we must worry about those that are open. What makes them open? The absence of a collider! Thus:

- $D \leftarrow Z \rightarrow W \rightarrow Y$ is an open backdoor and must be closed.
- $D \leftarrow Z \rightarrow W \leftarrow Y$ is a closed backdoor (W is a collider that closes it) and must not be touched.

With OLS, closing a backdoor typically means adjusting for a variable on an open backdoor. In the first example, it would mean including Z or W as a right-hand side covariate ($y = \alpha + \beta D + \gamma Z + \varepsilon$ or $y = \alpha + \beta D + \rho W + \varepsilon$).

Note:

- You *don't* want to adjust for (i.e., "control for") a variable that is a mediator on a path from D to Y and pointing away from D (Montgomery, Nyhan, and Torres 2018). For instance, in the path $D \rightarrow M \rightarrow Y$, you don't want to adjust for M. Doing so would "hide" part of the effect of D on Y.
- You *don't* want to adjust for a variable that is a collider on a path between D and Y (Elwert and Winship 2014). In the path $D \rightarrow W \leftarrow Y$, you don't want to adjust for W. If you do so, you open an otherwise closed backdoor.
- You *do* want to adjust for a variable that is a fork pointing toward D and toward Y.

Still, the problem is that we often don't know with any degree of certainty what the correct DAG is. Think of the following example: we want to measure the effect of an increase in interest rates (D) on consumer confidence (Y). How could we build a DAG? There could be all kinds of variables affecting both D and Y, including business cycles, trade shocks, unemployment, wars, etc. Of course, an expert in the domain can build more convincing DAGs than a random person. But will we ever be sure that we have the right model? Probably not.

6 Randomized controlled trials

This version: December 16, 2025

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>
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6.1 Readings

People interested in experimental methods in the social sciences should refer to Morgan and Winship (2014) (on causality writ large), Imbens and Rubin (2015) (Part II) (on randomized experiments), Athey and Imbens (2016) (on the econometrics of RCTs), and Gerber and Green (2012) (on field experiments).

6.2 Randomized Controlled Trials (RCTs)

6.2.1 Overview

To overcome the limitations caused by incomplete knowledge of DAGs, researchers can turn to **research designs that do not require full knowledge of the DAG**. What we need is to find a design in which we can confidently say that there are no open backdoors in our DAG. One way to do so is by taking control over the assignment of the treatment D: if we

can decide who gets the treatment and who doesn't, we can break the correlation between D and other variables Z that could confound D's effect.

That's where **randomization** comes in. The intuition is simple: if you randomize D (eg, use a coin toss to decide who gets the treatment and who doesn't), then by construction, nothing will be correlated with it (at least in expectation). Thus, but taking control over D's assignment, we break all the arrows pointing at D: the variation of D is not caused by anything else than our randomization.

Thus, **randomized controlled trials** are very popular: we can ensure that exogeneity is met by design, thereby increasing the odds that our estimates are unbiased and consistent.[1] It is popular in academic research, including many fields in the social sciences. Examples of papers using RCTs include (Duflo, Kremer, and Robinson 2009; Aklin et al. 2017; Asquith 2020). It is also popular in industry, where it's often referred to as **A/B testing**, where two groups receive different versions of an intervention, and outcomes are then compared. Types of RCT experiments include:

- **Lab experiment**: experiments conducted with individuals in a controlled lab environment. For instance, this includes experiments in which people playing with each others in a game theoretic setting.
- **Field experiments**: experiments conducted in a real-life setting. For instance, this includes experiments conducted with teachers in schools.
- **Survey experiments**: experiments conducted within surveys; people receive randomized content. For instance, people might receive competing ads and the analyst wants to know which one makes people more interested in a product.

Unfortunately, RCTs cannot always be used. There are ethical and practical constraints that limit their scope. More on this below. But this explains why people often switch gears

[1] Technically, we will still need our earlier assumptions, including a random sample and no perfect multicollinearity.

and use **quasi-experimental** methods. These methods try, by and large, to find exogenous variation in D that makes a given context *look* like an RCT. This includes methods such as **instrumental variables**. More on this in separate course notes.

6.2.2 The problem

Consider the following starting point. Suppose we had a dataset with a treatment $D \in \{0, 1\}$ and an outcome Y. Suppose that D is not exogenous. There is some variable out there that drives some units i to get the treatment and others not, and that this variable also causes Y. This is a typical example of confounding. What would be the meaning of our naive comparison of outcomes for treated units versus control units?

$$\begin{aligned}
\text{Naïve difference} =& E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 0] \\
=& E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 1] \\
&+ E[Y_i(0)|D_i = 1] - E[Y_i(0)|D_i = 0] \quad \text{adding and subtracting} \\
=& E[Y_i(1) - Y_i(0)|D_i = 1] \\
&+ E[Y_i(0)|D_i = 1] - E[Y_i(0)|D_i = 0] \quad \text{regrouping} \\
\equiv& \text{ATT} + \text{Selection (baseline) bias}
\end{aligned}$$

where **ATT** refers to the average treatment effect on the treated units.

The question is: how to get rid of selection bias?

6.2.3 Randomization of treatment

By randomizing D_i, we make D independent from $Y(0)$, $Y(1)$ and functions thereof. We typically write:

$$(Y(0), Y(1)) \perp D.$$

We call this the **ignorability assumption** (Rosenbaum and Rubin 1983) . This means that D is independent of $Y(0)$, $Y(1)$, and of functions of the two (such as $Y(1) - Y(0)$). Thus, learning whether a unit was treated doesn't yield any information

about the potential outcomes or the treatment effect.[2] A useful implication of ignorability is that: $E[Y(0)|D] = E[Y(0)]$ and $E[Y(1)|D] = E[Y(1)]$.

Note that we might sometimes write:

$$(Y(0), Y(1)) \perp D|\mathbf{X},$$

in which case we say that $Y(0)$ and $Y(1)$ are *conditionally* independent of D. This would be an assumption we would need to make if $\mathbf{X}$ is correlated with D and Y. We call this the **conditional ignorability assumption** .

Here's the neat part: randomization guarantees, in expectation, that this assumption is met. I say "*in expectation*" because we could by bad luck fail to have independence. For instance, imagine that I randomize assignment of aspirin to a sample of 100 patients, with 50 women and 50 men. I could accidentally randomize it in a way that all the men get the aspirin and all the women are in the control group. If gender is a confounder, then this random assignment won't help us.

Setting this threat aside, we can simplify:

$$\begin{aligned}
\text{Naïve difference} =& E[Y_i(1) - Y_i(0)|D_i = 1] \\
&+ E[Y_i(0)|D_i = 1] - E[Y_i(0)|D_i = 0] && \text{from above} \\
=& E[Y_i(1) - Y_i(0)|D_i = 1] && \text{by ignorability} \\
=& E[Y_i(1) - Y_i(0)] && \text{by ignorability} \\
\equiv& \text{Average treatment effect}
\end{aligned}$$

Thus, randomization removes selection bias *and* makes the ATT equal the ATE.

The latter makes intuitive sense: the ATE and ATT will diverge if, for instance, units can decide by themselves to take the treatment based on their knowledge of its effectiveness. But if treatment assignment is random, then there is no reason to expect that the population taking the treatment is different in a meaningful way from the population not taking it, and thus ATE and ATT will coincide.

[2] Of course, if the treatment effect is different from zero, then D and (observed) Y will be dependent.

6.2.4 Estimation

Since we can ignore selection and confounding, we could estimate the ATE by plugging in sample averages and taking the difference in means:

$$\widehat{\text{ATE}} := E[\widehat{Y_i(1) - Y_i(0)}] = \frac{1}{N_1} \sum_{i:D=1} Y_i - \frac{1}{N_0} \sum_{i:D=0} Y_i,$$

where N_1 and N_0 is the number of treated and control units.

We can then compute the standard error of this expression:

$$\begin{aligned} se(\text{ATE}) &= se\left(\frac{1}{N_1} \sum_{i:D=1} Y_i - \frac{1}{N_0} \sum_{i:D=0} Y_i\right) \\ &= \sqrt{\frac{\sigma_{y1}^2}{N_1} + \frac{\sigma_{y0}^2}{N_0}} \end{aligned}$$

We will need to plug in sample estimates of σ_{y1}^2 and σ_{y0}^2 (the variance of $Y(1)$ and $Y(0)$, respectively).

Alternatively, we can estimate a linear model with OLS:

$$Y = \hat{\alpha} + \hat{\beta} D + u,$$

where $\hat{\beta}$ is the estimate of the ATE. The advantage of using OLS is that it allows us to include additional covariates in the model. As long as these covariates are not (post-treatment) mediators (Montgomery, Nyhan, and Torres 2018), and as long as they have an effect on Y, their effect will be to reduce the size of the squared residuals and thus reduce standard errors. We can then also use whatever types of standard errors we want depending on the design of our study (see below on clustered standard errors).

6.3 RCT in practice

In practice, here are typical steps used to conduct an RCT:

1. **Define the population and unit of analysis**. Are you looking at firms? At consumers? At municipalities? Draw a representative sample from that population.
2. **Define the treatment and outcomes**. Specify the treatment of interest (e.g., a direct change or an encouragement to take action) and measurable outcomes.
3. **Measurement strategy**. Decide how you will gather data on both treatment and outcomes, such as surveys or administrative data.
4. **Hypothesis building** . Formulate hypotheses that state the expected relationship between the treatment and outcome (e.g., $H_0 : \beta = 0$ versus $H_1 : \beta \neq 0$).
5. **Conduct a power analysis** . Calculate the minimum sample size required to detect a meaningful effect. Power analysis helps ensure that the study is neither underpowered (too small to detect effects) nor overpowered (excessively costly). More below.
6. **Measure pre-treatment data**. These data can be used in several ways:
   - Check balance of covariates between treatment and control group: did your randomization really break correlations between D and potential confounders?
   - Use these variables later as controls to increase the precision of our estimates.
7. **Assign treatment**. Use a random number generators to treat units in your sample. Generally: $Pr(\text{Treatment}) = 0.5$ maximizes power. You can use Bernoulli trials.
8. **Implement treatment**. If needed, collect data on treatment compliance (more on this below).
9. **Measure outcome(s)**.
10. **Estimate treatment effects**. This may happen with or without adjustments for pre-treatment variables.

6.4 Power analysis

6.4.1 Statistical power and sample size selection

This step – selecting a sample size – differs from what we have seen so far. We didn't really consider the possibility until now that we could choose the size of our sample: we took whatever data we received as a given. But in RCTs, you can control (within your budget constraint) the size of the sample. This is very interesting: we know that standard errors are responsive to sample size (the larger the sample, the smaller the standard errors). Thus, we can ask ourselves: what N would I need to detect an effect of size β?

To get to this idea, we need a few definitions:

- **Type I error** : the probability of rejecting the null hypothesis when the null was in fact true, ie:

  $$\alpha = Pr(\text{reject } H_0 | H_0 \text{ true})$$

  and is the probability of a **false positive**. It's often referred to as α (not to be confused with the y-intercept in the linear model).

- **Type II error** : the probability of failing to reject the null hypothesis when the alternative is true, ie:

  $$\beta = Pr(\text{fail to reject } H_0 | H_0 \text{ false})$$

  and is the probability of a **false negative**. It's often referred to as β (not to be confused with the slope of the linear model).

- **Statistical power** : the probability of rejecting the null when the null was in fact false, ie:

  $$\kappa = Pr(\text{reject } H_0 | H_0 \text{ false}) = 1 - \beta$$

6.4.2 Minimum detectable effect (MDE)

In general, we would like power (κ) to be high: why bother do an RCT if we end up with low probability of detecting an effect? But how can we set it? To do so, we will need to come up with a reasonable guess regarding the anticipated treatment effect.

To see this, let us start with the following idea: given a research design and what we know about a given treatment, what is the required sample size to detect the **minimum detectable effect (MDE)**, which is the smallest effect size that a study can detect with a specified probability of achieving statistical significance? Let's work through an example.[3]

Consider the example in Figure 6.1. Suppose that we want to test the null hypothesis that $\beta = 0$. Yet suppose that the true effect is $\beta = 3$. If the true effect is $\beta = 3$, then our distribution of $\hat{\beta}$ will be centered on 3 (since it is unbiased). However, some $\hat{\beta}$'s that we will obtain over repeated samples will be smaller and some larger than 3. In fact, some will be so small that a null hypothesis test will deem then statistically insignificant.

In the example in Figure 6.1, all the draws of $\hat{\beta}/se(\hat{\beta})$ below 1.96 (the critical t value assuming we have a large sample) will lead us to commit a Type II error : we will fail to reject the null even though the null is false.

From there, the MDE is the smallest effect that, if true, has an p% probability of generating a statistically significant effect for a given α (H. S. Bloom 1995). For instance, we might say something like: "assuming we want 80% power (κ) and we want to set the Type I error at $\alpha = 0.05$, then 3.4 is the MDE that can be detected." In Figure 1, this would mean that the unshaded part of the blue distribution represents 80% of the mass (which requires to move the blue curve to the right).

Note: it's up to you to decide how to set κ and α. The convention is often to use $\kappa = 0.8$ and $\alpha = 0.05$ (or smaller), but this is a matter of how much you care about being wrong in one way or another.

[3]This section benefitted a lot from Cyrus Samii's notes on MDE.

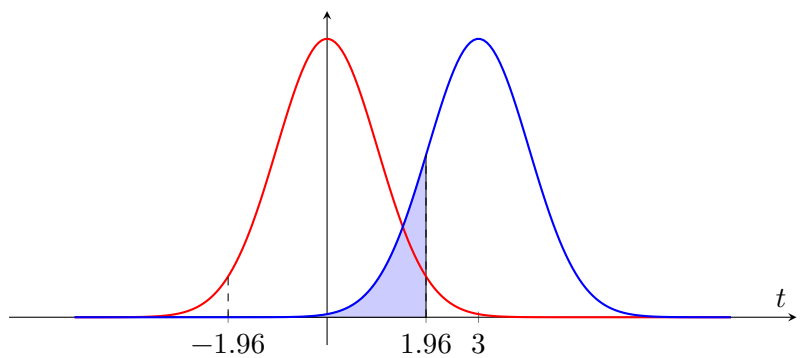

Figure 6.1: Minimum detectable effect problem. The red curve is the distribution of t values $(\widehat{\beta}/se(\widehat{\beta}))$ under the null hypothesis that $\beta = 0$. The blue one is the distribution of t values if $\beta = 3$. Eyeballing, about one-third of the draws from the $\beta = 3$ distribution will be statistically insignificant, that is, will have a t-value below 1.96 (mass under the blue curve below 1.96, shaded in blue). Thus, statistical power is about 2/3 here (unshaded mass under the blue curve above 1.96).

Type S and Type M errors

There is another type of errors that could emerge aside from Type I and Type II errors: you could get a significant estimate of β that is of the wrong sign. In Figure 2, this would be the share of outcomes under the blue curve that's smaller than -1.96 (on the left). These draws of $\hat{\beta}$ are significant but negative, even though β is positive. In Figure 1, this probability seems pretty small (we can barely see the share under the curve to the left of -1.96), but it can be large in other settings.
The share of the blue curve below the critical t statistic of the opposite sign to β and is referred to by Gelman et al. as a **Type S** ("sign") error (Gelman and Carlin 2014):

$$\text{Type S error} := Pr(\hat{\beta} \text{ has the wrong sign} | \hat{\beta} \text{ is statistically significant})$$

Gelman et al. also define the problem of **Type M** (for "magnitude") problem or **exaggeration ratio**. Consider Figure 2: if we condition on statistical significance, the only estimates of the correct sign that we will obtain are those above 1.96. But the distribution of what remains of the blue curve (truncated at 1.96) will have a different mean than 3 (the true treatment effect). We can then define the exaggeration ratio as:

$$\text{Type M ratio} := \frac{\int_{-\infty}^{t_{\alpha/2}} N(\hat{\beta}; \beta, 1) d\hat{\beta} + \int_{t_{\alpha/2}}^{\infty} N(\hat{\beta}; \beta, 1) d\hat{\beta}}{\beta}$$

where β is the ATE, ie, the true average treatment effect. Note that we standardized the variable so that $se(\hat{\beta} = 1)$; this is just for convenience.
The numerator is the average of all the estimates that are statistically significant: those that are significantly negative (first term) and those that are significantly positive (second term). The first term is evaluated from negative infinity up to $-t_{\alpha/2}$ (at that point, the estimates aren't significant anymore). Then the second term resumes from the moment estimates are significant and positive ($t_{\alpha/2}$) to infinity.

In Figure 1, the Type M ratio for $\alpha = 0.05$ corresponds to the expected value of (a) the mass under the blue curve below -1.96 plus (b) the mass under the curve above 1.96, divided by 3 (the true β).

6.4.3 Computing MDE

Formally, the MDE β^{MDE} is:

$$\text{MDE} \equiv |\beta^{\text{MDE}}_{\alpha,\kappa,se(\hat{\beta})}| = se(\hat{\beta})(t_{\alpha/2} + t_{1-\kappa})$$

In other words, for a given α, κ, and $se(\hat{\beta})$, the MDE will be the point at a distance of $se(\hat{\beta})(t_{\alpha/2} + t_{1-\kappa})$ to 0 (assuming that's our null hypothesis; otherwise, just shift this by a constant).

Think of it this way: if the true effect β were 1.96 standard errors away from 0, then our blue curve in Figure 1 would be centered at 1.96 (a bit to the left of where it is now). Let's assume we set $\alpha = 0.05$. Half the time, we would draw a $\hat{\beta}$ that's insignificant (ie, it would have a t statistic less than 1.96, which is the critical t value for $\alpha = 0.05$) and half the time it would be significant (t statistic above 1.96). Thus, our power would be 50%. That's the $t_{\alpha/2}$ in the equation of the MDE. If we want to increase our power (increase κ), then we need to move even further from 1.96. That's the additional $t_{1-\kappa}$ of the MDE equation. The higher κ, the smaller we want the blue mass to be, and thus the larger the MDE becomes.

MDE example

A quick example will clarify this: suppose we decide that we want to run an RCT with the following parameters:

- Type I error: $\alpha = 0.01$
- Type II error: $\beta = 0.1$, which implies that $\kappa = 0.9$.

I can then compute the critical values attached to it. First, our Type I critical t statistic:

```
> qt(0.01/2, 1000, lower.tail=F)
[1] 2.580755
```

Then then same for Type II:

```
> qt(0.1, 1000, lower.tail=F)
[1] 1.282399
```

This then yields:

$$\text{MDE} = se(\hat{\beta})(t_{\alpha/2}+t_{1-\kappa}) = se(\hat{\beta})(2.58+1.28) = 3.86 \cdot se(\hat{\beta})$$

Thus, fixing κ and α at 0.9 and 0.01, respectively, means that the MDE will be 3.86 standard errors away from 0. In other words, to achieve a power of 90% and using an α of 0.01 Type I error, then the smallest detectable effect will be 3.86 standard errors away from zero.
Note that this is more demanding that 'just' obtaining a statistically significant effect. You can get the latter with estimates only 2.58 standard errors away from zero (for $\alpha = 0.01$).

As you can see, the MDE will become larger if:

- We reduce our α (reduce the probability of committing a Type I error)
- We increase our κ (increase our power, ie, reduce the probability of a Type II error)
- We increase our standard errors (less precise estimates)

6.4.4 Optimal sample size

We now know what effect size we can detect for a given α, κ, $se(\hat{\beta})$. But we can flip the question and say: what kind of sample size do we need to detect a given MDE?

To do so, we will need to estimate the standard error of $\hat{\beta}$. Recall the definition of $\hat{\beta}$'s standard error:

$$se(\hat{\beta}) = \sqrt{\frac{\sigma_{y1}^2}{N_1} + \frac{\sigma_{y0}^2}{N_0}},$$

where σ_{y1}^2 and σ_{y0}^2 are the variances of the outcome under treatment and control conditions.

In an RCT, we don't know for sure *ex ante* how many units will be in the treatment and in the control group (since it's a random draw) and thus we don't know what N_1 and N_2 are. However, we can write this down in a probabilistic way, with probability of receiving a treatment p and sample size N:

$$se(\hat{\beta}) = \sqrt{\frac{1}{N}\left(\frac{\sigma_{y1}^2}{p} + \frac{\sigma_{y0}^2}{1-p}\right)}$$

Recall that MDE $= se(\hat{\beta})(t_{\alpha/2} + t_{1-\kappa})$. We can plug in the estimate for $se(\hat{\beta})$ and solve for N:

$$N = \frac{(t_{\alpha/2} + t_{1-\kappa})^2 \left(\frac{\sigma_{y1}^2}{p} + \frac{\sigma_{y0}^2}{1-p}\right)}{\text{MDE}^2}$$

The implications are straightforward:

- N decreases in the size of the MDE. The larger the effect we anticipate, the fewer observations we will need.
- N increases in $\sigma_{y1}^2, \sigma_{y0}^2$. The noisier our variables, the more observations we will need.
- N decreases when α increases, κ decreases.

Here's the way we can use this knowledge. In practice, we will do as follows.

1. We start by coming up with a plausible guess regarding the treatment effect of our intervention D. This guess can be informed by our knowledge. It could also be a threshold imposed by commercial reasons (eg, "my new website only makes sense if it increases consumer engagements by 2,000 users per month").

2. Then, we would plug in the MDE in the expression above, set $t_{\alpha/2}$ and $t_{1-\kappa}$ depending on our tolerance of Type I and Type II errors, plug in a reasonable estimate for σ_{y1}^2 and σ_{y0}^2 (eg from past data).

3. We would then compute the sample size N needed to detect such an effect, given all the parameters we just fixed.

This would then tell us how to set the sample size of this RCT.

6.5 Group-level treatment

Sometimes, it won't be possible to treat individual units. Instead, you might have to treat entire groups. For instance, imagine you are randomizing electrification programs at the village level: then the entire village will be treated (or not). Even though you might be interested in household- or individual-level outcomes, you won't be able to randomize at that level.

This, in itself, is not a problem. You are still randomizing, and thus ignorability will still hold. You can obtain unbiased and consistent estimates of treatment effects using the estimation approach discussed above. Good.

However, group-level treatment does raise issues for **inference**. Recall that one of the difficulty when writing down the standard errors of a parameter $\widehat{\beta}$ is the matrix $E[\varepsilon'\varepsilon]$. If you recall, this is the variance-covariance matrix of error terms. If we assume (a) homoskedasticity (constant variance of error terms conditional on $\mathbf{X}$) and (b) independence of error terms (zero covariance), then we could write:

$$E[\varepsilon'\varepsilon] = \begin{bmatrix} \sigma^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma^2 \end{bmatrix}$$

and $Var(\widehat{\beta})$ simplifies to $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$. Then, as we saw earlier (see notes on inference with OLS), we can estimate σ^2 with s^2:

$$s^2 = \frac{\mathbf{u}'\mathbf{u}}{n-k},$$

where n is the sample size and k the number of parameters that are estimated.

But now, the assumption of independence becomes harder to justify . For instance, individuals within a village may have abnormally high (or low) values of Y and thus larger (or smaller) error terms. Hence, we might have something like this (ignoring heteroskedasticity):

$$E[\varepsilon'\varepsilon] = \begin{bmatrix} \sigma^2 & a & a & 0 & 0 & 0 \\ a & \sigma^2 & a & 0 & 0 & 0 \\ a & a & \sigma^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma^2 & b & b \\ 0 & 0 & 0 & b & \sigma^2 & b \\ 0 & 0 & 0 & b & b & \sigma^2 \end{bmatrix}$$

where a and b are the covariances of error terms within group A and group B. For instance, it could be that each unit is an individual ($i = 1, 2, ..., n$), and we have two villages in our sample (A and B). We would then say that the errors are correlated within groups (aka there is nonzero **intracluster correlation**) (Moulton 1986, 1990).

Fortunately, we can overcome this problem. We will typically use **clustered standard errors** for inference (Kloek 1981; Liang and Zeger 1986; A. Colin Cameron and Miller 2015; Abadie et al. 2023) . These standard errors will allow within-group (or "within-cluster") correlation of error terms. And typically, since we assume heteroskedasticity anyway, we will also let the variance vary.

Concretely, we estimate clustered standard errors, using G to indicate groups (or clusters) and u as the residuals:

$$\text{Var}_{\text{cluster}}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{g=1}^{G} \mathbf{X}_g' \mathbf{u}_g \mathbf{u}_g' \mathbf{X}_g \right) (\mathbf{X}'\mathbf{X})^{-1}$$

Informally, we take cluster by cluster and compute the middle bit using the residuals of that group. We then sum up over all groups and divide by the square of $\mathbf{X}'\mathbf{X}$ (since it appears twice).

6.6 Compliance

6.6.1 Compliance problems

One of the big problems in RCTs is compliance: what happens if the treated unit refuses to take the treatment? We call these units non-compliers. For instance:

- An RCT in which people are asked to watch YouTube videos selected by the analyst: people might watch other videos instead (or none at all).
- An RCT in which people are encouraged to buy a new item: people might not be interested.

This problem affects our analysis. We can think of it this way:

1. Treatment is assigned to all units i ($T \in \{0, 1\}$). Eg: the analyst decides who will be asked to take an aspirin and who won't.
2. Each unit i decides whether to comply or not ($D \in \{0, 1\}$). Eg: the patients decide whether to swallow the aspirin they received or not.
3. If $T_i = 1$ and $D_i = 1$: we say that unit i is a complier. (For now, we assume that nobody in the control group cheated and took the treatment, such that $T_i = 0$ implies $D_i = 0$.)

The problem is that this makes things a bit hazy regarding the average treatment effect (ATE): we now have a form of selection where some units comply and others don't, and we don't have tools to address this. Fortunately, there are other estimands that are of help here: the intent to treat effect (ITT) and the local average treatment effect (LATE) .

6.6.2 Intent to treat and local average treatment effects

The solution is to think of two alternative treatment effects that we might be interested in.

- **Local average treatment effect** (LATE) is the effect of the treatment *among compliers*. That is, for those who complied, how did Y change on average?
- **Intent to treat** (ITT) is the effect of treatment averaging the outcomes Y of those who were treated and accepted the treatment ($T = 1$ and $D = 1$) and those who didn't accept it ($T = 1$ and $D = 0$), compared to the rest ($T = 0$).

Consider the following example from (Aklin et al. 2017): we encouraged a random set of Indian households to rent solar panels. We considered several outcomes of interest, including consumption of kerosene (solar panels should reduce the need for kerosene lamps), household income (electricity can be used to create small businesses), time spent studying (solar panels can be used for night light), etc.

Essentially, we randomized where the sales representatives of the solar company we worked with would go. Whether a household received a visit or not is what we really randomized and this corresponds to T ($T = 1$ if a household received a sales pitch from a representative). Of course, out of all the households they visited, only a small number of households rented these solar panels (thus, we had many $T = 1$ and $D = 0$). This made it natural for us to ask two kinds of questions:

- What is the average effect of solar panels on those who rented them (the compliers)? That's the LATE.
- What is the average effect of solar panels overall regardless of whether they rented them or not? That's the ITT.

You might wonder: why do we care about ITT? We do care about it from a policy (or business strategy) perspective. If my question is "I'm the Prime Minister and I want to roll out solar panels across an entire state. I want to know what kind of effect this program will have!" In such a case, I cannot limit myself to focus solely on the households that actually end up using them. I need to account for the fact that some will not comply.

Thus, we computed both the ITT and the LATE. The results are reported in Figure 2 for the case of kerosene. As you can

see, our intervention reduced kerosene consumption a lot for people who complied (-48 rupees per month). But once we take into account non-compliers, then we note that the overall effect is watered down (-14 rupees per month). This makes sense: since they didn't start using solar panels, non-compliers will not experience a decrease of kerosene consumption. Thus, they will drag the effect of solar panels down.

Table 2. Effect of MGP solar microgrids on household kerosene spending. Effects are shown for spending in the private market (A), the PDS (B), and overall (C). The SEs are clustered by habitation and are shown in parentheses. All dependent variables are measured in rupees per month. n = 3825; number of households, 1281. *p < 0.10, **p < 0.05, ***p < 0.01.

	ITT (OLS)		LATE (IV)	
	(1)	(2)	(3)	(4)
(A) Kerosene spending on private market				
Treatment	−14.01*** (5.28)	−14.49** (6.91)	−47.49** (19.83)	−49.36** (24.62)
Household FE		Yes		Yes
Wave FE	Yes	Yes	Yes	Yes
Pretreatment mean for control group = 72				
First-stage estimate			0.29	0.29
First-stage F statistic			10.15	10.01

Figure 6.2: Source: (Aklin et al. 2017)

6.6.3 Estimation of ITT and LATE

The ITT can be estimated with OLS as the following linear model:

$$Y_i = \alpha + \beta T_i + \varepsilon_i,$$

where $\hat{\beta} = \widehat{\text{ITT}}$. Note that we use T (the assignment to treatment) rather than D (the treatment itself). Since T is randomized, this model can be consistently estimated with OLS.

In our example, this means that we used data on whether a household was visited by a sales representative (T) rather than whether they rented solar panels (D).

For the LATE, we must estimate it using an **instrumental variable** approach , in which T serves as an instrument for D. These are then estimated with two-stage least squares. The long story short is that the two stages are:

$$D_i = \phi + \gamma T_i + \mu_i \qquad \text{1st stage}$$
$$Y_i = \alpha + \beta_{IV} \hat{D}_i + \varepsilon_i \qquad \text{2nd stage}$$

where $\hat{\beta}_{IV} = \widehat{\text{LATE}}$. More on this in the section devoted to instrumental variables.

6.7 Limitations of experimental methods

Despite their benefits, RCTs cannot always be used:

- **Cost**. Implementing an RCT can be resource-intensive due to the need for large sample sizes, data collection, and intervention costs.
- **Feasibility**. Some variables cannot be randomized in practice. For instance, I cannot randomize how religious someone feels. Nor can I instruct all the central banks in the world to randomize their interest rates.
- **Ethical concerns**. Some treatments, such as access to life-saving drugs, raise ethical issues around fairness and harm.

See Teele (2014) for a discussion on field experiments, in particular. See Nardini (2014) on ethics in clinical trials (including some horrible cases).

7 Instrumental variables

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> **Caveat**
>
> These notes are offered in support of *Introduction to Econometrics* (MGT 581). They were written to help students understand the logic behind important topics in econometrics. But they don't (always) replace the insights and depth that textbooks can provide. This is why we connect each session to chapters of our core textbooks. See the syllabus for more details.

7.1 Readings

People interested in instrumental variables may read (Imbens and Rubin 2015), (Joshua D. Angrist, Imbens, and Rubin 1996) (one of the papers that kickstarted the use of IV in economics), (Sovey and Green 2011) (a short practical guide), and (Imbens and Rubin 2015) (ch.24-26) (an in-depth discussion of IVs). Additional useful papers include (Heckman, Urzua, and Vytlacil 2006), (Peter M. Aronow and Carnegie 2013).

7.2 Introduction to instrumental variables

When the exogeneity assumption cannot be met, then OLS estimates are biased and inconsistent. Instrumental variables (IV) provide an alternative estimation method by using an external variable (the **instrument**) that helps isolate the exogenous variation in the endogenous treatment (D) that we care about.

A variable Z must satisfy two critical conditions to serve as an instrument for our endogenous variable of interest, D (for reasons that will shortly become clear):

1. **Relevance**: Z must be correlated with the endogenous explanatory variable ($Cov(Z, D) \neq 0$). This ensures that Z captures the variation in D necessary for estimating its effect.
2. **Exclusion restriction**: Z must not be correlated with the error term in the model ($Cov(Z, \varepsilon) = 0$). This ensures that the instrument does not independently affect the outcome variable, aside from its effect through X.

For example, in studying the effect of education on earnings, the distance to the nearest college can be a valid instrument if it affects the likelihood of attending college (relevance) but does not directly influence earnings other than through education (exogeneity).

7.3 Mathematic justification

To see how such a variable Z would help, we must place it in the same equation as β. We can write:

$$\begin{aligned} Cov(Y, Z) &= Cov(\alpha + \beta D + \varepsilon, Z) \\ &= Cov(\alpha, Z) + Cov(\beta D, Z) + Cov(\varepsilon, Z) \\ &= \beta Cov(D, Z) && \text{by prop of cov and by excl restr} \end{aligned}$$

Then, solving for β:

$$\beta_{IV} = \frac{Cov(Y, Z)}{Cov(D, Z)}$$

Note how the two assumptions already came into play:

- **Relevance** is needed for the estimation of β_{IV}; if $Cov(D, Z) = 0$, then the denominator of β_{IV} wouldn't be defined.
- **Exclusion restriction** is needed for consistency: otherwise, we couldn't get rid of $Cov(\varepsilon, Z)$ (more below).

We can even go a bit further. If we divided both the numerator and denominator by $Var(Z)$, we find that:

$$\begin{aligned}\beta_{IV} &= \frac{Cov(Y,Z)/Var(Z)}{Cov(D,Z)/Var(Z)} \\ &= \frac{\gamma}{\pi}\end{aligned}$$

where γ and π are parameters from the following two equations:

$$\begin{aligned}Y &= \alpha_0 + \gamma Z + \varepsilon_0 \\ D &= \alpha_1 + \pi Z + \varepsilon_1\end{aligned}$$

This definition allows us to rewrite the IV in a slightly different way:

$$\beta_{IV} = \frac{\text{Reduced form}}{\text{First stage}}$$

Terminology

People often refer to the following expressions as...

$$\begin{aligned}\text{Reduced form equation: } & Y = \alpha_0 + \gamma Z + \varepsilon_0 \\ \text{First stage equation: } & D = \alpha_1 + \pi Z + \varepsilon_1 \\ \text{Structural form equation: } & Y = \alpha_2 + \beta D + \varepsilon_2\end{aligned}$$

Note: the content of this note assumes that the treatment effect β is a constant until Section 6, when we explore the problem of *heterogeneous* treatment effects.

7.4 Estimation

7.4.1 Wald IV estimator

Note that the estimated treatment as defined above is a **population** quantity:

$$\beta_{IV} = \frac{Cov(Y,Z)}{Cov(D,Z)}$$

This being said, it can straightforwardly be estimated with the **Wald estimator** (Wald 1940; Morgan and Winship 2014, 296), which applies if the instrument Z is a dummy variable. Start with the structural equation and then condition on Z:

$$\begin{aligned} Y &= \alpha + \beta D + \varepsilon && \text{structural equation} \\ E[Y|Z=1] &= E[\alpha|Z=1] + E[\beta D|Z=1] + E[\varepsilon|Z] && \text{taking cond expectation} \\ &= \alpha + \beta E[D|Z=1] && \text{constants and by excl rest} \end{aligned}$$

Likewise, if we condition on $Z = 0$, we obtain:

$$E[Y|Z=0] = \alpha + \beta E[D|Z=0]$$

We can then take the difference between the two and rearrange:

$$\begin{aligned} E[Y|Z=1] - E[Y|Z=0] &= [\alpha + \beta E[D|Z=1]] - [\alpha + \beta E[D|Z=0]] \\ &= \beta[E[D|Z=1] - E[D|Z=0]] && \text{simplifying} \\ \beta_{\text{Wald}} &= \frac{E[Y|Z=1] - E[Y|Z=0]}{E[D|Z=1] - E[D|Z=0]} && \text{isolating } \beta \end{aligned}$$

We can then compute the sample conditional means of all four components:

$$E[\widehat{Y|Z=1}] = \frac{\sum_i Z_i Y_i}{\sum_i Z_i} \qquad E[\widehat{Y|Z=0}] = \frac{\sum_i (1-Z_i) Y_i}{\sum_i (1-Z_i)}$$

$$E[\widehat{D|Z=1}] = \frac{\sum_i Z_i D_i}{\sum_i Z_i} \qquad E[\widehat{D|Z=0}] = \frac{\sum_i (1-Z_i) D_i}{\sum_i (1-Z_i)}$$

$$\text{Then: } \hat{\beta}_{\text{Wald}} = \frac{E[\widehat{Y|Z=1}] - E[\widehat{Y|Z=0}]}{E[\widehat{D|Z=1}] - E[\widehat{D|Z=0}]}$$

The limitation is that this doesn't neatly suit itself to the inclusion of additional covariates nor of continuous instruments. Yet we may still want to block backdoor that could generate confounding and we may want to use other types of instruments. As a result, we often use two-stage least squares as an estimation method.

7.4.2 Two-Stage Least Squares (2SLS)

The 2SLS method is a two-step procedure used to estimate the coefficients in an IV setting. As its name indicates, we will use OLS twice.

Let's start with the case of one instrument and one endogenous treatment. We can build the 2SLS from first principles. Recall we said that the IV estimator is the ratio of the effect of Z on Y divided by the effect of Z on D. Thus, we can write this as:

$$\begin{aligned}\beta_{\text{IV}} &= \frac{(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}}{(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D}} \\ &= (\mathbf{Z}'\mathbf{D})^{-1}\mathbf{Z}'\mathbf{y}\end{aligned}$$

The 2SLS estimator in case of an over-identified model (a model with more instruments than endogenous variables) is given by:

$$\hat{\beta}_{\text{2SLS}} = [\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D}]^{-1}[\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}]$$

Some extra math of IV

Here is how to get from:

$$\beta_{\text{2SLS}} = [\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D}]^{-1}[\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}]$$

to:

$$(\mathbf{Z}'\mathbf{D})^{-1}\mathbf{Z}'\mathbf{y}$$

when the model is just-identified (i.e., we have the same number of instruments as endogenous variables; see box).

Just-identified model

We write that a model is…

- **under-identified** if the number of instruments is strictly smaller than the number of endogeneous variable(s);
- **just-identified** if the number of instruments is equal to the number of endogeneous variable(s);
- **over-identified** if the number of instruments is strictly greater to the number of endogeneous variable(s).

First, the IV optimization problem is the following minimization problem:

$$\min_{\beta} = (\mathbf{y} - \mathbf{D})'\mathbf{P}_Z(\mathbf{y} - \mathbf{D}),$$

where the **projection matrix** $\mathbf{P}_Z$ is:

$$\mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'$$

(Note: the projection matrix is the same $\mathbf{Z}$ times the OLS estimator once you add something at the end of this expression. See it by yourself: add $\mathbf{y}$ and then you see that it's the same as the OLS estimator of the effect of $\mathbf{Z}$ on $\mathbf{y}$ times $\mathbf{Z}$.)
Solving this problem, we obtain (as above):

$$\beta_{IV} = [\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D}]^{-1}[\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}]$$

Note that $\mathbf{D}'\mathbf{Z}$ is square under just-identified conditions (as is $\mathbf{Z}'\mathbf{D}$). As a result, it is also invertible (which would not be the case when we have an over-identified model). Then we can rewrite:

$$[\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D}]^{-1} = (\mathbf{Z}'\mathbf{D})^{-1}(\mathbf{Z}'\mathbf{Z})(\mathbf{D}'\mathbf{Z})^{-1}$$

so that:

$$(\mathbf{Z}'\mathbf{D})^{-1}(\mathbf{Z}'\mathbf{Z})(\mathbf{D}'\mathbf{Z})^{-1}(\mathbf{D}'\mathbf{Z})(\mathbf{Z}'\mathbf{Z})^{-1}(\mathbf{Z}'\mathbf{y})$$

Simplifying, we obtain:

$$\beta_{IV} = (\mathbf{Z}'\mathbf{D})^{-1}(\mathbf{Z}'\mathbf{y})$$

But this simplification hinges on the invertibility of $\mathbf{D}'\mathbf{Z}$ and $\mathbf{Z}'\mathbf{D}$, which, recall, is only possible because the two matrices are of same dimensions. In substance, this means that we have as many endogenous as instrumental variables. If we had more instruments than endogenous variables, then we are stuck with the earlier, longer expression. Not to worry: this is where 2SLS will help us, since we can move ahead with over-identified models with this estimator.
At last, we can note that:

$$\begin{aligned}
\beta_{\text{IV}} &= (\mathbf{Z}'\mathbf{D})^{-1}(\mathbf{Z}'\mathbf{y}) \\
&= [\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D}]^{-1}[\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}] && \text{from above} \\
&= [\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D}]^{-1}\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y} && \text{multiply/divide by } \mathbf{Z}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1} \\
&= (\hat{\mathbf{D}}'\hat{\mathbf{D}})^{-1}\hat{\mathbf{D}}'\mathbf{y} \\
&:= \hat{\beta}_{\text{2sls}}
\end{aligned}$$

The last bit is the consequence of:

$$\hat{\mathbf{D}} = \mathbf{Z}\underbrace{(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D}}_{\Pi_{ols}}$$

where Π_{ols} is the OLS estimate of the effect of $\mathbf{Z}$ on $\mathbf{D}$ (recall the formula $\beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$). This corresponds to the first stage of 2SLS.

In practice, estimation happens as follows.

1. **First stage**: regress the endogenous explanatory variable (D) on the instrument (Z) and other exogenous variables to obtain the predicted values of D ($\hat{D}$) with OLS:

   $$D_i = \pi_0 + \pi_1 Z_i + \mu_i$$

   In matrix notation, this corresponds to generating esti-

mates to generate the predicted value:

$$\begin{aligned}\hat{\mathbf{D}} &= \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D} \\ &= \mathbf{Z}\Pi\end{aligned}$$

2. **Second stage**: replace D with its predicted values ($\hat{D}$) from the first stage and estimate the outcome equation with OLS:

$$Y_i = \alpha + \beta_{2sls}\hat{D}_i + \varepsilon_i,$$

with the corresponding matrix notation indicated above.

$\widehat{\beta_{2sls}}$ is a consistent estimator of β. This procedure ensures that the variation in D used to estimate its effect on Y comes only from the exogenous variation provided by Z.

7.5 Inference

Let us check what the performance of IV is with respect to our key criteria of **bias** and **consistency**. We will leave **efficiency** aside.

7.5.1 Consistency

Issue: I said that valid IVs yield **consistent** estimates, not that they yield **unbiased** estimates. Let's work through it, and let us start to check consistency. We had:

$$\beta_{IV} = \frac{Cov(Z,Y)}{Cov(Z,D)}$$

Knowing that $Y = \beta D + \varepsilon$, we can write:

$$\begin{aligned}Cov(Z,Y) &= Cov(Z, \beta D + \varepsilon) \\ &= \beta Cov(Z,D) + Cov(Z,\varepsilon)\end{aligned}$$

Thus:

$$\beta_{IV} = \beta \frac{Cov(Z,D)}{Cov(Z,D)} + \frac{Cov(Z,\varepsilon)}{Cov(Z,D)}$$

$$\beta_{IV} = \beta + \frac{Cov(Z,\varepsilon)}{Cov(Z,D)} \qquad \text{assuming that } Cov(Z,D) \neq 0 \text{ (relevance)}$$

Then, taking probability limits to assess consistency:

$$\begin{aligned} plim\hat{\beta}_{IV} &= plim\left[\beta + \frac{Cov(Z,\varepsilon)}{Cov(Z,D)}\right] \\ &= \beta + plim\left[\frac{Cor(Z,\varepsilon)}{Cor(Z,D)}\frac{sd(\varepsilon)}{sd(D)}\right] \quad \text{by def of Cov and Cor} \end{aligned}$$

Under the **exclusion restriction** assumption, $Cov(Z,\varepsilon)$ is zero and the second term disappears. As a result, we get consistency.

We can obtain the same result with matrix notation:

$$\begin{aligned} \hat{\beta}_{\text{2sls}} &= (\mathbf{Z}'\mathbf{D})^{-1}\mathbf{Z}'\mathbf{y} && \text{by definition} \\ &= (\mathbf{Z}'\mathbf{D})^{-1}\mathbf{Z}'(\mathbf{D}\beta + \varepsilon) && \text{by def of } y \\ &= \beta + (\mathbf{Z}'\mathbf{D})^{-1}\mathbf{Z}'\varepsilon && \text{distributing} \\ &= \beta + (N^{-1}\mathbf{Z}'\mathbf{D})^{-1}N^{-1}\mathbf{Z}'\varepsilon && \text{stabilizing by } 1/N \end{aligned}$$

Then 2SLS is consistent if:

$$\begin{aligned} \text{plim} N^{-1}\mathbf{Z}'\varepsilon &= 0 && \text{exclusion restriction} \\ \text{plim} N^{-1}\mathbf{Z}'\mathbf{X} &\neq 0 && \text{relevance} \end{aligned}$$

Note: we could also have assessed consistency by noting that the Wald estimator is a ratio of two sample covariances, which are consistent estimates of population covariances.

Thus: the IV estimator is consistent (under **exclusion restriction** and **relevance**).

7.5.2 Bias

However, is IV unbiased? No Andrews, Stock, and Sun (2019).

The intiution is the following (see (Keane and Neal 2023) for the full derivation of these results): when you estimate β_{2sls}, you are using $\hat{\pi}$ from the first stage instead of the true value π. But by doing so, we are inducing bias. We can show this mathematically. Like before, we have:

$$\begin{aligned} \text{Structural equation: } y &= \beta D + \varepsilon \\ \text{First stage: } D &= \pi Z + \mu. \end{aligned}$$

If D is endogenous, and if π and Z are exogenous, then the endogeneity must manifest itself via μ, and as a result $Cor(\mu, \varepsilon) \neq 0$. If $Cor(\mu, \varepsilon) = 0$, then D would actually be exogenous and we would not need IV.

If we plug in the second equation in the first, we obtain:

$$y = \beta\pi Z + (\beta\mu + \varepsilon)$$

If we knew π, we could estimate this expression with OLS, since πZ is exogenous: it is uncorrelated with μ and with ε, the composite error term.

But we don't know π and instead use the estimated $\hat{\pi}$, which must be different from π ($Pr(\pi = \hat{\pi}) = 0$). We have:

$$\begin{aligned} D &= \hat{D} + (D - \hat{D}) && \text{by definition} \\ &= \hat{\pi}Z + [(\pi Z + \mu) - \hat{\pi}Z] && \text{plugging in for all terms} \\ &= \hat{\pi}Z + [(\pi Z + \mu) - \hat{D}] \\ \underbrace{D - \hat{D}}_{\hat{\mu}} &= Z(\pi - \hat{\pi}) + \mu && \text{rearranging} \end{aligned}$$

Thus, the residual of the first stage is the error term μ plus the estimation error. What we are really estimating is (once we replace μ by $\hat{\mu}$, the residual of the first stage):

$$y = \beta\hat{\pi}Z + \underbrace{[Z\beta(\pi - \hat{\pi}) + \beta\mu + \varepsilon]}_{\text{composite error}}.$$

The composite error term is now correlated with $\hat{\pi}Z$, and thus the estimates are biased. However, IV is consistant since the difference $\pi - \hat{\pi}$ goes to zero as $n \to \infty$.

We can show this in a different way, building off the following intuition. Recall that β_{IV} is a ratio:

$$\beta_{IV} = \frac{\text{Reduced form}}{\text{First stage}} = \frac{\gamma}{\pi}$$

where π is the effect of Z (instrument) on D (endogenous treatment), and γ is the effect of Z on Y. In 2SLS, both paramaters are estimated with OLS. As we know, OLS parameters are asymptotically normally distributed. But if $\hat{\pi}$ (or, more accurately, $t_\pi = \hat{\pi}/se(\hat{\pi})$) is small, then the numerator will often be 'dangerously' close to zero. As a result, $\hat{\beta}_{IV}$ could have a very uneven distribution and $E[\hat{\beta}_{IV}] \neq \beta_{IV}$. Think of it this way: assume that $\hat{\gamma}$ is fixed at 10. Now imagine that $\hat{\pi}$ is centered at 1 with (say) a standard error of 1. Imagine you get a bunch of values for $\hat{\pi}$ from five different sample:

$$\begin{aligned}
\hat{\pi} = 1.99 &\to \hat{\beta}_{IV} = 5.02 \\
\hat{\pi} = 1.9 &\to \hat{\beta}_{IV} = 5.2 \\
\hat{\pi} = 1 &\to \hat{\beta}_{IV} = 10 \\
\hat{\pi} = 0.1 &\to \hat{\beta}_{IV} = 100 \\
\hat{\pi} = 0.01 &\to \hat{\beta}_{IV} = 1000
\end{aligned}$$

See how small values of $\hat{\pi}$ make $\hat{\beta}_{IV}$ increase massively? The values of $\hat{\pi}$ that are one standard deviation above its true value yield a $\hat{\beta}_{IV}$ that's 50% the size of the true β. But values of $\hat{\pi}$ that are almost one standard deviation below its true value yield estimates that are 10,000% too large. Thus, this distribution is not going to be nicely centered on β.[1] But as N increases, the standard errors will be more precise, and the problem disappears.

We can also visualize this in Figure Figure 7.1.

[1] Note that it may be that β_{IV} has no expected value, depending on the distribution of the numerator and denominator (Keane and Neal 2023).

```r
library(ggplot2)

# Create data for plotting, excluding pi = 0
pi_values <- seq(-1.2, 1.2, length.out = 500)
pi_left <- pi_values[pi_values < 0]   # pi < 0
pi_right <- pi_values[pi_values > 0]  # pi > 0
y_left <- 1 / pi_left
y_right <- 1 / pi_right

# Combine data into two separate segments
data_left <- data.frame(pi = pi_left, y = y_left)
data_right <- data.frame(pi = pi_right, y = y_right)

# Plot
ggplot() +
  geom_line(data = data_left, aes(x = pi, y = y),
  color = "blue", size = 1) +
  geom_line(data = data_right, aes(x = pi, y = y),
  color = "blue", size = 1) +
  geom_point(data = data.frame(pi = 1, y = 1),
  aes(x = pi, y = y), color = "red", size = 3) +
  labs(
    title = "Graph of beta.hat vs. pi.hat",
    x = "pi.hat",
    y = "beta.hat (:=gamma.hat/pi.hat)"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5)
  ) +
  coord_cartesian(ylim = c(-20, 20))
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```

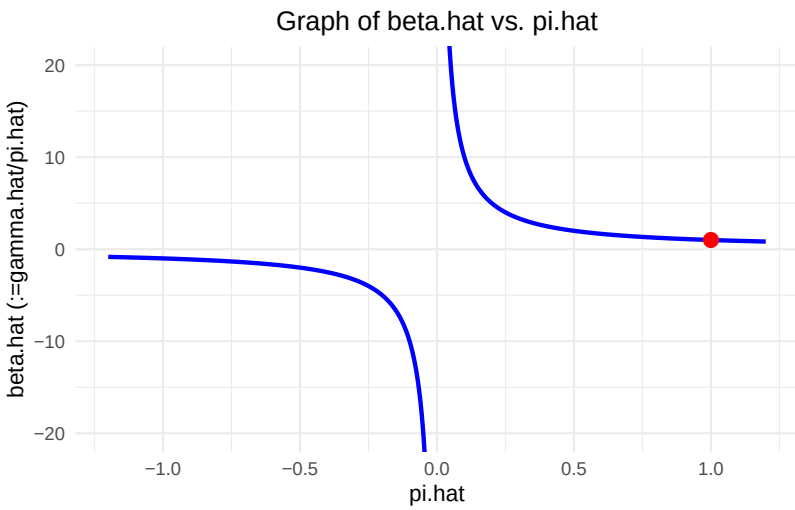

Figure 7.1: Small errors in estimation of pi can cause large errors in estimation of beta. Red dot: population value of beta if gamma=1 and pi=1. Blue lines: set of estimated values beta.hat, as pi.hat goes from -1.5 to 1.5 (fixing gamma at 1). As pi.hat gets close to zero, the size of hat.beta becomes extremely large (tending toward positive or negative infinity) and far from the true value of beta (1).

7.5.3 Standard errors and hypothesis testing

We want to do something similar to our derivation of the distribution of the OLS estimator, namely to get it closer to something we are familiar with: $\sqrt{N}(\hat{\beta}-\beta)$, which we know is *asymptotically* normally distributed.

To do so, we start with:

$$\hat{\beta}_{IV} = (\hat{\mathbf{D}}'\hat{\mathbf{D}})^{-1}\hat{\mathbf{D}}\mathbf{y}$$

Plug in the structural equation instead of $\mathbf{y}$:

$$\hat{\beta}_{IV} = (\hat{\mathbf{D}}'\hat{\mathbf{D}})^{-1}\hat{\mathbf{D}}(\mathbf{D}\beta + \varepsilon)$$

This yields:

$$
\begin{aligned}
\hat{\beta}_{IV} &= \beta + (\hat{\mathbf{D}}'\hat{\mathbf{D}})^{-1}(\hat{\mathbf{D}}\varepsilon) \\
&= \beta + \left(\frac{1}{N}\hat{\mathbf{D}}'\hat{\mathbf{D}}\right)^{-1}\left(\frac{1}{N}\hat{\mathbf{D}}\varepsilon\right) \quad \text{multiply/divide by 1/N} \\
\hat{\beta}_{IV} - \beta &= \left(\frac{1}{N}\hat{\mathbf{D}}'\hat{\mathbf{D}}\right)^{-1}\left(\frac{1}{N}\hat{\mathbf{D}}\varepsilon\right) \\
\sqrt{N}(\hat{\beta}_{IV} - \beta) &= \left(\frac{1}{N}\hat{\mathbf{D}}'\hat{\mathbf{D}}\right)^{-1}\left(\frac{1}{\sqrt{N}}\hat{\mathbf{D}}\varepsilon\right) \qquad \text{multiply by } \sqrt{N}
\end{aligned}
$$

From here:

$$
\begin{aligned}
\frac{1}{N}\hat{\mathbf{D}}'\hat{\mathbf{D}} &\xrightarrow{p} E[\hat{\mathbf{D}}'\hat{\mathbf{D}}] && \text{by the LLN} \\
\left(\frac{1}{\sqrt{N}}\hat{\mathbf{D}}\varepsilon\right) &\xrightarrow{d} \mathcal{N}(\mathbf{0}, \hat{\mathbf{D}}'\varepsilon\varepsilon'\hat{\mathbf{D}}) && \text{by the CLT}
\end{aligned}
$$

where LLN is the **law of large number** and CLT the **central limit theorem**.

And we can conclude with:

$$
\hat{\beta}_{IV} \sim \mathcal{N}(\beta, E[\hat{\mathbf{D}}'\hat{\mathbf{D}}]^{-1}E[\hat{\mathbf{D}}'\varepsilon\varepsilon'\hat{\mathbf{D}}]E[\hat{\mathbf{D}}'\hat{\mathbf{D}}]^{-1}).
$$

This quantity can be estimated by plugging in the residuals $\mathbf{u}$ for ε, just like we did for robust standard errors (Eicker-Huber-White standard errors) with OLS. From there, you can, in principle, do the usual hypothesis testing and build confidence intervals like we did with OLS.

Note: what I just wrote holds *if* the instrument(s) is strong. If not, then inference will not be valid – not just because of the bias stemming from weak instruments, but also because the standard errors will not be accurate: recent work suggests that usual tests are underpowered if the instrument is weak (Keane and Neal 2023).

In case of weak instruments, then one can use the Anderson-Rubin (robust) confidence intervals instead ((Theodore W. Anderson and Rubin 1949); for a textbook discussion, see (Stock and Watson 2011), p.472).

- Compute the residual of the endogenous regression: $u = Y - \beta D$.
- Regress u on the exogenous variables (if any) and the instrument(s).
- The AR statistic is the F-statistic on the parameters attached to the instruments.
- Invert this test: the 95% AR confidence interval is the set of β not rejected at the 5% level by the AR test.

7.5.4 Testing for instrument weakness

As we saw above, the validity of inference *also* depends on the strength of the instrument, at least with finite samples (in asymptotia, things become easier because of the precision of the instrument increases).

The strength of the instrument is measured via the F statistic of the joint effect of all instruments on the endogenous treatment(s).

Now, what is a good F statistic? People used to believe that an F statistic greater than 10 should do the trick (Staiger and Stock 1997).[2] Unfortunately, recent work suggests that you may need an F statistic that is considerably larger. A recent study suggests an F statistic closer to 104 to avoid the problems caused by weak instruments (Lee et al. 2022)… This is still a moving field, so let's see how this debate gets settled.

7.5.5 Testing for instrument exogeneity

It is not possible to test the exogeneity (exclusion restriction) assumption *if* the model is just-identified. If, however, the model is over-identified (more instruments than endogenous treatments), then we can test the exogeneity assumption via the **J-test** (for a textbook explanation, see (Stock and Watson 2011) p.448).

[2] In the special case with a single instrument and a single endogenous variable, the F statistic corresponds to the square of the t statistic. An F statistic of 10 therefore corresponds to a t statistic of about 3.2 in that case.

The intuition is the following. Suppose we have two instruments (Z_1 and Z_2) and one endogenous variable (D). Then you should be able to estimate β_{IV} with Z_1 alone, with Z_2 alone, and with both. If all instruments are exogenous, then we should (on average) not see large differences between our estimates. If we do see large differences, then presumably one instrument isn't actually exogenous.

The procedure for this test is, for m instruments:

1. First estimate the equation of interest using 2SLS and all m instruments; compute the predicted values Y , using the data D (not the $\widehat{D}$ used to estimate the second stage).
2. Compute the residuals $u_i = y_i - \widehat{y}_i$.
3. Regress u_i against instruments $Z_1, Z_2, ...$ and other exogenous (but not instrumental) variables $\mathbf{W}$.
4. Compute the F statistic testing the hypothesis that the coefficients on $Z_1, Z_2, ...$ are all zero (if not: then at least one of your instrument is endogenous).
5. The J-statistic is $J = mF$. J has a X^2 distribution with $m - k$ degrees of freedom (number of instruments minus number of endogenous variables).

7.6 Heterogenety and LATE

When we discussed RCTs (see notes on RCT), we had brought up two notions:

1. Treatment effects might vary by subgroups instead of being a universal constant.
2. Units in the treatment group might decide not to take the treatment, and units in the control group might decide not to stay in the control group. Units that don't comply were called "non-compliers."

One natural question is: does the treatment effect vary by groups based on compliance?

To give a motivating example, imagine that the treatment is a TV ad and we have a sample of 1,000 people. Let's say that half is randomly allocated to the treatment, namely watching

an ad for 10 minutes per day. Whether people watch or not is beyond our control: some might close their eyes, not listen, be mentally checked out. Likewise, people in the control group might 'cheat' and watch TV anyway.

The good news is that IV can be used to estimate the **local average treatment effect** (LATE): the ATE for compliers.

To refresh your memory:

- Z is the assignment to treatment (i.e. whether unit i is in the treatment or the control group).
- D is the treatment (i.e. whether unit i too the treatment).
- For instance, Z could be the assignment to receive an aspirin or not, and D is whether individuals actually take an aspirin or not.

Then we have four cases:

- A **complier** is a unit i for which: (a) $Z = 1 \rightarrow D = 1$, whereas (b) $Z = 0 \rightarrow D = 0$. We write $D(1) - D(0) = 1$.
- A **defier**: (a) $Z = 1 \rightarrow D = 0$, whereas (b) $Z = 0 \rightarrow D = 1$. We write $D(1) - D(0) = -1$.
- An **always-taker**: (a) $Z = 1 \rightarrow D = 1$, and (b) $Z = 0 \rightarrow D = 1$. We write $D(1) - D(0) = 0$.
- A **never-taker**: (a) $Z = 1 \rightarrow D = 0$, and (b) $Z = 0 \rightarrow D = 0$. We write $D(1) - D(0) = 0$,

where $D(Z)$ is the value of D for $Z = 0$ or $Z = 1$.

We can now see more clearly what is going on (Joshua D. Angrist, Imbens, and Rubin 1996). From above, we had:

$$\beta_{IV} = \frac{E[Y|Z = 1] - E[Y|Z = 0]}{E[D|Z = 1] - E[D|Z = 0]}$$

We can work through the earlier expression. Let's go bit by bit and start with the denominator. Since D is a dummary variable (as is Z), we can rewrite the expectations as probabilities $P()$:

$$E[D|Z = 1] = P(D|Z = 1) \quad \text{and} \quad E[D|Z = 0] = P(D|Z = 0)$$

We can thus also write:

$$\beta_{IV} = \frac{E[Y|Z=1] - E[Y|Z=0]}{P(D|Z=1) - P(D|Z=0)}.$$

Here it gets a tiny bit technical, so bear with me. $P(D|Z=1)$ is the proportion of units taking the treatment when $Z=1$ and $P(D|Z=0)$ is the proportion of units taking the treatment when $Z=0$.

Here, we will make the simplifying assumption that we don't have defiers (which are kind of a weird bunch of units anyway…). We call this the **monotonicity assumption**: $D_i(Z=1) - D_i(Z=0) \geq 0$ for all units i. All units become more likely to take the treatment if they are assigned to it.

As a result, we find that the share of compliers is:

$$P(D|Z=1) - P(D|Z=0) = \text{Share of compliers}$$

The reason is that always-takers don't matter here: the probability of taking the treatment is invariant in Z. Thus, in the absence of defiers, this difference in probability is entirely driven by compliers.

Then we realize that the denominator of the IV is the same as the share of compliers:

$$\beta_{IV} = \frac{E[Y|Z=1] - E[Y|Z=0]}{\text{Share of compliers}},$$

where we assume that the share of compliers is strictly positive (which is the same as the relevance assumption).

Now, about the numerator: it's the effect of our assignment to treatment (Z) on the outcome, ignoring whether units complied or not. We encountered this as well: it's the definition of the **intent to treat effect** (ITT). Thus:

$$\beta_{IV} = \frac{\text{ITT}}{\text{Share of compliers}}.$$

Notice that if everyone complies (share of compliers equals 1), then LATE = ITT. This makes sense: our intent-to-treat

ended up treating everyone we wanted, and thus evyerone complied, and thus whether I run a regression with Z or D doesn't matter (the two are perfectly correlated).

Conversely, if a small number of people comply (share of compliers smaller than 1), then ITT $< \beta_{IV}$. This also makes sense: the ITT gets weakened by the fact that some units didn't comply and they didn't actually get treated. In our earlier example, it corresponds to the case where people in the treatment group don't take aspirin. As they won't be affected by it, they will not experience the boost in well-being that those who take the aspirin (the compliers) get.

Appendix

Code for the IV bias figure

```
library(ggplot2)

# Create data for plotting, excluding pi = 0
pi_values <- seq(-1.2, 1.2, length.out = 500)
pi_left <- pi_values[pi_values < 0]   # pi < 0
pi_right <- pi_values[pi_values > 0]  # pi > 0
y_left <- 1 / pi_left
y_right <- 1 / pi_right

# Combine data into two separate segments
data_left <- data.frame(pi = pi_left, y = y_left)
data_right <- data.frame(pi = pi_right, y = y_right)

# Plot
ggplot() +
  geom_line(data = data_left, aes(x = pi, y = y), color = "blue", size = 1) +
  geom_line(data = data_right, aes(x = pi, y = y), color = "blue", size = 1) +
  geom_point(data = data.frame(pi = 1, y = 1), aes(x = pi, y = y), color = "red", size = 3) +
  labs(
    title = "Graph of beta.hat vs. pi.hat",
    x = "pi.hat",
    y = "beta.hat (:=gamma.hat/pi.hat)"
```

```
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5)
  ) +
  coord_cartesian(ylim = c(-20, 20))
```

8 Panel data and fixed effects models

This version: December 16, 2025

> ⚠ **Caveat**
>
> These notes are offered in support of *Introduction to Econometrics* (MGT 581). They were written to help students understand the logic behind important topics in econometrics. But they don't (always) replace the insights and depth that textbooks can provide. This is why we connect each session to chapters of our core textbooks. See the syllabus for more details.

8.1 Readings

People interested in panel data may refer to (Wooldridge 2002) (a comprehensive textbook), (A. Colin Cameron and Trivedi 2005) (ch. 21-22) (another textbook), (Morgan and Winship 2014) (ch.11) (causal inference with repeated observations), (J. Angrist and Pischke 2008) (ch.5), (B. E. Hansen 2024) (ch.17). For difference-in-differences, see Baker et al. (2025).

8.2 Baseline bias

You may recall that we tried to understand what, if anything, we can learn from a naïve difference in means of Y between a treatment and a control group. We worked out the following

result:

$$
\begin{aligned}
\text{Naïve difference} =& \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0] \\
=& \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 1] \\
&+ \mathbb{E}[Y_i(0)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0] \quad \text{adding and subtracting} \\
=& \mathbb{E}[Y_i(1) - Y_i(0)|D_i = 1] \\
&+ \mathbb{E}[Y_i(0)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0] \qquad \text{regrouping} \\
\equiv& \text{ATT} + \text{Selection (baseline) bias}
\end{aligned}
$$

Here, I want to focus a little bit on what this **baseline bias** means. The idea is pretty simple: it might be misleading to compare $\mathbb{E}[Y|D = 1]$ to $\mathbb{E}[Y|D = 0]$ (or rather, their sample analogues). The reason is that we are missing counterfactuals. Consider my treatment group. I don't see what Y would be for these units *had they not been treated*. The naïve difference is only informative if $\mathbb{E}[Y|D = 0]$ among the control group (which I do see) can be used as a counterfactual. If not, then my naïve difference will contain the effect of the treatment on Y for the treatment group plus the difference in baseline between the two groups.

Let me work out a simple example. Suppose that D is access to ChatGPT (0 or 1) and Y is a student's grade. We then have (I'm inventing numbers):

- $\mathbb{E}[Y|D = 1] = 5.3$. On average, students using ChatGPT obtain a grade of 5.3/6.
- $\mathbb{E}[Y|D = 0] = 4.5$. On average, students not using ChatGPT obtain a grade of 4.5/6.

The naïve estimate of the ATE is thus 0.8. But is this accurate? Maybe, but only if students who used ChatGPT would also have had 4.5 on average without ChatGPT. Now, we could imagine a world in which students most tempted to use ChatGPT are those who are struggling in class. Maybe their baseline without ChatGPT might have been a 4.2/6. In that case, our naïve estimate understates the effect of ChatGPT (naïve

estimate < ATT):

$$\begin{aligned}\text{ATT} &= 5.3 - 4.2 = 1.1\\ \text{Baseline bias} &= 4.2 - 4.5 = -0.3\\ \text{Naïve estimate} &= \text{ATT} + \text{Baseline bias} \Rightarrow 1.1 - 0.3 = 0.8\end{aligned}$$

(Or maybe it's the opposite: students who are smarter in the first place are quicker to learn how to use a new tech like ChatGPT. In such case, the naïve estimate would overstate the effect of ChatGPT (naïve estimate > ATT). This would be closer to the image in Figure 1 below.)

Can be get rid of the baseline bias? We don't observe it directly, since one of its constituent term is an unobservable counterfactual (we don't see how the treated units would have behaved without treatment). We saw in earlier notes that RCT equalize the two quantities (in expectation) by randomizing treatment allocation. But we also saw that you can't always used RCTs (whether for ethical or practical reasons).

Fortunately, there is another solution – assuming we can access temporal data with repeated observations.

8.3 Temporal data

Until now, we had models such as:

$$y_i = \beta D_i + \varepsilon_i,$$

where i indicates units of analysis, such as firms, consumers, voters, countries, etc. So far, we could limit ourselves to a single subscript, because i was uniquely identifying each row. We call this **cross-sectional** data .

Sometimes, though, we might be able to get access to **temporal data**. For instance, we could be modeling GDP data in Switzerland. In the simplest case, rows aren't identified by unit (here: Switzerland). Instead, they are identified by time. We talk then of a **time series** data structure . We then would write:

$$y_t = \beta D_t + \varepsilon_t,$$

where t denotes time (days, months, etc.). An example of time dataset, which is entirely structured by time is a year-quarter dataset (here: for Switzerland alone):

Ctry (i)	**Year (t)**	**Y**	**D**	**W**
Switzerland	2000Q1	13	0	6
Switzerland	2000Q2	12	1	7
...	...	...	...	...
Switzerland	2020Q4	17	1	6

Table 8.1: **Time series data structure**. Here: a year-quarter dataset for Switzerland.

Sometimes, we have variation both across units and across time. Such data is typically called **panel data** or **longitudinal data** . Such a data is structured by repeated observations of the same units (e.g., a set of 100 consumers who receive a survey every quarter).[1]

Units i nor time periods t uniquely identify rows anymore; instead, it is both that jointly identify each row:

$$y_{i,t} = \beta D_{i,t} + \varepsilon_{i,t}.$$

In our matrix notation, we now have:

$$\mathbf{y}_{i,t} = \begin{bmatrix} y_{1,1} \\ y_{1,2} \\ \vdots \\ y_{n,t} \end{bmatrix}_{(nt)\times 1} \qquad \mathbf{X}_{i,t} = \begin{bmatrix} 1 & D_{1,1} & x_{1,1} & \cdots & w_{1,1} \\ 1 & D_{1,2} & x_{1,2} & \cdots & w_{1,2} \\ 1 & D_{2,1} & x_{2,1} & \cdots & w_{2,1} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & D_{n,t} & z_{n,t} & \cdots & w_{n,t} \end{bmatrix}_{(nt)\times k} \qquad \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \vdots \\ \beta_k \end{bmatrix}_{k\times 1} \qquad \varepsilon = \begin{bmatrix} \varepsilon_{1,1} \\ \varepsilon_{1,2} \\ \vdots \\ \varepsilon_{n,t} \end{bmatrix}_{(nt)\times 1}$$

Thus, $\mathbf{X}$ is a stacked matrix where each row is a unique combination $i \times t$. For instance, we may have the following dataset.

[1] Sometimes, you might come across studies talking about **cross-section times series**. This is almost identical to a panel dataset, except that the units can change from time period to time period. For instance, think of a dataset of countries with average consumer confidence scores: we have repeated data for Switzerland, France, etc., but who is surveyed might change every year.

Note how each row is uniquely defined by a country-year. The matrix $\mathbf{X}$ stacks vectors for each country-year, such that $\mathbf{X}$ ($\mathbf{X}$ is constituted of stacked X_i').

Country	**Year**	$\mathbf{Y_{i,t}}$	$\mathbf{D_{i,t}}$	$\mathbf{W_{i,t}}$
Switzerland	2000	3	1	6
Switzerland	2001	2	0	6
...	...	...	...	...
Switzerland	2020	5	1	5
France	2000	15	0	4
France	2001	12	0	3
...	...	...	...	...
Zambia	2020	4	1	5

Table 8.2: **Cross-section time-series data structure**. Here: a country-year dataset.

Note that the size of out matrices increases, though the number of parameters remains the same as before.

Here's the thing. Adding time as a structure in our dataset can have three implications.

1. It can make some analysis unreliable if we are not careful. Most straightforwardly: we often assumed independence of error terms ε; this is unlikely to hold in a temporal setting.
2. We can leverage time to implement **quasi-experimental designs** . This will be the case of difference-in-differences and of fixed effects models.
3. We can leverage time to model **dynamics**. For instance, we can ask interesting questions such as: "how quickly does an increase in interest rates affect bank lending?" We obviously couldn't do this with cross-sectional data.

8.4 Difference-in-differences (DiD)

8.4.1 DiD intuition

Consider Figure 8.1. This represents the problem of baseline bias as stated above. We can do a naïve estimate of the treatment effect by comparing the red (T) minus the blue (C) points. However, we don't know where the units in the treatment group would have ended up if they hadn't been treated ($\mathbb{E}[Y(0)|D=1]$ in orange). The treatment effect (at least on the treated) is given by the vertical space between the red dot and the orange dot. But of course, the orange dot is not observable.

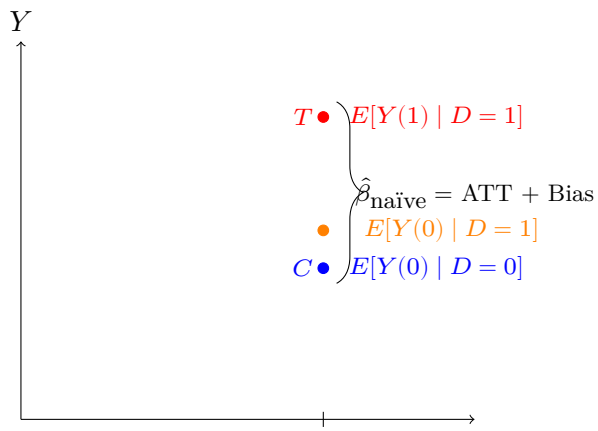

Figure 8.1: Baseline bias with cross-sectional data. Red and blue points are observable average outcomes for the treatment (red) and control (blue) groups. Orange is the unobservable counterfactual for the treatment group, had it not been treated.

Thus the question is: is there a way to estimate the gap between the red dot and the lower black dot? If yes, we could then compute:

$$\text{ATT} = \underbrace{[\mathbb{E}[Y(1)|D=1] - \mathbb{E}[(Y(0)|D=0)]]}_{\text{naïve}} - \underbrace{[\mathbb{E}[Y(0)|D=1] - \mathbb{E}[Y(0)|D=0]]}_{\text{baseline diff}}$$

The intuition of the **difference-in-differences** (**DiD**) estimator is the following: if we have data pre-treatment (at time

$t-1$), we could use the difference in $\mathbb{E}[Y]$ between the two groups then as a reasonable guess for that baseline difference (Lechner, n.d.). We could then use this difference to get rid of the baseline difference. This idea is captured by Figure 8.2.

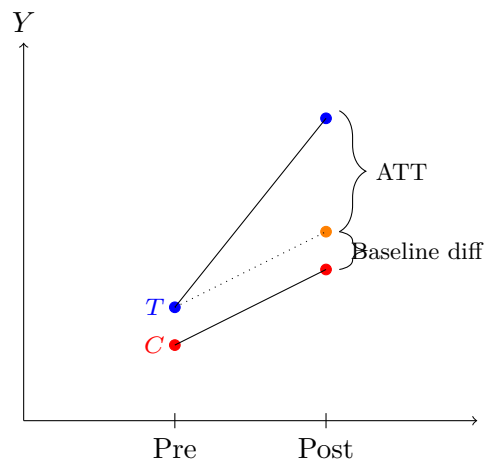

Figure 8.2: Difference-in-difference estimator.

Another way to describe DiD is as the difference of two comparisons:

1. Treated and untreated: as we saw, this is misleading for the ATT if there is a different baseline.
2. Before and after: this is misleading if other variables affect the trend of Y

Instead, by combining the two, DiD removes each source of errors.

Typically, DiD will be useful when:

1. The treatment occurs as an exogenous shock, but we worry that the treatment and the control group were different prior to the shock already.
2. We have panel data both pre- and post-treatment.

8.4.2 Estimation

We start with the simplest case. We have two time periods $T \in \{0, 1\}$. Treatment is assigned between the two periods. We refer to "pre" as $t = 0$ and "post" as $t = 1$. There is an indicator for each unit whether it is in the treatment or the control group. Call this $D \in \{0, 1\}$ (where 1 denotes treatment). Note that this indicator is time-invariant: it simply flags each observation in the dataset.

We can then define the ATT from the DiD framework as:

$$\text{ATT}_{\text{DiD}} = [\mathbb{E}[Y|D = 1, t = 1] - \mathbb{E}[Y|D = 0, t = 1]] - [\mathbb{E}[Y|D = 1, t = 0] - \mathbb{E}[Y|D = 0, t = 0]]$$

There are two ways to estimate DiD. One is to plug in averages and take the differences:

$$\begin{aligned}\widehat{\text{ATT}_{\text{DiD}}} = &\left[\frac{1}{N_{D=1}}\left[\sum Y_i|D = 1, t = 1\right] - \frac{1}{N_{D=0}}\left[\sum Y_i|D = 0, t = 1\right]\right] \\ &- \left[\frac{1}{N_{D=1}}\left[\sum Y_i|D = 1, t = 0\right] - \frac{1}{N_{D=0}}\left[\sum Y_i|D = 0, t = 0\right]\right]\end{aligned}$$

There are three reasons why we might want to estimate a linear model with OLS instead:

1. We might want to add additional covariates in the model.
2. In some cases, we might have differences in timing of treatment and more than 2 time periods. See section on fixed effects models below.
3. We might want to use more complicated standard errors (including clustered standard errors; see below).

With a linear model, can then write the DiD model to estimate the ATT as:

$$y_{i,t} = \alpha + \gamma D_{g(i)} + \tau T_t + \beta D_{g(i)} T_t.$$

D is a time-invariant dummy and takes value 1 if unit i is in the treatment group g, 0 otherwise. T is a cross-sectionally-invariant dummy that takes value 1 in the post-treatment period and 0 in the pre-treatment period. The interaction $D_{g(i)} T_t$ can vary both over time and across units.

Note the interpretation of the parameters:

- $\alpha = \mathbb{E}[Y|D=0, \text{pre}]$: $\bar{Y}$ for untreated, pre-treatment
- $\alpha + \tau = \mathbb{E}[Y|D=0, \text{post}]$: $\bar{Y}$ for untreated, post-treatment
- $\alpha + \gamma = \mathbb{E}[Y|D=1, \text{pre}]$: $\bar{Y}$ for treated, pre-treatment
- $\alpha + \gamma + \tau + \beta = \mathbb{E}[Y|D=1, \text{post}]$: $\bar{Y}$ for treated, post-treatment

So what D really does is to estimate the gap between T and C pre-treatment (the baseline bias). What T does is to estimate the change that happened between the two time periods (the red line in Figure 8.2 isn't flat). And the interaction effect recovers an estimate of the ATT :

$$\widehat{\text{ATT}} = \hat{\beta}$$

To see this:

$$\begin{aligned}
\text{Naïve estimate } &= \mathbb{E}[Y|D=1, \text{post}] - \mathbb{E}[Y|D=0, \text{post}] \quad \text{by definition} \\
&= [\alpha + \gamma + \tau + \beta] - [\alpha + \tau] \\
&= \gamma + \beta
\end{aligned}$$

Then:

$$\begin{aligned}
\text{Baseline bias } &= \mathbb{E}[Y|D=1, \text{pre}] - \mathbb{E}[Y|D=0, \text{pre}] \\
&= [\alpha + \gamma] - \alpha \\
&= \gamma
\end{aligned}$$

Recall that:

$$\text{Naïve estimate} = \text{ATT} + \text{Baseline bias},$$

which implies that the ATT is the naïve estimate minus the baseline bias. We thus get:

$$\begin{aligned}
\text{ATT} &= \text{Naïve estimate} - \text{Baseline bias} \\
&= [\gamma + \beta] - \gamma \\
&= \beta
\end{aligned}$$

The DiD model can be estimated with least squares. Note that it is a **saturated model** : there is a combination of parameters

for each combination of independent variables (here: D and T). As a result, we don't have to worry about linearity.

As for data, here is the structure of a 2-period, municipality-based dataset:

Country	**Year**	$\mathbf{Y_{i,t}}$	$\mathbf{D_{g(i)}}$	$\mathbf{T_t}$	$\mathbf{D_{g(i)}T_t}$
New York City	2019	3	1	0	0
New York City	2020	2	1	1	1
Pittsburgh	2019	5	0	0	0
Pittsburgh	2020	15	0	1	0
...	...	...	...	...	...
Philadelphia	2019	4	1	0	0
Philadelphia	2020	2	1	1	1

Table 8.3: **DiD data structure**. Here: a city-year dataset, with two periods, time-invariant treatment status, and time-variant interaction effect. New York and Philadelphia are in the treatment group; Pittsburgh isn't.

Illustration

Example 8.1 (Minimum wage)**.** Card and Krueger (Card and Krueger 1994) are often credited to develop the DiD framework. They wanted to study the effect of a minimum wage increase (D) on employment rate (Y) (measured as number of full time-equivalence (FTE) per restaurant). Their design was the following:

- They had data on 410 fast-food restaurants for two US neighboring states: New Jersey and Pennsylvania
- New Jersey experienced an increase in minimum wage from \$4.25 to \$5.05 in 1882.
- Pennsylvania did not experience a minimum wage change during that time.

Thus, we have two groups: 331 restaurants in the treatment group (i.e., in New Jersey) and 79 in the control group (Pennsylvania). We don't necessarily expect the two to have the same level of employment per restaurant pre-treatment (baseline bias). Maybe New Jersey restaurants employ more people on average than restaurants in Pennsylvania. Thus a comparison after the change in minimum wage would not be informative. Instead, a DiD might be particularly suited.
The data they had included the average number of full-time employees per restaurant. From their paper, we can identify each component of the analysis.

	$\bar{Y}_{\text{PA}}$ (D=0)	$\bar{Y}_{\text{NJ}}$ (D=1)	**Difference**
Pre-treatment	23.33	20.44	-2.89
Post-treatment	21.17	21.03	-0.14
$\widehat{\textbf{ATT}}$			2.75

We can reconstruct the ATT as:

$$\widehat{\text{ATT}} = (21.03 - 21.17) - (20.44 - 23.33) = 2.75$$

Thus, minimum wage seems on average to have slightly *increased* employment (instead of decreasing it, as many feared).
You can already guess what some of the identifying assumptions will be. Here, we must be sure that the minimum-wage hike in New Jersey had no effect on Pennsylvania. We must also ponder whether Pennslyvania is a good counterfactual for New Jersey. This will lead us to the **parallel trend** assumption. Finally, we must also be sure that New Jersey's restaurants didn't fire people *before* the law came into force – the problem of **anticipation**.

8.4.3 Identifying assumptions

Figure 8.2 makes also very obvious what one of the key identifying assumption of the difference-in-difference estimator is: for the gap between treatment and control pre-treatment to be a valid estimate of the baseline difference, it must be that the two groups would have evolved in a similar way if no treatment had been assigned. This is the dotted line in Figure 8.2. Formally, define the baseline bias:

$$\Delta_{D,post} := \mathbb{E}[Y(0)|D = 1, \text{post-T}] - \mathbb{E}[Y(0)|D = 0, \text{post-T}]$$

I use Δ as the difference operator. Here, it's the difference across group, post-treatment.

We then estimate $\Delta_{D,post}$ with $\widehat{\Delta}_{D,pre}$, that is:

$$\widehat{\Delta}_{pre} = \mathbb{E}[Y(0)|D = \widehat{1, \text{pre-treatment}}] - \mathbb{E}[Y(0)|D = \widehat{0, \text{pre-treatment}}]$$

Finally, we compute two sets of differences (hence the name):

$$\text{Population ATT} = \underbrace{\mathbb{E}[Y(1)|D = 1] - \mathbb{E}[Y(0)|D = 0]}_{\text{naïve diff}} - \Delta_{D,post}$$
$$\widehat{ATT} = \mathbb{E}[Y(\widehat{1)|D} = 1] - \mathbb{E}[Y(\widehat{0)|D} = 0] - \widehat{\Delta}_{D,pre}$$

Thanks to the longitudinal data, both of the right-hand side quantities are observable. But that gap is a valid estimate as long as it doesn't change over time. In other words, Δ_D has to be a constant. Or, put another way, the two quantities must co-move for us to use it as a way to get rid of the baseline bias. We call this the **parallel trend assumption**:

$$\mathbb{E}[\Delta_t Y(0)|D = 1] = \mathbb{E}[\Delta_t Y(0)|D = 0],$$

that is, the change in Y over time (Δ_t) is the same for the treatment and the control group, on average.

The parallel trend assumption cannot be tested when you have data for a single pre-treatment period. However, if we do have multiple pre-treatment periods, we can in principle see whether this assumption is plausible. To be clear: we cannot actually

test it, because we will never observe the counterfactual of the treatment group had it not be treated. But if we have multiple pre-treatment periods, and if Δ remains roughly constant throughout, then we might be reassured enough to project this difference in the future.

Another assumption that we need is the absence of **anticipation effect**: Y shouldn't change prior to the assignment of D. Here's a motivating example. As an analyst, you want to model household spending as a function of wages. Suppose that you are told that a bunch of salary of a bunch of people in your sample will increase next year. These people might start spend more now. But in a DiD framework, the treatment only starts in January of next year, yet it has already had an effect in months prior.

8.5 One-Way and Two-Way Fixed Effects models

8.5.1 Fixed effect models

The canonical DiD uses two groups (treated versus control) and two time periods (pre and post-treatment). We can generalize this in two ways:

- We can use multiple time periods $t = 1, 2, ...$, if data are available.
- We can define groups in a more fine-grained way.

Concretely, we can write:

$$y = \kappa_i + \tau_t + \beta D_{i,t} + \varepsilon_{i,t}.$$

> **Note**. κ_i is short for $\sum_i^n \kappa_i$, a set of n dummy indicators (where n is the number of units, not the total dataset). For instance, κ_1 would equal 1 for all observations from Switzerland, and zero otherwise. Next, κ_2 would equal 1 for observations from France, and zero otherwise. We can then include as many parameters κ's as there are units, at least as long as we don't include a y-intercept.

(If we want a y-intercept, we need to suppress one κ – otherwise we would have perfect multicollinearity. The suppressed unit would then serve as baseline and be captured by α, the y-intercept ($\mathbb{E}[Y|\text{baseline group}] = \alpha$).)

This is different from the standard DiD model, where we had a *single* time-invariant indicator $D_{g(i)}$ that was equal to 1 for *all* units in the treatment group. In the generalized model, we have n (or $n-1$) indicators.

In the same spirit, we have t dummy indicators τ for each time period. If our dataset is structured as country-years, with years from 2010 to 2020, we would have an dummy indicator for 2010 (equal 1 across all countries for observations measured in 2010), another for all observations in 2011 (equal 1 across all countries for observations measured in 2011), etc.

An illustrative dataset is reproduced here.

Ctry	**Year**	$\mathbf{Y_{i,t}}$	$\mathbf{D_{i,t}}$	**SWI**	**FRA**	**2019**	**2020**	**2021**
Swi	2019	3	0	1	0	1	0	0
Swi	2020	2	1	1	0	0	1	0
Swi	2021	5	1	1	0	0	0	1
Fra	2019	15	0	0	1	1	0	0
Fra	2020	12	0	0	1	0	1	0
Fra	2021	12	0	0	1	0	0	1
...	...	...	...	...	...	...	...	...

Table 8.5: **TWFE data structure**. Here: a country-year dataset, with three periods, with country and year dummies. Switzerland is treated in 2020 and 2021. France is never treated.

Finally, note that D is now varying over time and over units. As such, it plays the role of the interaction term in the standard DiD model (where we interacted a time-invariant $D_{g(i)}$ with a unit-invariant T_t).

With this hypothetical country-year data, a TWFE model would be something like:

$$\begin{aligned} y_{i,t} = \beta D_{i,t} + \kappa_1 \mathbb{I}(\text{CH})_i + \kappa_2 \mathbb{I}(\text{FRA})_i \\ + \tau_1 \mathbb{I}(2019)_t + \tau_2 \mathbb{I}(2020)_t + \tau_3 \mathbb{I}(2021)_t + \varepsilon_{i,t}, \end{aligned}$$

where $\mathbb{I}$ is an indicator operator (ie, a dummy that equals yes if the condition is met). (Note the absence of intercept. Why?)

We call this generalized model the **(two-way) fixed effects** model (or **TWFE**) . The "two-ways" refers to the two sets of paraterms: n parameters (or fixed effects) for units and t parameters (fixed effects) for time. And in the case in which D is a dichotomous treatment, the TWFE will estimate the DiD ATT.

Aside from the fact that there might a lot of parameters κ and τ (for instance: 150 countries measured over ten years would mean 160 such parameters), there is nothing magical about them. These are dummies like other dummies we saw before. However, they get at something interesting from a design perspective. Here's the key: when you include a fixed effect, then the identification of β stems from variation coming from *within* the fixed effect.

To see what this means, consider a country-year dataset and a two-way fixed effects model. Focus on the country fixed effects. As stated, these will be a series of dummy indicators for each country (Switzerland, France, Germany, etc.). What this means for β is: $\widehat{\beta}$ will be based on changes in D *within* countries rather than variation in D *across* countries. Think of this from a counterfactual perspective: when I introduce country fixed effects, I will no longer compare Y between Switzerland and France. Instead, I will compare Switzerland when it's in the control group to Switzerland when it's in the treatment group. Then I'll do the same in France, in Germany, etc. The resulting β will be a weighted comparison of these *within-unit* changes in Y caused by changes in D.

To show what kind of payoff can be obtained from within-variation, consider Figure 8.3, with a continuous treatment x and two units (A, B) measured at three levels of x. For instance, think here of each unit as a customer and y the amount

they spend after watching x minutes of ads. The figure depicts **Simpson's paradox**: if we pool all observations without fixed effects, then the estimated treatment effect is negative (black dotted line). However, it's clear that *within* units, the treatment effect is positive.

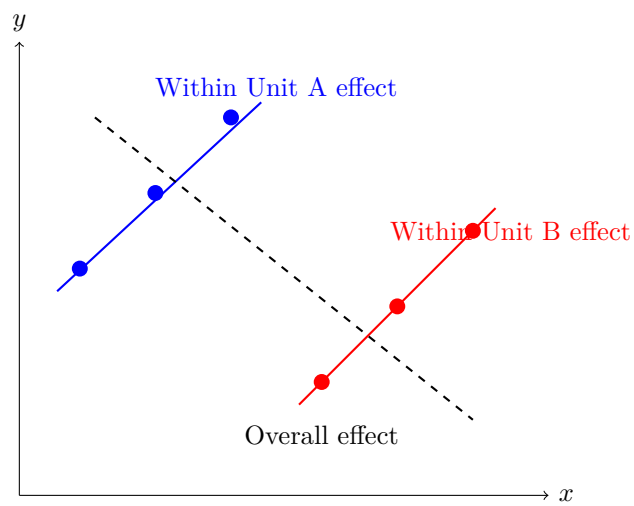

Figure 8.3: Simpson's paradox. Here: each unit has its own y-intercept (i.e., fixed effect), which changes the slope of the treatment effect from negative (dotted line) to positive (red and blue lines).

The dotted black line is what you obtain if you estimate:

$$y_i = \alpha + \beta x_i$$

If you use a (one-way) fixed effect model, instead, you'd estimate (more on one-way FE models in a second):

$$y_i = \phi_i + \gamma x_i.$$

What you would find is that $\hat{\gamma} > 0$ (the slope of the red and blue lines) whereas $\hat{\beta} < 0$ (the slope of the dotted line). The FE model is intuitively more convincing in general. Think of it this way: imagine you were asked to guess what the treatment effect would be for a new 3rd unit C. Would you put money on the treatment effect to be positive or negative?

What's very nice about this is that the fixed effects do what we were looking for in DiD: they remove the baseline differences across units (i.e., they give each unit a different y-intercept). In contrast to the canonical DiD, where we only wanted to remove the differences between treatment and control, here we will remove the differences between each unit (eg, countries).

Likewise, time fixed effects will remove time-specific effects. For instance, if all countries in a given year suffer from a common shock, this shock will be absorbed by the τ for this year.

Note that we could also estimate a **one-way fixed effect model**:

$$y_{i,t} = \beta D_{i,t} + \kappa_i + \varepsilon_{i,t}.$$

We would still be relying on within-unit variation to estimate β (since we include unit fixed effects, i.e., n parameters κ_i). However, this specific model would not take into account time fixed effects. Thus, we wouldn't know if the results are driven by something correlated with time and with our treatment. We could try to get at this by including a linear or nonlinear parameter(s) for time – this would capture **time trends**. For instance, here's a one-way fixed effect model that adjusts for a quadratic time trend:[2]

$$y_{i,t} = \beta D_{i,t} + \tau_1 T + \tau_2 T^2 + \kappa_i + \varepsilon_{i,t}.$$

But as a general rule, it's hard to justify using time trends (linear or not) over time fixed effects. Maybe the most convincing reason is if you want to do **forecasting**: a time fixed effect is not predictable (since it captures something stochastic about time). Thus, if you want to make a forecast for 2027, you won't be able to account for time trends. However, with a time trend, one can make out-of-sample forecasts that account for time.

Another nice feature of panel data is that we can compute the change in the treatment effect by year. Imagine we have t from -2 to 3, with treatment happening homogenously between $t = 0$ and $t = 1$. We could estimate:

$$\begin{aligned} y_{i,t} = \alpha &+ \beta_1 D_{i,t}\mathbb{I}(t=1) + \beta_2 D_{i,t}\mathbb{I}(t=2) + \beta_3 D_{i,t}\mathbb{I}(t=3) \\ &+ \tau_2 + ... + \tau_5 + \kappa_1 + ... + \kappa_n + \varepsilon_{i,t} \end{aligned}$$

[2]Instead of a counter T, you could also use years (eg $\tau_1 \text{Year}_t + \tau_2 \text{Year}_t^2$).

We interact the treatment variable with an indicator for time periods. Thus, β_1 gives us the treatment effect in the first post-treatment period, β_2 for the second period, and β_3 for the last period.

Doing so relaxes the assumption that treatment effects have to be immediate and constant over time. For instance, imagine the effect of a new patent on a firm's profit: it won't jump immediately and forever to a new level Y. Instead, it will gradually increase profits over time.

8.5.2 Complications with FE and TWFE models

8.5.2.1 Time-invariant treatments

The most obvious shortcoming of TWFE is its inability to estimate parameters of time-invariant variables. Intuitively, it's pretty clear why: since we are relying on *within-unit* variation, and since such a variable does not exhibit within variation, it cannot be estimated. Another way to see this is to note that a time-invariant parameter will be perfectly correlated with unit fixed effects. Likewise (but that's rarer), a treatment that is assigned to all units in the same year cannot be estimated either.

Thus, important treatments cannot be estimated with TWFE. For instance, imagine a model trying to estimate whether a country has access to the sea as a predictor of its growth rate (landlocked countries might struggle to capitalize on trade). Assuming no change in borders, a country either has access to the sea or it doesn't. And thus, a variable $D_i \in \{\text{landlocked, sea access}\}$ could not be estimated using a TWFE model (or even a single unit FE model).

8.5.2.2 Staggered treatment and heterogeneous treatment effects

Worse, in the last few years, several papers raised concerns about the validity of the TWFE model as a way to estimate the DiD estimator (Goodman-Bacon 2021; Imai and Kim

2021; Callaway and Sant'Anna 2021; De Chaisemartin and d'Haultfoeuille 2020; Arkhangelsky et al. 2024; Baker et al. 2025). The concerns (and solutions) vary, but one of the problem relates to the staggered timing of treatment as well as its heterogeneity (ie, the notion that the treatment effect may change across units or over time).

Let's start with the following case. Imagine that you have a set of $N = 10$ units and $t = 10$ time periods. Suppose that 4 units get treated at time $t = 2$, 4 units get treated at time $t = 6$, and 2 units never get treated. Thus, we have three groups:

- A: Never treated
- B: Treated at time $t = 2$
- C: Treated at time $t = 6$

The TWFE/DiD model will combine four types of comparisons:

- B with A at $t = 2$
- B with C at $t = 2$.
- C with A at $t = 6$
- C with B at $t = 6$.

This is the 'simple' case with two different treatment slots. You could easily imagine that the treatment could happen any time in your data (for instance, some countries might be treated at in 2000, others in 2001, others in 2003, etc.). Then, the TWFE model yields a weighted combination of all 2x2 comparisons possible (Goodman-Bacon 2021).

There are several problems with this. First, some comparisons don't necessarily make sense. In the example above, is group B really a good comparison group for C at time $t = 6$? By that time, B has already been treated. Thus, these comparisons could be weird.

The bigger problem is that these comparisons are only valid if the treatment effect is constant over time. That is: once a group of units are treated, their average converges immediately to the new value of Y and stays there. This is often unlikely: many treatments will take time to have an effect. Drugs may necessitate months or years to change health. A change in

Central Bank interest rate might take months before having an effect on real estate investments. Etc.

Time-varying treatment effects – a form of heterogeneity – will create problems in the kind of comparisons shown above. When B gets treated, it first moves from Y to Y' (immediate effect of the treatment at time $t = 2$). Then let's suppose that it continues to increase year after year to $Y'' > Y'$. Then, when B is used again as a comparison with C (this time, at $t = 6$), its value will be much higher.

Warning: this literature is generating dozens of papers as I write these lines. Things haven't completely settled just yet. The latest review is Baker et al. (2025).

8.5.3 Inference

In principle, we can do the same kind of inferential work as before. But the temporal and grouped nature of the dataset creates new problems that we didn't have so far.

8.5.3.1 Clusters

Recall that we assumed generally so far that error terms were heteroskedastic and yet independent across units. This made generally sense in the context of conventional random sampling.

However, now, our sample isn't random in the same way and thus observations are not independent. We repeatedly get information from the same units. We may thus worry that error terms could be correlated.

Not taking such temporal correlation into account generally leads us to have overly tight confidence intervals, thereby committing more Type I errors (falsely rejecting the null hypothesis that $\beta = 0$) (Bertrand, Duflo, and Mullainathan 2004). Fortunately, there is a way out of this by allowing correlation of error terms *within* units. Each unit (eg, each country) represents a

cluster or group. We can then write the variance-covariance matrix of $\mathbb{E}[\varepsilon\varepsilon']$ as (here, with two units a and b, each measured over 3 time periods):

$$\mathbb{E}[\varepsilon\varepsilon'] = \begin{bmatrix} \sigma^2_{a,1} & \varepsilon_{a,1}\varepsilon_{a,2} & \varepsilon_{a,1}\varepsilon_{a,3} & 0 & 0 & 0 \\ \varepsilon_{a,2}\varepsilon_{a,1} & \sigma^2_{b,2} & \varepsilon_{a,2}\varepsilon_{a,3} & 0 & 0 & 0 \\ \varepsilon_{a,3}\varepsilon_{a,1} & \varepsilon_{a,3}\varepsilon_{a,2} & \sigma^2_{a,3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma^2_{b,4} & \varepsilon_{b,4}\varepsilon_{b,5} & \varepsilon_{b,4}\varepsilon_{b,6} \\ 0 & 0 & 0 & \varepsilon_{b,5}\varepsilon_{b,4} & \sigma^2_{b,5} & \varepsilon_{b,5}\varepsilon_{b,6} \\ 0 & 0 & 0 & \varepsilon_{b,6}\varepsilon_{b,4} & \varepsilon_{b,6}\varepsilon_{b,5} & \sigma^2_{b,6} \end{bmatrix}$$

For instance, $\varepsilon_{a,2}\varepsilon_{a,1}$ denotes the correlation of the error terms of unit a at time $t = 1$ and $t = 2$. What we have here is a form of **serial correlation** :

$$Cor(\varepsilon_{i,t-1}, \varepsilon_{i,t}) \neq 0.$$

Note: in the DiD case, the clusters are defined by the *temporal* structure of the data. That is, we expect correlation to happen within units. For instance, for a dataset of consumers over two periods, a consumer's error term at time t might be predictive of their error term at time $t+1$. But clusters could be driven by other forces. They could be geographic: households from the same municipalities might have correlated error terms. Or it could be based on industries: firms from the same sector might have correlated errors. And sometimes, it could be both: there is dependence of firms within sectors and years.

Thus, the more general way to write down the problem is to state that there is a nonzero correlation within groups g:

$$Cor(\varepsilon_{g,j}, \varepsilon_{g,k}) \neq 0 \text{ for } j \neq k.$$

Ignoring these nonzero correlations may lead to seriously biased estimated standard errors (Moulton 1990). In general, estimated standard errors will be too small, leading to an excessive rate of rejection of the null hypothesis.

We can relax the independence assumption by acknowledging the grouped nature of the data and using what we call **clustered standard errors** (allowing you to make **cluster-robust inference**) (Liang and Zeger 1986; Bertrand, Duflo,

and Mullainathan 2004; A. Colin Cameron and Miller 2015; Abadie et al. 2023; De Chaisemartin and Ramirez-Cuellar 2024). Clustered standard errors can then be used for hypothesis testing or to build confidence intervals. For a review on various ways in which standard errors could be computed for DiD and FE models more generally, see A. Colin Cameron and Miller (2015) and B. E. Hansen (2024). See Abadie et al. (2023) for in-depth guidance to select the optimal clustering strategy.

What's happening under the hood is the following. Concretely, using residuals u from the OLS regression, and assuming we have G groups (such as countries or firms):

$$\widehat{\text{Var}}_{\text{clus}}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{g=1}^{G} \mathbf{X}_g' \mathbf{u}_g \mathbf{u}_g' \mathbf{X}_g \right) (\mathbf{X}'\mathbf{X})^{-1}$$

All three terms yield $k \times k$ matrices.[3] The new part is the middle term, where we go cluster/group by cluster/group. For a given group g of size N_g, we estimate $\varepsilon\varepsilon'$ with the sample residuals u:

$$\mathbf{u}_g \mathbf{u}_g' = \begin{bmatrix} u_{g,1}^2 & u_{g,1}u_{g,2} & \cdots & u_{g,1}u_{g,N_g} \\ u_{g,2}u_{g,1} & u_{g,2}^2 & \cdots & u_{g,2}u_{g,N_g} \\ \vdots & \vdots & \ddots & \vdots \\ u_{g,N_g}u_{g,1} & u_{g,N_g}u_{g,2} & \cdots & u_{g,N_g}^2 \end{bmatrix}$$

You then pre- and post-multiply this by $\mathbf{X}_g$ (the data for group g). Next, you sum this up for all groups $g \in G$. This gives you the middle part, which is then pre- and post-multiplied by $(\mathbf{X}'\mathbf{X})^{-1}$. This gives the estimate of clustered standard errors. You would then read something like: "the standard errors are clustered by firms" (assuming groups are defined by firms). Note: if the number of groups is small, then the estimates of these variances can be quite poor (Donald and Lang 2007).

[3] The first and last terms $(\mathbf{X}'\mathbf{X})^{-1}$ are $k \times k$. The middle term contains: $(\mathbf{u}_g \mathbf{u}_g')$ (which is $n \times n$). It is first pre-multiplied by an $k \times n$ matrix $\mathbf{X}_g'$ and second it is post-multiplied by an $n \times k$ matrix $\mathbf{X}_g$. We thus obtain a $k \times k$ matrix for the middle term.

9 Other models and estimators

This version: December 16, 2025

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>
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9.1 Readings

To be added.

9.2 Synthetic control

The idea of **synthetic control** is intuitive and is best illustrated by an example (Abadie, Diamond, and Hainmueller 2010; Doudchenko and Imbens 2016; Ferman and Pinto 2019; Athey et al. 2021). Consider a given unit that received a treatment. For instance, it could be a canton: Vaud, alone, decided to implement a new taxation policy in 2020 (this is an imaginary example). For years 2021, 2022, ..., the question is: what would have happened to Vaud's tax revenues (Y) in the absence of the new tax policy (D)? What we are asking is to identify the ATT (average treatment effect on the treated), namely the effect of the tax policy on tax revenues for the canton that implemented it (i.e., Vaud). The intuition of synthetic control is to say: *the*

counterfactual for Vaud after 2020 is likely some kind of mix of other cantons. What we need to do is to create a *synthetic* treated unit that imputes its outcome *had it not been treated*, using data from control units.

The question then becomes: how can we weight other cantons to recreated the counterfactual outcome Y after the treatment was assigned? Probably, Vaud is going to be more like Geneva than, say, Niedwald. But how can we identify the right sets of weights?

To solve this, we need a bit of notation. Consider a panel dataset with N units measured T periods, an outcome $Y_{i,t}$, and a dichotomous treatment D. WLOG, we say that that the treated unit is N, and that there are $[t, t^*)$ periods until the treatment, and $[t^*, T]$ periods after. We can write the matrix of outcomes *under control* as:

$$\begin{bmatrix} y_{1,1} & y_{1,2} & \cdots & y_{1,T-1} & y_{1,T} \\ y_{2,1} & y_{2,2} & \cdots & y_{2,T-1} & y_{2,T} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ y_{N,1} & y_{N,2} & \cdots & y_{N,T-1} & y_{N,T} \end{bmatrix}$$

Using our example, we can write:

$$\begin{bmatrix} y_{\text{ZH},1990} & y_{\text{ZH},1991} & \cdots & y_{\text{ZH},2019} & y_{\text{ZH},2020} & y_{\text{VD},2021} \\ y_{\text{GE},1990} & y_{\text{GE},1991} & \cdots & y_{\text{GE},2019} & y_{\text{GE},2020} & y_{\text{VD},2021} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \\ y_{\text{VD},1990} & y_{\text{VD},1991} & \cdots & y_{\text{VD},2019} & y_{\text{VD},2020} & y_{\text{VD},2021} \end{bmatrix}$$

But note that some of these observations are missing, as they are the unobserved counterfactual after t^*:

where ZH is Zurich, GE is Geneva, VD is Vaud, etc. You'd have 26 rows in this dataset.

$$\begin{bmatrix} y_{\text{ZH},1990} & y_{\text{ZH},1991} & \cdots & y_{\text{ZH},2019} & y_{\text{ZH},2020} & y_{\text{ZH},2021} \\ y_{\text{GE},1990} & y_{\text{GE},1991} & \cdots & y_{\text{GE},2019} & y_{\text{GE},2020} & y_{\text{GE},2021} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \\ y_{\text{VD},1990} & y_{\text{VD},1991} & \cdots & y_{\text{VD},2019} & ? & ? \end{bmatrix}$$

Now, synthetic control suggests to replace the missing observations by a weighted average of the control units. The challenge, then, is to obtain the weights. Using our example: if tax revenues of Vaud are a combination of the tax revenues of Geneva,

Valais, Zurich, etc., then how should we decide what the weights are?

One way to do so is to use pre-treatment data and regress $Y_{N,t<t^*}$ on $\mathbf{Y}_{D=0,t<t^*}$:

$$y_{N,t} = \gamma_0 + \sum_{i=1}^{N-1} \gamma_i y_{i,t} | t < t^*.$$

This is a generalized version by Doudchenko and Imbens (2016) of the original approach by Abadie, Diamond, and Hainmueller (2010). Because of this matrix structure, people sometimes refer to this as a *vertical* regression (Athey et al. 2021).

Estimate this model via OLS to obtain $\hat{\gamma}_0, \hat{\gamma}_1, ..., \hat{\gamma}_{N-1}$ (note that each control unit receives a time-invariant weight γ_i; these are, in effect, fixed effects based on pre-treatment data). Then, use these estimates (multiplied by the observed control outcomes $y_{i,t>t^*}$) to generate predicted values $\hat{y}_N$ for post-treatment periods $t^*, ..., T$ for the treated unit N.

$$\begin{bmatrix} y_{\text{ZH},1990} & y_{\text{ZH},1991} & \cdots & y_{\text{ZH},2019} & y_{\text{ZH},2020} & y_{\text{ZH},2021} \\ y_{\text{GE},1990} & y_{\text{GE},1991} & \cdots & y_{\text{GE},2019} & y_{\text{GE},2020} & y_{\text{GE},2021} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \\ y_{\text{VD},1990} & y_{\text{VD},1991} & \cdots & y_{\text{VD},2019} & \hat{y}_{\text{VD},2020} & \hat{y}_{\text{VD},2021} \end{bmatrix}$$

The ATT, then, is the averaged difference between y_N (observed outcome for treated unit N) and $\hat{y}_N$ (synthetic counterfactual value). In the case of a single post-treatment period (T), this boils down to:

$$\widehat{\text{ATT}}_{\text{synth}} = y_{N,T} - \hat{y}_{N,T}$$

If we have multiple post-treatment period, the treatment effect can be plotted on a period-per-period basis to see how the effect materializes (or not) over time. Strictly speaking, the ATT would then be the average over all post-treatment periods (from t^* to T):

$$\widehat{\text{ATT}}_{\text{synth}} = \frac{1}{T - t^*} \sum_{t=t^*}^{T} \left[y_{N,t} - \hat{y}_{N,t} \right]$$

Note. The quality of this approach depends on how well you can model weights. This, in turn, typically depends on the amount of pre-treatment data you have. The more observations, the better. Typically, you want many pre-treatment observations. You could presumably also improve the quality of

predictions by including covariates $\mathbf{X}$ in the pre-treatment weight equation. We then typically plot the predicted Y's and the observed Y's for the pre-treatment period and hope that they are good approximation. If not, it means that our treated unit isn't well predicted by a combination of other units.

Note. It's not self-evident how to construct confidence intervals or do hypothesis testing. One solution, suggested by Abadie, Diamond, and Hainmueller (2015), is to reassign treatment to each of the other $N-1$ units, obtain $\widehat{\text{ATT}}$ for each permutation. In other words, you create fake treated units (hence them calling this "in-space placebos") and then see if similar ATTs obtain in these cases. This gives us a distribution of N $\widehat{\text{ATT}}$ (including $\widehat{\text{ATT}}_N$, the ATT for the 'real' treated unit N). We can then use percentiles of this distribution as p values and can assess how likely our $\widehat{\text{ATT}}_N$. Note that when treatment is randomized, this boils down to randomization inference. If D isn't randomized, then this p value only tells us the likelihood of obtaining $\widehat{\text{ATT}}_N$ or larger among the distribution of all placebo $\widehat{\text{ATT}}$. This is different from p values evaluating the plausibility of the data given a null hypothesis. Other approaches for inference have been developed in the meantime (Xu 2017).

Note. The underlying assumption is that the weights γ_i don't change over time. This may or may not hold.

Note. A different approach would be to run a "horizontal" regression, in which you take the set of control units, regress the last period of the control outcomes are regressed on their lagged values, creating a set of time effects. These can then be plugged in with pre-treatment y for the treated unit(s) to generate forecasts of what y would have been.

Synthetic control: illustration

Example 9.1 (Synthetic control)**.** The literature on synthetic controls took off with Abadie, Diamond, and Hainmueller (2010). They use the following example: California launched a tobacco control policy in 1988 (aka "Proposition 99") (that's the treatment). Did it have an effect on per-capita cigarette sales (outcome)?
If we plot the trend lines, it's hard to tell, as Figure 9.1 shows. There's a general downward trend both in California and in the rest of the US.

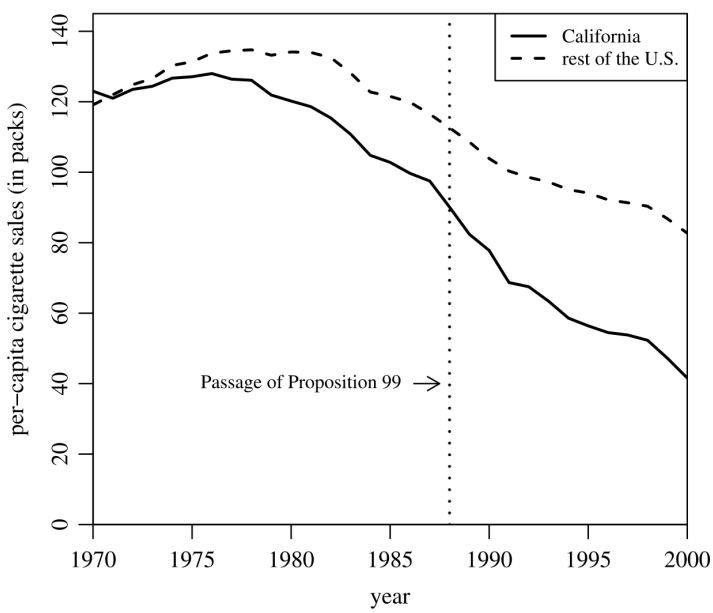

Figure 9.1: Trend lines in cigarette sales (per capita). Source: Abadie, Diamond, and Hainmueller (2010).

But the question is: what would have happened to California in the absence of this tobacco-control policy? That's where we can use the synthetic control approach. What we need to do is to create a synthetic California. Once we do this, we obtain Figure 9.2. We see that tobacco sales would have remained much higher, had California

not passed Proposition 99 (assuming all underlying assumptions are met).

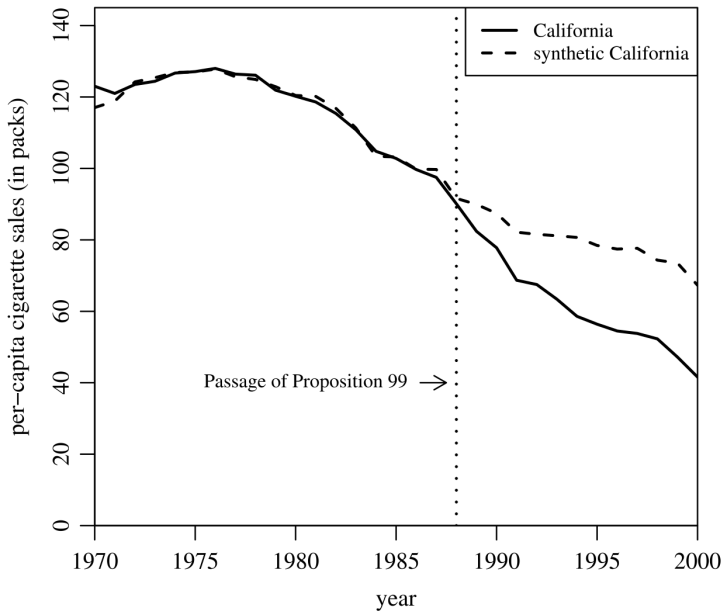

Figure 9.2: The path of observable and of synthetic California. Source: Abadie, Diamond, and Hainmueller (2010).

We can then plot the implied ATT, namely the vertical difference between the two curves (Figure 9.3). We see that the effect grew from about 1988 to about 1995 and has remained stable since.

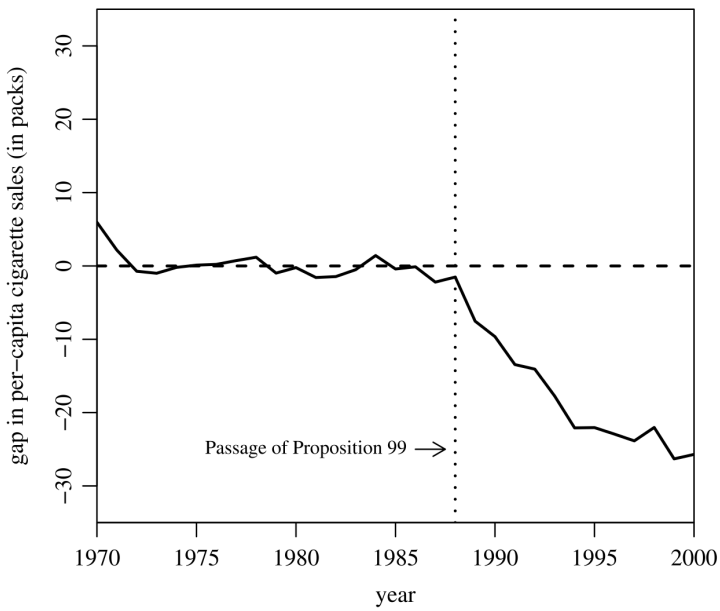

Figure 9.3: Plotting the ATT year-by-year. Technically, the ATT would be the average of the treatment effects after 1988. Note that we want a flat line pre-treatment. Source: Abadie, Diamond, and Hainmueller (2010).

And we can also check which states are used to generate this combination by looking at the largest weights (Table 9.1). This is a good way to ensure that the mix makes sense. In our earlier example, I would raise an eyebrow is Vaud was a combination of Niedwald and Glaris with zero weight on cantons that are similar (geographically, language-wise, in terms of size, etc.).

State	Weight	State	Weight
Alabama	0	Montana	0.199
Alaska	–	Nebraska	0
Arizona	–	Nevada	0.234
Arkansas	0	New Hampshire	0
Colorado	0.164	New Jersey	–
Connecticut	0.069	New Mexico	0
Delaware	0	New York	–
District of Columbia	–	North Carolina	0
Florida	–	North Dakota	0
Georgia	0	Ohio	0
Hawaii	–	Oklahoma	0
Idaho	0	Oregon	–
Illinois	0	Pennsylvania	0
Indiana	0	Rhode Island	0
Iowa	0	South Carolina	0
Kansas	0	South Dakota	0
Kentucky	0	Tennessee	0
Louisiana	0	Texas	0
Maine	0	Utah	0.334
Maryland	–	Vermont	0
Massachusetts	–	Virginia	0
Michigan	–	Washington	–
Minnesota	0	West Virginia	0
Mississippi	0	Wisconsin	0
Missouri	0	Wyoming	0

Table 9.1: Table of weights used to generate a synthetic California. Smoking-wise, California appears to be a mix of Utah, Nevada, Montana, Colorado, and Connecticut (in order of importance). Source: Abadie, Diamond, and Hainmueller (2010).

10 Maximum likelihood

This version: December 16, 2025

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10.1 Readings

People interested in nonlinear models should refer to Eliason (1993), A. Colin Cameron and Trivedi (2005).

10.2 Theory of maximum likelihood

The origins of **maximum likelihood** (ML – not to be confused with *machine learning*) are linked to the work of R.A. Fisher (Fisher 1922). Suppose we have a set of data and an unknown vector of parameters regarding their probability density function (pdf) (for a continuous variable) or probability mass function (for a discrete variable), γ. The idea of ML, in a nutshell, is to select as estimate(s) of γ the vector $\hat{\gamma}_{ml}$ that maximizes the likelihood of observing this particular set of data.

For instance, suppose we want an estimate the probability of obtaining a head p in a coin toss. Suppose further we obtain

a dataset of 10 coin tosses, 7 of which are head. Then (trivially, I admit), the estimate p that maximizes the likelihood of obtaining such a dataset is $\hat{p}_{ml} = 0.7$.

Formally, for a sample of N *independent* observations of variable y, we define the **likelihood function**:

$$L(\gamma) = \prod_{i=1}^{N} f(y; \gamma)$$

where L is the likelihood, $\prod$ the product (over N observations) of probability density functions $f()$ of y, as a function of k parameters γ.

Since L can grow too large quickly and thus make computation difficult, we often take the log of the likelihood function instead. Call this the **log-likelihood** $\mathcal{L}$ (or sometimes LL):

$$\mathcal{L}(\gamma) = \sum_{i=1}^{N} log\left[f(y; \gamma)\right].$$

Then, ML selects $\hat{\gamma}_{ml}$ that maximizes $\mathcal{L}$:

$$\arg\max_{\hat{\gamma}} \mathcal{L}(\gamma) := \hat{\gamma}_{ml}.$$

The key challenge will be (a) to define the likelihood function, and (b) to obtain a solution to this maximization problem. Part (a) will primarily be a matter of theory combined with the nature of the data you are looking at. Part (b) will sometimes involve closed-form solutions, sometimes not. As such, ML estimation may have involve iteration over the space of potential values for γ until a maximum is reached (hopefully a global one).

ML with a linear model

While we often use ML to overcome the shortcoming of OLS when outcomes are nonlinear, you can use for linear models too.
Let's start with the usual model $y = \alpha + \beta D + \varepsilon$. However, in the ML approach, we must make assumptions

about the likelihood function of y. To do so, we must make an assumption about ε, which is the only source of variation. Let us assume that it is a iid stochastic noise distributed normal: $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. Note: this is not a free assumption: the validity of our analysis here depends on this assumption!
From this starting point, we can state that:

$$y \sim \mathcal{N}(\alpha + \beta D, \sigma^2)$$

The likelihood function, given n observations, is:

$$L(\alpha, \beta, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \alpha - \beta D_i)^2}{2\sigma^2}\right)$$

That is, it is the product n times of normal distributions. Note that we derive this statement from our assumption about the error term, which we assume to be normal. This is distinct from OLS, where we didn't make any such assumption (we relied on asymptotic properties of OLS estimators).
We take the log to simplify, generating the log-likelihood $\mathcal{L}$ function (with three arguments):

$$\mathcal{L}(\alpha, \beta, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \alpha - \beta D_i)^2$$

Maximum likelihood implies the maximization of the log-likelihood $\mathcal{L}(\cdot)$. If we look at the last expression, we note that the only quantity that changes in α and β is the last term (we don't control σ^2). This term is negative. Thus, to maximize $\mathcal{L}$, we must minimize the last term.
But here, you should note that we have seen $\sum_{i=1}^{n}(y_i - \alpha - \beta D_i)^2$ before: it's the sum of the squared residuals, which is what we already minimize with OLS. We can minimize this quantity with the general expression $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$. And then, if we take the second derivative, we obtain:

$$\frac{\partial^2 \mathcal{L}}{\partial\beta\partial\beta} = (\mathbf{X}'\mathbf{X})(-\sigma^{-2}),$$

which is negative definite, and thus $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ is the max of the likelihood.

Thus, ML coincides with OLS for the linear model: $\hat{\beta}_{ols} = \hat{\beta}_{ml}$. ML uses a different starting point: it starts by making an assumption about the distribution of ε and uses it to derive a likelihood function for Y. But despite its different starting point, it ends with the same result.

10.3 Inference with maximum likelihood

10.3.1 Standard error with ML

Asymptotically, ML estimators exhibit all kinds of nice properties. First, and useful for inference, we can establish that the MLE is asymptotically normally distributed:

$$\hat{\beta}_{ml} \sim \mathcal{N}(\beta, \mathbf{I}(\beta)^{-1}),$$

where $\mathbf{I}(\beta)$ is the **information matrix**. It is defined as the negative of the matrix of the expected values of the second derivative (or **Hessian**) of the LL (i.e., $\mathcal{L}$) with respect to β:

$$\mathbf{I}(\beta) = -E\left[\frac{\partial^2 \mathcal{L}(\beta)}{\partial\beta\partial\beta'}\right]$$

We can then conduct similar kind of procedures, including hypothesis testing, as in the linear world.

10.3.2 Confidence intervals for quantities of interest

In many cases, we are not that interested in classical hypothesis testing, mostly because the point estimates don't tell us much. Instead, we compute a quantity of interest, such as a change in probability. But then, we need to find a way to assess our confidence in these quantities. In other words, we need a confidence interval. How can we build them? They don't always exist analytically. Instead, we can simulate them. The most common procedure is the following:

- Obtain the ML estimate of β. Define $\hat{\beta}$ the vector of point estimates and $\hat{V}$ the estimated variance-covariance matrix of the parameters.
- Repeat the following procedure 1,000 times (or so):
  - Randomly draw β's from the multivariate normal distribution $\mathcal{N}(\hat{\beta}, \hat{\mathbb{V}})$.
  - Compute the quantity of interest (e.g. average $\hat{P}_2 - \hat{P}_1$). That's generally one number based on that particular set of parameters.
  - Save the quantity of interest
- You then have 1,000 observations of the quantity of interest. You can then construct a 95% confidence interval.

10.3.3 Properties of MLE

What about MLE's general properties? We can establish two results.

1. MLE is **consistent**:

$$\lim_{n\to\infty} \mathbb{P}[|\hat{\beta}_{ml} - \beta| > \varepsilon] = 0$$

2. MLE is **efficient**, in the sense that it achieves the **Cramér-Rao lower bound**. We can see this directly from its asymptotic distribution. The Cramér-Rao inequality indicates that (assuming we have an unbiased estimator):

$$\mathbb{V}(\hat{\beta}) \geq \frac{1}{-\mathbb{E}\left[\frac{\partial^2 \mathcal{L}(\beta)}{\partial\beta\partial\beta'}\right]},$$

that is, $\mathbb{V}(\hat{\beta})$ cannot be smaller than the quantity of the left-hand side. But that's exactly the variance of the the MLE. Thus, MLE is efficient.

10.4 Goodness of fit

Goodness of fit can be assessed in various ways with maximum likelihood.

- Lagrange multiplier test
- Wald Test
- Likelihood-ratio test
- Pseudo-R^2
- AIC and BIC

Other tests can be used on an *ad hoc* basis. For instance, in the case of a nonlinear model with a dichotomous outcome, we may use the percentage of correctly predicted values as a measure of quality of model. I'll discuss these wherever relevant.

10.4.1 Likelihood-ratio test

Likelihood-ratio test (LR). This test is similar in spirit to the F test in linear models. The test compares the difference in log-likelihood of two models: one that is unrestricted (including all independent variables) and one that is restricted (where we remove q variables). This test is computed as follows:

$$\text{LR} = 2(\mathcal{L}_u - \mathcal{L}_r) \sim \chi_q,$$

where $\mathcal{L}_u$ is the unrestricted model and $\mathcal{L}_r$ is the restricted model (the one removing q parameters). LR is distributed χ with q degrees of freedom. We can then apply the usual hypothesis testing with the null being that these q parameters are indeed equal to zero.

10.4.2 Pseudo-R^2

Pseudo-R^2. There are various measures that mimic the linear model's R^2. One is to compute:

$$\text{pseudo-}R^2 = 1 - \frac{\mathcal{L}_u}{\mathcal{L}_o}$$

where $\mathcal{L}_u$ is the log-likelihood of the model and $\mathcal{L}_o$ is the log-likelihood if only the intercept was included. If the predictions of the model were perfect, $\mathcal{L}_u = 1$ and pseudo-$R^2 = 0$, but that will never be the case.

11 Nonlinear Models

This version: December 16, 2025

> **Caveat**
>
> These notes are offered in support of *Introduction to Econometrics* (MGT 581). They were written to help students understand the logic behind important topics in econometrics. But they don't (always) replace the insights and depth that textbooks can provide. This is why we connect each session to chapters of our core textbooks. See the syllabus for more details.

11.1 Readings

People interested in nonlinear models should refer to Maddala (1983) (on limited dependent variables), Eliason (1993) (on maximum likelihood), A. Colin Cameron and Trivedi (2005) (ch.5 for maximum likelihood), Section IV for nonlinear models). See also McFadden (1976) (on random utility models) and Train and Weeks (2005) (for discrete choice models in general).

11.2 Linear probability model

11.2.1 Logic of the linear probability model

So far, we wrote down linear models under the assumption that…

- Treatments could be continuous or categorical (with dummy variables being a special case of the latter with two categories)
- Outcomes were defined on the real line (or at least a sizeable subset thereof; for instance, GDP is defined on the nonnegative segment of the real line, so we treat it 'as if continuous').

We did not raise the question of: **what happens if the outcome is discrete?**

Let us start with the special case where $y \in \{0, 1\}$. Some examples:

- Does treatment D determine whether a Central Bank uses a fixed or a floating exchange rate?
- Does treatment D determine whether a consumer buys an app or not?
- Does treatment D determine whether a firm enters a market or not?

We could wonder whether this creates any kind of issues? Can we still use OLS to estimate such a model?

The answer is yes and no. Let's start with the motivation for a 'yes.' Consider the following model:

$$y_i = \alpha + \beta D_i + \varepsilon_i.$$

Now, *nothing* prevents us from estimating these population parameters with OLS (assuming no perfect multicollinearity). We would then obtain $\hat{\alpha}$, $\hat{\beta}$.

The interpretation of these parameters is slightly different but quite intuitive. Note that until now, we modeled the expected value of Y, $\mathbb{E}[Y|D]$. But note that we can also rewrite this as the probability of $y = 1$ given D:

$$\mathbb{E}[Y|D] = \mathbb{P}[Y = 1|D],$$

and thus:

$$\begin{aligned} \text{Probability: } & \mathbb{P}[Y = 1] = \alpha + \beta D_i + \varepsilon_i \\ \text{Predicted probability: } & \mathbb{P}[\widehat{Y = 1}] = \hat{\alpha} + \hat{\beta} D_i \end{aligned}$$

We call this the **linear probability model** (LPM). In this case, we can now interpret our parameters as follows:

- α: probability that $y = 1$ when all covariates are set at zero.
- β: the marginal effect of D on the probability that $y = 1$.

For instance, suppose we have a dataset on consumer choices over a period of one year:

$$\mathbb{P}[\text{Buy new laptop } (=1)_i] = 0.3 + 0.2 \cdot \text{Has a job}_i + \varepsilon_i,$$

then the probability that someone buys a new laptop in the last twelve months is 0.3 (or 30%) for people without a job, and 0.5 (or 50%) for those who have a job.

11.2.2 Inference with LPM

There are several challenges with LPM.

First, it will sometimes generate predicted probabilities that are inconsistant with probability theory. For instance, imagine a model:

$$\mathbb{P}[\text{Buy new laptop } (=1)_i] = 0.4 + 0.1\text{Monthly income in kCHF} + \varepsilon_i.$$

Suppose that you have someone with an income of CHF7k, then we would have:

$$\mathbb{P}[\text{Buy new } \widehat{\text{laptop}} \ (=1)_i] = 0.4 + 0.1 \cdot 0.7 = 1.1.$$

This is not the end of the world *per se*: if you don't care about predictions, and instead only care about marginal treatment effects, then you can probably ignore this. Instead, you will only wonder if the estimate of the treatment effect ($\widehat{\beta}$) is meaningful. And here, some believe that it's a perfectly fine approximation in many cases (J. Angrist and Pischke 2008).

Second, LPM wil be construction be heteroskedastic. As X gets very large (in either direction), the average error will increase, and thus the variance of the error term cannot be constant. See

Figure 11.1. Thus, we must use robust standard errors (White 1980).

Beyond this, the model is just fundamentally misspecified when treatments are continuous. The expected value of the error term, when X becomes very large or very small, cannot be zero. For very large negative values of X, and assuming a positive relation between X and Y, the values of ε will all be positive, since the predicted line with be below all values, whether they are 0 or 1. Likewise, for very large positive values of X (and still assuming a positive relation), the regression line will be above all observations, and thus $\varepsilon < 0$.

Note that what I just wrote applies to continuous treatments. If we had a dummy treatment D, then all values $\hat{P}$ would be bounded between 0 and 1, and there would be no problem of the kind I just discussed.

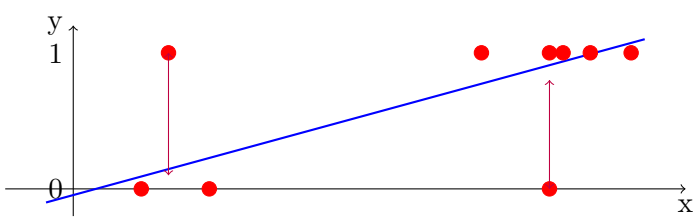

Figure 11.1: Heteroskedasity with linear probability models.

11.2.3 OLS' shortcomings

OLS has several shortcomings when analyzing discrete outcomes.

- If Y is a dummy, it may generate predictions outside the [0,1] interval. It is thus poorly suited for predictions.
- If Y is ordinal with >2 categories, it implicitly assumes that Y is continuous. And it can again generate predictions beyond the interval of values that Y can take.
- OLS cannot handle categorical variables that are not ordinal. For instance, imagine a model in which the outcome is the color of the car being bought. How do you translate {Red, Green, Blue} into a numerical variable? There is no order.

11.3 Logit and probit

Linear probability models, by definition, will eventually lead to predictions that are outside the $[0, 1]$ interval (except in the trivial case where parameters equal 0). What we want is a model that constrains predicted values on the $[0, 1]$ interval.

If we want a model between 0 and 1, there are in theory an endless number of models that could do the trick. But not all models are principled. Instead, what we will do is start with a **random utility model**. This allows us to justify the models that we often use: the **logit** and the **probit**.

11.3.1 Models

So how do logit/probit work? There are a number of ways to get to a nonlinear model, but the best starting point is to think it terms of **latent variable model** (or **random utility model**). Instead of beginning with an observable outcome y (which equals to zero or 1), we start with a latent version of that variable y^*. This variable captures the underlying decision process that eventually leads y to be zero or 1. What we want, then, is to write this latent variable as a linear model:

$$y^* = \alpha + \beta x + \varepsilon$$

As of now, the only assumption we make (besides that we have the *correct model*) is that ε is *symmetrically* distributed and *uncorrelated* with X. Note that ε is the only random stuff in there, the rest (α, β, X) being fixed.

We claim that:

$$\begin{aligned} y &= 1 \text{ iff } y^* > 0 \\ &= 0 \text{ iff } y^* \leq 0 \end{aligned}$$

That is, if the latent variable is above some arbitrary cutoff (zero), then the outcome will be 1. Otherwise, it's zero. We

can then write:

$$\begin{aligned}\mathbb{P}[y=1] &= \mathbb{P}[y^* > 0] \\ &= \mathbb{P}[\alpha + \beta x + \varepsilon > 0] \\ &= \mathbb{P}[\varepsilon > -(\alpha + \beta x)] \\ &= 1 - \mathbb{P}[\varepsilon < -(\alpha + \beta x)]\end{aligned}$$

We can write this because ε is random and therefore the model is not deterministic.

Now, we need to think a bit more about ε. The second expression asks us something like: what is the likelihood that ε is smaller than some quantity $-(\alpha + \beta x)$. We have seen similar questions fairly regularly, for instance when we talked about things such as t tests. What we need to answer them is the cumulative distribution function for ε. For the time being, let's call it $F(\cdot)$. We can then continue:

$$\begin{aligned}\mathbb{P}[y=1] &= 1 - \mathbb{P}[\varepsilon < -(\alpha + \beta x)] && \text{(from above)} \\ &= 1 - F[-(\alpha + \beta x)] \\ &= F(\alpha + \beta x) && (\varepsilon \text{ is symmetric})\end{aligned}$$

So we have made some progress. Now the question is: what is F? Remember that it is the cumulative distribution function. The corresponding *density function* tells us how ε is distributed. As said, so far we only assumed that ε is uncorrelated to $\mathbf{X}$ and that it is symmetric. Now, we need to go a bit further and decide what its distribution is.

- If F is the logistic distribution, then we have the **logit** model.
- If F is the (cumulative) normal distribution, then we have the **probit** model.

The logit looks like this:

$$F(x) = \frac{1}{1 + exp(-x)}$$

In our case, this yields:

$$\begin{aligned}
\mathbb{P}[y=1|X] &= F(\alpha+\beta x) && \text{(from above)} \\
&= \frac{1}{1+exp(-(\alpha+\beta x))} \\
&= \frac{exp(\alpha+\beta x)}{1+exp(\alpha+\beta x)}
\end{aligned}$$

For the probit:

$$\begin{aligned}
\mathbb{P}[y=1|X] &= F(x) \\
&= \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} exp\left(\frac{-x^2}{2}\right) dv
\end{aligned}$$

11.3.2 Estimation

The challenge comes when we try to estimate these models. As of right now, we only have β, not $\hat{\beta}$. We will use **maximum likelihood** to obtain these estimates. Recall that the idea behind maximum likelihood is the following: given a set of assumptions about the underlying model, and given the data we have, what are the parameters that are the most likely?

For a sequence of draws of the dichotomous outcome $y_1, y_2, ..., y_n$ ($y \in \{0,1\}$), what we want to maximize is the joint probability of y given $\mathbf{X}$:

$$\mathbb{P}[y_1, y_2, ..., y_n|\beta, \mathbf{X}]$$

If the draws are IID, then we can rewrite this as:

$$\mathbb{P}[y_1, y_2, ..., y_n|\beta, \mathbf{X}] = \Pi_{y_i=1}\mathbb{P}[y_i=1|\beta, \mathbf{X}] \cdot \Pi_{y_i=0}\mathbb{P}[y_1=0|\beta, \mathbf{X}]$$

But we know what $P(y_i=1)$ is! We can write:

$$\begin{aligned}
\ell &= \Pi_{y_i=1} F(\mathbf{X}\beta|\beta, \mathbf{X}) \cdot (1-F(\mathbf{X}\beta|\beta, \mathbf{X})) \\
&= \Pi_{y_i=1} F(\mathbf{X}\beta|\beta)^{y_i} \cdot (1-F(\mathbf{X}\beta|\beta))^{1-y_i},
\end{aligned}$$

where ℓ is called the **likelihood function**. Since multiplicative stuff becomes quickly cumbersome (think of what happens if $n = 1,000$), we generally take the log, i.e. the **log-likelihood**:

$$\ell\ell = \sum y_i ln(F(\mathbf{X}\beta)) + \sum (1 - y_i) ln(1 - F(\mathbf{X}\beta))$$

The last step is to maximize this function. First, we must decide whether $F()$ is the logit or the probit distribution. Then, we take the first derivative with respect to β, set it to zero, and solve for our maximum likelihood estimate $\hat{\beta}_{ml}$.

Note that ML's estimation of these models is, *under regularity conditions*:

- Consistent
- Asymptotically normal (more on this below)
- Asymptotically efficient
- But it is biased in small samples
- But it is inconsistent in the presence of heteroskedasticity.
- ML is also biased+ inconsistent when using fixed effects in panel data (**incidental parameters** problem).

The incidental parameter problem is the following: suppose your model is $y = f(\alpha_i + \beta x_i)$. The α's are fixed effects and β is a parameter common to all units. Now, suppose that T is fixed and N varies. As $N \to \infty$, β will be better estimated; however, that's not true of the α's, because we still only have the finite T to estimate each of them. In other words: as the number of units goes up, this does nothing to help estimating our α's, because we are just adding more units. What is worse: the bias can spill over to the estimation of the β's. See A. Colin Cameron and Trivedi (2005) (p.781).

11.3.3 Interpretation of logit and probit

The interpretation of logit/probit is tricky. In the linear model, the partial derivative with respect to a variable x was simply its parameter β. Easy enough. That's not the case anymore. To see this, just visualize the logit/probit curve: the derivative is not constant. In full notation:

$$\frac{\partial \mathbb{P}[\mathbf{X}]}{\partial x_j} = f(\mathbf{X}\beta)\beta_j,$$
$$\text{where } f(z) = \frac{F}{dz}(z).$$

Note. F is a cumulative distribution function (e.g. normal), and f is the probability density function.

To find the marginal effect of x, then, we need to compute the following:

$$\begin{aligned}\Delta\hat{\mathbb{P}} &= \hat{\beta} f(\mathbf{X}\beta)\Delta x \\ &= \hat{\beta} f(\mathbf{X}\beta) && \text{(if } \Delta x = 1\text{)}\end{aligned}$$

In other words, the marginal effect is $\hat{\beta}$ *times* a function of "where" we are (where the departure is from). Note that the effect will be higher "in the middle" (where we go from $\hat{P}$ close to zero to close to one) and smaller when $\mathbf{X}\beta$ is very negative or very positive. Generally, we often pick values of $\mathbf{X}$ that are meaningful, such as the sample median or the mean.

For our cases, we have:

$$\begin{aligned}\Delta\hat{\mathbb{P}} &= \hat{\beta} f(\mathbf{X}\beta) && \text{from above} \\ &= \hat{\beta}\frac{exp(\mathbf{X}\beta)}{(1+exp(\mathbf{X}\beta))^2} && \text{logit} \\ &= \hat{\beta}\frac{1}{\sqrt{2\pi}exp\left(\frac{-(\mathbf{X}\beta)^2}{2}\right)} && \text{probit}\end{aligned}$$

(You can see why people who do this manually prefer logit.)

An identical approach would be to do compute:

- Obtain the ML estimate of $\hat{\beta}$.
- Compute $\hat{\mathbb{P}}_1$ for the case you're interested in, such a the median observer.
- Compute $\hat{\mathbb{P}}_2$ when x is replaced by $x + c$ (where c is a constant).
- Compute confidence intervals (see below).

Another (and different!) possible quantity of interest would be:

- Obtain the ML estimate of $\hat{\beta}$.
- Compute $\hat{\mathbb{P}}_1$ for each existing observation using $\hat{\beta}$.
- Replace variable x_i by $z = x_i + c$ (where c is a constant).
- Compute $\hat{\mathbb{P}}_2$ using the new dataset (where z replaces x).
- Compute $\hat{\mathbb{P}}_2 - \hat{P}\mathbb{P}_1$ and take the average.

- Compute confidence intervals (see below).

This is fairly rare, partly because the effect will ultimately depend on the entire sample and will not be based on a specific individual.

Marginal effect with probit

Example 11.1 (Marginal effect with probit)**.**

Suppose we have two independent variables Z and W (and call $\mathbf{X}$ the joint vectors of Z and W as well as a column of 1's). We obtain ML estimates of the probit model:

$$\begin{aligned}\mathbb{P}[y_i = 1|\mathbf{X}] &= \Phi(\mathbf{X}\beta) \\ &= \Phi(\alpha + \beta_1 Z + \beta_2 W) \\ &= \Phi(2.3Z - 1.1W) \qquad \text{setting } \alpha = 0 \text{ for simplicity}\end{aligned}$$

Let us then assume that $Z = 1.5$ and $W = 2$ (imagine that these are the values of a particularly interesting unit). We then have:

$$\mathbf{X}\beta = 2.3 \cdot 1.5 + (-1.1)(2) = 1.25.$$

Next, we want to find out the marginal effect. We need to compute the PDF ϕ of the normal distribution evaluated at $\mathbf{X}\beta$:

$$\phi(\mathbf{X}\beta) = \phi(2.5) \approx 0.183$$

We can then compute the marginal effects of Z and W for this unit as:

$$\begin{aligned}\frac{\partial \mathbb{P}}{\partial Z} &= 0.183 \cdot 2.3 = 0.42 \\ \frac{\partial \mathbb{P}}{\partial W} &= 0.183 \cdot (-1.1) = -0.2\end{aligned}$$

Concretely:

- A one-unit increase in Z increases the probability that $Y = 1$ by 0.42 (or 42 percentage points).
- A one-unit increase in W reduces the probability that $Y = 1$ by 0.2 (or 20 percentage points).

11.3.4 Hypotesis testing

Once we have $\hat{\beta}_{ml}$, we can construct standard errors and confidence intervals. We can then conduct t tests, just like with OLS. The standard errors are given by the asymptotic variance-covariance matrix:

$$\widehat{V[\hat{\beta}_{ml}]} = \left[\sum_i \frac{[f(\mathbf{X}\hat{\beta}_{ml})]^2 \mathbf{X}'\mathbf{X}}{F(\mathbf{X}\hat{\beta}_{ml})[1 - F(\mathbf{X}\hat{\beta}_{ml})]} \right]^{-1}$$

The standard errors are given by the square root of the diagonals.

11.3.5 Goodness of fit

Goodness of fit can be assessed in various ways. See the notes on maximum likelihood for a general overview (likelihood-ratio test, etc.).

In the specific case of a nonlinear model with dichotomous outcomes, we can add a few additional tests.

Percentage correctly predicted (PCP). This is straightforward:

- Estimate the ML model of $\mathbb{P}[y = 1|\mathbf{X}]$.
- Compute the predicted probability $\hat{y}$.
- Generate a variable that is the prediction of your model; the variable equals 0 if $\hat{y} < 0.5$ and 1 if $\hat{y} > 0.5$.
- Calculate the percentage of correct prediction.

  Note. The issue is that if the sample proportion of 1's, $\tilde{p}$ is very low or very high, and that we end up predicting only zeros or only 1's, then our predictions will be outstanding, but we will be wrong on the few important cases (these rare 1's or 0's). One solution is to replace the threshold of 0.5 ($\hat{y} < 0.5$) by $\tilde{p}$, such that $\hat{y} < p \to 0$ and 1 otherwise.

You can report your results in a cross tab, in which you want to minimize the numbers in the off-diagonals. For instance, assume you have a model that seeks to

PCP

Example 11.2 (PCP)**.**

Suppose you are a consultant for a US political party. You get a sample of 100 adults with information about their partisanship (Democrat or Republican), their gender, and their income. You estimate a probit or logit model:

$$\text{Republican}_i = f(\alpha + \beta \text{Gender}_i + \gamma \text{Income}_i)$$

You then obtain estimated probabilities of being a Republican, $\widehat{\text{Republican}}$. And since anyone who is not a Republican is a Democrat in this sample, you also obtain $\widehat{\text{Democrat}} = 1 - \widehat{\text{Republican}}$. You can then build a cross-tab to see if your model is good at making predictions. Here, the percent of correct predictions is 91% $((48 + 43)/100)$. See Table Table **??**.

	Democrat	Republican
$\widehat{\text{Democrat}}$	48	6
$\widehat{\text{Republican}}$	3	43
Total	**51**	**49**

Expected PCP. See Herron (2000). The idea here is to reward good predictions. If you predict $y = 1$ based on $\hat{p} = 0.9$, that's better than if you had done so with $\hat{p} = 0.51$. Herron developed the notion of expected PCP:

$$\text{ePCP} = \frac{1}{N}\left(\sum_{y_i=1} \hat{\mathbb{P}}_i + \sum_{y_i=0}(1 - \hat{\mathbb{P}}_i)\right)$$

References

Abadie, Alberto, Susan Athey, Guido W Imbens, and Jeffrey M Wooldridge. 2023. "When Should You Adjust Standard Errors for Clustering?" *Quarterly Journal of Economics* 138 (1): 1–35.

Abadie, Alberto, Alexis Diamond, and Jens Hainmueller. 2010. "Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program." *Journal of the American Statistical Association* 105 (490): 493–505.

———. 2015. "Comparative Politics and the Synthetic Control Method." *American Journal of Political Science* 59 (2): 495–510.

Aitken, Alexander C. 1936. "IV.—on Least Squares and Linear Combination of Observations." *Proceedings of the Royal Society of Edinburgh* 55: 42–48.

Aklin, Michaël, Patrick Bayer, S. P. Harish, and Johannes Urpelainen. 2017. "Does Basic Energy Access Generate Socioeconomic Benefits? A Field Experiment with Off-Grid Solar Power in India." *Science Advances* 3 (5): e1602153.

Altman, Naomi, and Martin Krzywinski. 2017. "Points of Significance: P Values and the Search for Significance." *Nature Methods* 14 (1): 3–4.

Anderson, Theodore W. 2003. *An Introduction to Multivariate Statistical Analysis, 3rd Edition*. Wiley.

Anderson, Theodore W, and Herman Rubin. 1949. "Estimation of the Parameters of a Single Equation in a Complete System of Stochastic Equations." *Annals of Mathematical Statistics* 20 (1): 46–63.

Andrews, Isaiah, James Stock, and Liyang Sun. 2019. "Weak Instruments in IV Regression: Theory and Practice." *Annual Review of Economics* 11 (727-753).

Angrist, Joshua D., Guido W. Imbens, and Donald B. Rubin. 1996. "Identification of Causal Effects Using Instrumental

Variables." *Journal of the American Statistical Association* 91 (434): 444–55.

Angrist, Joshua D, and Alan B Krueger. 1999. "Empirical Strategies in Labor Economics." In *Handbook of Labor Economics*, edited by Orley C. Ahsenfelter and David Card. Vol. 3.

Angrist, Joshua D, and Jörn-Steffen Pischke. 2010. "The Credibility Revolution in Empirical Economics: How Better Research Design Is Taking the Con Out of Econometrics." *Journal of Economic Perspectives* 24 (2): 3–30.

Angrist, Joshua, and Steffan J. Pischke. 2008. *Mostly Harmless Econometics*. Princeton University Press.

Arkhangelsky, Dmitry, Guido W. Imbens, Lihua Lei, and Xiaoman Luo. 2024. "Design-Robust Two-Way-Fixed-Effects Regression for Panel Data." *Quantitative Economics* 15 (4): 999–1034.

Aronow, Peter M, and Allison Carnegie. 2013. "Beyond LATE: Estimation of the Average Treatment Effect with an Instrumental Variable." *Political Analysis* 21 (4): 492–506.

Aronow, Peter M., and Benjamin T. Miller. 2019. *Foundations of Agnostic Statistics*. New York: Cambridge University Press.

Asquith, Nigel. 2020. "Large-Scale Randomized Control Trials of Incentive-Based Conservation: What Have We Learned?" *World Development* 127: 104785.

Athey, Susan, Mohsen Bayati, Nikolay Doudchenko, Guido Imbens, and Khashayar Khosravi. 2021. "Matrix Completion Methods for Causal Panel Data Models." *Journal of the American Statistical Association* 116 (536): 1716–30.

Athey, Susan, and Guido W. Imbens. 2016. "The Econometrics of Randomized Experiments."

Athey, Susan, Julie Tibshirani, and Stefan Wager. 2019. "Generalized random forests." *The Annals of Statistics* 47 (2): 1148–78.

Baker, Andrew, Brantly Callaway, Scott Cunningham, Andrew Goodman-Bacon, and Pedro Sant'Anna. 2025. "Difference-in-Differences Designs: A Practioner's Guide."

Bertrand, Marianne, Esther Duflo, and Sendhil Mullainathan. 2004. "How Much Should We Trust Differences-in-Differences Estimates?" *Quarterly Journal of Economics* 119 (1): 249–75.

Billingsley, Patrick. 1995. *Probability and Measure*. Wiley.

Bloom, Howard S. 1995. "Minimum Detectable Effects: A Simple Way to Report the Statistical Power of Experimental Designs." *Evaluation Review* 19 (5): 547–56.

Bloom, Nicholas, James Liang, John Roberts, and Zhichun Jenny Ying. 2015. "Does Working from Home Work? Evidence from a Chinese Experiment." *Quarterly Journal of Economics* 130 (1): 165–218.

Bloom, Nicholas, Raffaella Sadun, and John Van Reenen. 2012. "Americans Do IT Better: US Multinationals and the Productivity Miracle." *American Economic Review* 102 (1): 167–201.

Bound, John, David A Jaeger, and Regina M Baker. 1995. "Problems with Instrumental Variables Estimation When the Correlation Between the Instruments and the Endogenous Explanatory Variable Is Weak." *Journal of the American Statistical Association* 90 (430): 443–50.

Brambor, Thomas, William R. Clark, and Matt Golder. 2006. "Understanding Interaction Models: Improving Empirical Analyses." *Political Analysis* 14 (1): 63.

Breusch, Trevor S, and Adrian R Pagan. 1979. "A Simple Test for Heteroscedasticity and Random Coefficient Variation." *Econometrica*, 1287–94.

Callaway, Brantly, and Pedro HC Sant'Anna. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics* 225 (2): 200–230.

Cameron, A Colin, and Douglas L Miller. 2015. "A Practitioner's Guide to Cluster-Robust Inference." *Journal of Human Resources* 50 (2): 317–72.

Cameron, A. Colin, and Pravin K. Trivedi. 2005. *Microeconometrics: Methods and Applications*. London: Cambridge University Press.

Cao, Yanni, Ria Martins, and Jianyong Wu. 2025. "Unveiling Complexity in Blue Spaces and Life Expectancy." *Environmental Research* 281: 121981.

Card, David, and Alan B. Krueger. 1994. "Minimum Wages and Employment." *American Economic Review*.

Casella, George, and Roger L Berger. 2021. *Statistical Inference*. Cengage Learning.

Chernozhukov, Victor, Mert Demirer, Esther Duflo, and Iván Fernández-Val. 2018. "Generic Machine Learning Inference

on Heterogeneous Treatment Effects in Randomized Experiments, with an Application to Immunization in India."

Chernozhukov, Victor, Christian Hansen, Nathan Kallus, Martin Spindler, and Vasilis Syrgkanis. 2024. *Applied Causal Inference Powered by ML and AI*. Authors.

Colnet, Bénédicte, Julie Josse, Gaël Varoquaux, and Erwan Scornet. 2023. "Risk Ratio, Odds Ratio, Risk Difference... Which Causal Measure Is Easier to Generalize?"

DasGupta, Anirban. 2008. *Asymptotic Theory of Statistics and Probability*. Springer.

Davidson, Russell, and James G. MacKinnon. 2004. *Econometric Theory and Methods*. Oxford University Press.

De Chaisemartin, Clément, and Xavier d'Haultfoeuille. 2020. "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review* 110 (9): 2964–96.

De Chaisemartin, Clément, and Jaime Ramirez-Cuellar. 2024. "At What Level Should One Cluster Standard Errors in Paired and Small-Strata Experiments?" *American Economic Journal: Applied Economics* 16 (1): 193–212.

De Finetti, Bruno. 1975. *Theory of Probability: A Critical Introductory Treatment*. John Wiley & Sons.

DeGroot, Morris H., and Mark J. Schervish. 2002. *Probability and Statistics*. Addison-Wesley.

Donald, Stephen G, and Kevin Lang. 2007. "Inference with Difference-in-Differences and Other Panel Data." *Review of Economics and Statistics* 89 (2): 221–33.

Doudchenko, Nikolay, and Guido W Imbens. 2016. "Balancing, Regression, Difference-in-Differences and Synthetic Control Methods: A Synthesis."

Duflo, Esther, Michael Kremer, and Jonathan Robinson. 2009. "Nudging Farmers to Use Fertilizer: Theory and Experimental Evidence from Kenya." Working Paper. *NBER Working Paper*, Working paper series, 15131. http://www.nber.org/papers/w15131.

Durrett, Rick. 2019. *Probability: Theory and Examples*. Cambridge University Press.

Eliason, Scott R. 1993. *Maximum Likelihood Estimation: Logic and Practice*. Newbury Park, CA: Sage Publications.

Elwert, Felix, and Christopher Winship. 2014. "Endogenous Selection Bias: The Problem of Conditioning on a Collider

Variable." *Annual Review of Sociology* 40 (1): 31–53.

Fang, Hanming, Michael P Keane, and Dan Silverman. 2008. "Sources of Advantageous Selection: Evidence from the Medigap Insurance Market." *Journal of Political Economy* 116 (2): 303–50.

Ferman, Bruno, and Cristine Pinto. 2019. "Inference in Differences-in-Differences with Few Treated Groups and Heteroskedasticity." *Review of Economics and Statistics* 101 (3): 452–67.

Filoso, Valerio. 2013. "Regression Anatomy, Revealed." *Stata Journal* 13 (1): 92–106.

Fisher, Ronald A. 1922. "On the Mathematical Foundations of Theoretical Statistics." *Philosophical Transactions of the Royal Society of London: Series A* 222 (594-604): 309–68.

Freedman, David A. 2006. "On the so-Called 'Huber Sandwich Estimator' and 'Robust Standard Errors'." *The American Statistician* 60 (4): 299–302.

Frisch, Ragnar, and Frederick V. Waugh. 1933. "Partial Time Regressions as Compared with Individual Trends." *Econometrica* 1 (4): 387–401.

Gailmard, Sean. 2014. *Statistical Modeling and Inference for Social Science*. Cambridge University Press.

Gauss, Carolo Friderico. 1857. *Theoria Motus Corporum Coelestium*. Little, Brown; Company.

Gelman, Andrew. 2013. "Commentary: P: Values and Statistical Practice." *Epidemiology* 24 (1): 69–72.

Gelman, Andrew, and John Carlin. 2014. "Beyond Power Calculations: Assessing Type s (Sign) and Type m (Magnitude) Errors." *Perspectives on Psychological Science* 9 (6): 641–51.

Gelman, Andrew, and Jennifer Hill. 2007. *Data Analysis Using Regression and Multilevel–Hierarchical Models*. Cambridge University Press.

Gelman, Andrew, and Eric Loken. 2013. "The Garden of Forking Paths."

Gelman, Andrew, and Hal Stern. 2006. "The Difference Between 'Significant' and 'Not Significant' Is Not Itself Statistically Significant." *The American Statistician* 60 (4): 328–31.

Gerber, Alan S, and Donald P Green. 2012. *Field Experiments: Design, Analysis, and Interpretation*. Norton.

Goldberger, Arthur S. 1991. *A Course in Econometrics*. Harvard University Press.

Goodman-Bacon, Andrew. 2021. "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics* 225 (2): 254–77.

Greene, William H. 2008. *Econometric Analysis*. 6th ed. Pearson.

Grimmer, Justin, Margaret E Roberts, and Brandon M Stewart. 2022. *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press.

Hald, Anders. 2008. *A History of Parametric Statistical Inference from Bernoulli to Fisher, 1713-1935*. Springer Science.

Hamilton, James D. 2020. *Time Series Analysis*. Princeton University Press.

Hansen, Bruce. 2022a. *Econometrics*. Princeton University Press.

———. 2022b. *Probability and Statistics for Economists*. Princeton University Press.

Hansen, Bruce E. 2024. "Standard Errors for Difference-in-Difference Regression."

Hastie, Trevor, Robert Tibshirani, Jerome H Friedman, and Jerome H Friedman. 2009. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer.

Hayashi, Fumio. 2000. *Econometrics*. Princeton University Press.

Heckman, James J, Sergio Urzua, and Edward Vytlacil. 2006. "Understanding Instrumental Variables in Models with Essential Heterogeneity." *Review of Economics and Statistics* 88 (3): 389–432.

Heckman, James J, and Edward Vytlacil. 2005. "Structural Equations, Treatment Effects, and Econometric Policy Evaluation 1." *Econometrica* 73 (3): 669–738.

Herron, Michael C. 2000. "Postestimation Uncertainty in Limited Dependent Variable Models." *Political Analysis* 8 (1): 83–98.

Holland, P. W. 1986. "Statistics and Causal Inference." *Journal of the American Statistical Association* 81 (396): 945–60.

Huber, Martin. 2023. *Causal Analysis*. MIT Press.

Huber, Peter J. 1967. "The Behavior of Maximum Likelihood Estimates Under Nonstandard Conditions." In *Proceedings*

of the Fifth Berkeley Symposium on Mathematical Statistics and Probability, 221–33.

Imai, Kosuke, and In Song Kim. 2021. "On the Use of Two-Way Fixed Effects Regression Models for Causal Inference with Panel Data." *Political Analysis* 29 (3): 405–15.

Imbens, Guido W, and Donald B Rubin. 2015. *Causal Inference in Statistics, Social, and Biomedical Sciences*. Cambridge University Press.

Jackman, Simon. 2009. *Bayesian Analysis for the Social Sciences*. Chichester: Wiley.

Jensen, Johan Ludwig William Valdemar. 1906. "Sur Les Fonctions Convexes Et Les inégalités Entre Les Valeurs Moyennes." *Acta Mathematica* 30 (1): 175–93.

Keane, Michael, and Timothy Neal. 2023. "Instrument Strength in IV Estimation and Inference: A Guide to Theory and Practice." *Journal of Econometrics* 235 (2): 1625–53.

Kennedy, Peter. 2003. *A Guide to Econometrics*. 5th Edition. Malden, MA: Blackwell Publishing.

Kloek, Teunis. 1981. "OLS Estimation in a Model Where a Microvariable Is Explained by Aggregates and Contemporaneous Disturbances Are Equicorrelated." *Econometrica*, 205–7.

Kolmogorov, Andrey. 1956. *Foundations of the Theory of Probability*. 2nd Edition. Chelsea Publishing.

Leamer, Edward E. 1983. "Let's Take the Con Out of Econometrics." *American Economic Review* 73 (1): 31–43.

Lechner, Michael. n.d. "The Estimation of Causal Effects by Difference-in-Difference Methods."

Lee, David S, Justin McCrary, Marcelo J Moreira, and Jack Porter. 2022. "Valid t-Ratio Inference for IV." *American Economic Review* 112 (10): 3260–90.

Legendre, Adrien Marie. 1806. *Nouvelles méthodes Pour La détermination Des Orbites Des Comètes*. Courcier.

Lewbel, Arthur. 2019. "The Identification Zoo: Meanings of Identification in Econometrics." *Journal of Economic Literature* 57 (4): 835–903.

Lewis, David. 1974. "Causation." *Journal of Philosophy* 70 (17): 556–67.

Liang, Kung-Yee, and Scott L Zeger. 1986. "Longitudinal Data Analysis Using Generalized Linear Models." *Biometrika* 73

(1): 13–22.

Lovell, Michael C. 1963. "Seasonal Adjustment of Economic Time Series and Multiple Regression Analysis." *Journal of the American Statistical Association* 58 (304): 993–1010.

Lundberg, Ian, Rebecca Johnson, and Brandon M. Stewart. 2021. "What Is Your Estimand? Defining the Target Quantity Connects Statistical Evidence to Theory." *American Sociological Review* 86 (3): 532–65.

Maddala, Gangadharrao Soundalyarao. 1983. *Limited-Dependent and Qualitative Variables in Econometrics*. Cambridge: Cambridge University Press.

Mankiw, N. Gregory, David Romer, and David N. Weil. 1992. "A Contribution to the Empirics of Economic Growth." *Quarterly Journal of Economics* 107 (2): 407–37.

Manski, Charles F. 1990. "Nonparametric Bounds on Treatment Effects." *American Economic Review* 80 (2): 319–23.

McElreath, Richard. 2018. *Statistical Rethinking: A Bayesian Course with Examples in r and Stan*. Chapman; Hall/CRC.

McFadden, Daniel. 1976. "Quantal Choice Analysis: A Survey." *Annals of Economic and Social Measurement* 5 (4): 363–90.

Merriman, Mansfield. 1877. "On the History of the Method of Least Squares." *The Analyst* 4 (2): 33–36.

Montgomery, Jacob M, Brendan Nyhan, and Michelle Torres. 2018. "How Conditioning on Posttreatment Variables Can Ruin Your Experiment and What to Do about It." *American Journal of Political Science* 62 (3): 760–75.

Morgan, Steven L., and Christopher Winship. 2014. *Counterfactuals and Causal Inference: Methods and Principles for Social Research*. 2nd Edition. Cambridge University Press.

Moulton, Brent R. 1986. "Random Group Effects and the Precision of Regression Estimates." *Journal of Econometrics* 32 (3): 385–97.

———. 1990. "An Illustration of a Pitfall in Estimating the Effects of Aggregate Variables on Micro Units." *Review of Economics and Statistics*, 334–38.

Murphy, Kevin P. 2022. *Probabilistic Machine Learning: An Introduction*. MIT Press.

Nardini, Cecilia. 2014. "The Ethics of Clinical Trials." *Ecancermedicalscience* 8.

Nelson, Charles, and Richard Startz. 1988. "The Distribu-

tion of the Instrumental Variables Estimator and Its t-Ratio When the Instrument Is a Poor One."

Oster, Emily. 2019. "Unobservable Selection and Coefficient Stability: Theory and Evidence." *Journal of Business & Economic Statistics* 37 (2): 187–204.

Paul, L. A., Edward J. Hall, and John Collins, eds. 2004. *Causation and Counterfactuals*. MIT Press.

Pearl, Judea. 2009. *Causality*. 2nd Edition. Cambridge University Press.

Pearl, Judea, and Elias Bareinboim. 2024. "External Validity: From Do-Calculus to Transportability Across Populations." *Statistical Science* 29 (4): 579–95.

Pearl, Judea, Madelyn Glymour, and Nicholas P Jewell. 2016. *Causal Inference in Statistics: A Primer*. John Wiley & Sons.

Ramsey, F. 1926. "Truth and Probability." *Studies in Subjective Probability*.

Reisach, Alexander G, Alberto Suárez, Sebastian Weichwald, and Antoine Chambaz. 2025. "The Case for Time in Causal DAGs."

Rosenbaum, Paul R, and Donald B Rubin. 1983. "The Central Role of the Propensity Score in Observational Studies for Causal Effects." *Biometrika* 70 (1): 41–55.

Rozanov, Y. A. 1969. *Probability Theory*. Dover.

Rubin, Donald B. 1974. "Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies." *Journal of Educational Psychology* 66 (5): 688–701.

Savage, Leonard J. 1951. "The Theory of Statistical Decisions." *Journal of the American Statistical Association* 46 (253): 55–67.

Solow, Robert M. 1956. "A Contribution to the Theory of Economic Growth." *Quarterly Journal of Economics* 70 (1): 65–94.

Sovey, Allison J., and Donald P. Green. 2011. "Instrumental Variables Estimation in Political Science: A Readers' Guide." *American Journal of Political Science* 55 (1): 188–200.

Stachurski, John. 2016. *A Primer in Econometric Theory*. MIT Press.

Staiger, Douglas, and James H. Stock. 1997. "Instrumental Variables Regression with Weak Instruments." *Economet-*

rica 65 (3): 557–86.

Stock, James H., and Mark W. Watson. 2011. *Introduction to Econometrics, 3rd Edition*. Pearson.

Tamer, Elie. 2010. "Partial Identification in Econometrics." *Annual Review of Economics* 2 (1): 167–95.

Tao, Terence. 2011. *An Introduction to Measure Theory*. Vol. 126. American Mathematical Society.

Teele, Dawn Langan. 2014. *Field Experiments and Their Critics: Essays on the Uses and Abuses of Experimentation in the Social Sciences*. Yale University Press.

Train, Kenneth, and Melvyn Weeks. 2005. *Discrete Choice Models in Preference Space and Willingness-to-Pay Space*. Springer.

Van der Vaart, Aad W. 2000. *Asymptotic Statistics*. Cambridge University Press.

Verbeek, Marno. 2018. *A Guide to Modern Econometrics 5th Edition*. Wiley.

Wald, Abraham. 1940. "The Fitting of Straight Lines If Both Variables Are Subject to Error." *Annals of Mathematical Statistics* 11 (3): 284–300.

White, Halbert. 1980. "A Heteroskedasticity-Consistent Covariance Matrix Estimator and a Direct Test for Heteroskedasticity." *Econometrica*, 817–38.

Wooldridge, Jeffrey M. 2002. *Econometric Analysis of Cross Section and Panel Data*. MIT Press.

———. 2009. *Introductory Econometrics*. 4t ed. South-Western Cengage Learning.

Xu, Yiqing. 2017. "Generalized Synthetic Control Method: Causal Inference with Interactive Fixed Effects Models." *Political Analysis* 25 (1): 57–76.

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